High Voltage Technology and Electrical Fields

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Introduction

The aim of this course is to understand the phenomenon of High Voltage (HV) Technology with respect to dimensioning rules, physical effects and application conditions. Based on a repetition of the theoretical basics, the specific conditions of high voltage engineering and development process will be explained more in detail. In this context, practical arrangements are considered in detail, examples are calculated and practical rules in the field of HV Technology will be given.
Why is HV Technology getting more and more important ? Considering the steadily increasing energy consumption in the world, the efficient energy transmission is already a technical task to be solved. To keep the ohmic losses small, the voltage level has to be increased. Especially in foreign countries with long transmission distances, the need for extra high voltage levels exists and actual developments lead to AC transmission line in the range of 800 kV to 1200 kV, which are applied for example in Canada, India, Russia and China. In Europe, the existing 420 kV-voltage level seems to be sufficient.

As alternative, the High Voltage Direct Current (HVDC) Transmission is very promising, since the dielectric capacitive losses are prevented. Via AC-DC converter, the 3-phase voltages are transferred to DC, afterwards the power is transferred by DC power and finally reconverted to AC again (via DC-AC converter). This principle requires a combination of different voltage types and wave forms. This course shows the main methods to generate the different voltage sources like AC, DC and lightning impulse voltages.
Furthermore the basics of the Electrical Field as main dimensioning parameter is explained and calculation methods are described based on the Maxwell equations.
The practical effects of discharges in gaseous, liquid and solid insulations are presented and calculated for special arrangements.
Additionally, the development process to construct high voltage equipment is discussed.
Finally the physical effect of lightning and thunder is explained together with the process of lightning discharge. As a result, various methods to protect humans and structures from being damaged by lightning are presented and calculated.

Learning Targets

This course is structured to illustrate the peculiarities of High Voltage Systems and to get you familiar with basic dimensioning rules. For this reason, theoretical and practical exercises will explain the discharge behavior of insulating materials.
Based on practical examples, the frame of a development process (pre-studies, feasibility studies, design comparison, development and type tests) is demonstrated.

With aid of this course, you will learn about:

- Basic dimensioning rules of HV-Systems, practical limits
- Development process: knowledge and verification
- Difference between suited and bad design
- Theoretical and practical sequence in case of HV-Tests
- Relationship between theory (Mathematics, Physics, Electrical) and Practical limits

With the discussed methods, you will be capable of making a design concept, apply dimensioning rules and create a development- and type- test plan. You will find related exercises within this course, which are being calculated during the lectures.

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The lecturer wish you many success !

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1 Introduction into High Voltage Technology

Within this chapter, you will get information, how the European power grid is structured in principle and which actual trends towards extra high voltages (EHV, UHV) are foreseen. Furthermore an overview over different voltage stresses in HV networks is given. The related definitions are considered.
Before starting into the field of high voltage technology, a short pre-chapter is dealing with some fundamental aspects of 3-phase power systems.

1.1 Basics with regard to 3-phase power systems

The 3-phase power system consists of three sinusoidal voltage sources (phase voltages) with a phase shift of three times 120°. Additionally a neutral and an earth conductor are included.

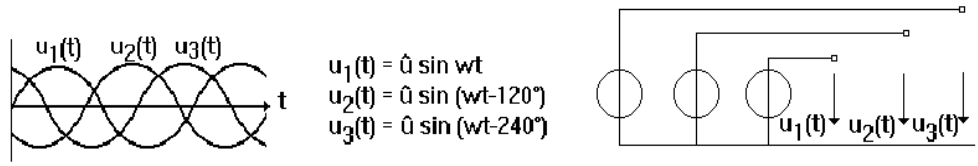


Fig. 1.1: Voltage Sources for 3-phase power system

Usually, the 3-phase power system is stressed symmetrically. In case of unsysmmetrical loads in one phase, the neutral conductor secures an exchange current back to the earthing point. The result out of this is, that the other phase voltages are not affected. The green-yellow marked conductor is representing the earthing connection.

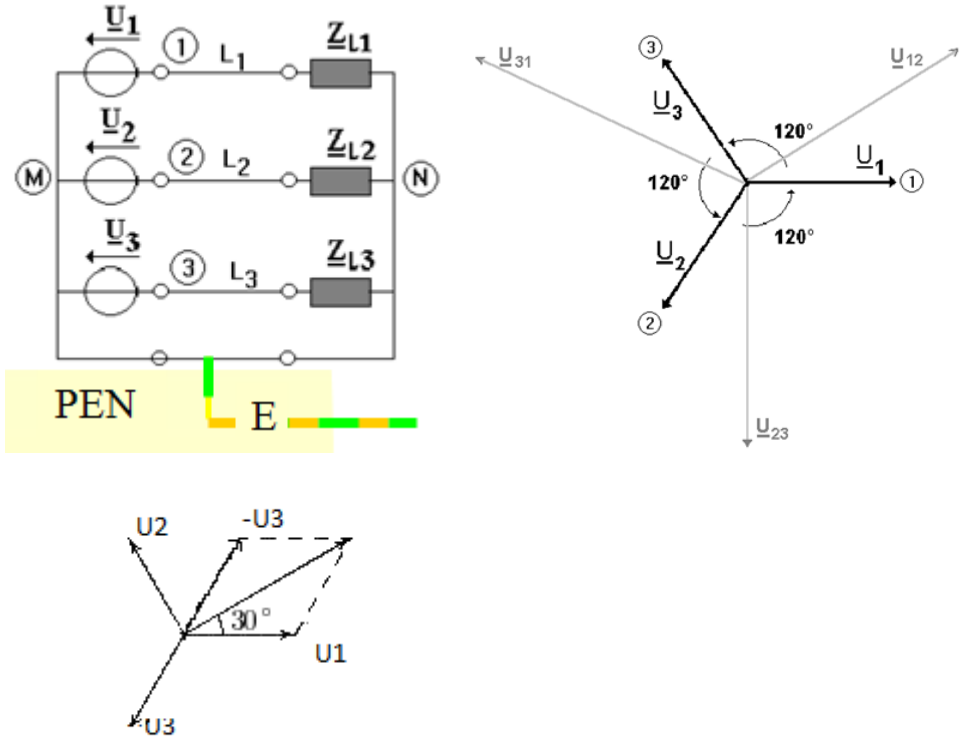


Fig. 1.2: Sources for 3-phase power system and example for superposition

For calculation reasons, it is recommended to fix the phase voltage U_1 onto the real axis. All other phase voltages (vectors) are referring to this real axis. In polar coordinates the following will result:

$$\underline{U}_1 = U \cdot e^{j0^\circ} \quad with \quad U = U_1 = U_2 = U_3$$

$$\underline{U}_2 = U \cdot e^{-j120^\circ}$$
$$\underline{U}_3 = U \cdot e^{-j240^\circ}$$

The **phase-to-phase voltages** are defined by:

$$\underline{U}_{12} = \underline{U}_1 - \underline{U}_2 = U(e^{j0^\circ} - e^{-j120^\circ})$$

Calculation Basics (Bronstein):

$$a = \alpha + j\beta \qquad b = \gamma + j\lambda$$

$$a \pm b = (\alpha \pm \gamma) + j(\beta \pm \lambda)$$

$$\underline{U}_{12} = U[(\cos 0^\circ + j \sin 0^\circ) - (\cos(-120^\circ) + \sin(-120^\circ))]$$

$$\underline{U}_{12} = U[(1 + j0) - (-0{,}5 + (-0{,}867))]$$

$$\underline{U}_{12} = U \cdot (1{,}5 + j(0 + \frac{\sqrt{3}}{2}))$$

with $$\left|1{,}5 + j\frac{\sqrt{3}}{2}\right| = \sqrt{(1{,}5)^2 + (\frac{\sqrt{3}}{2})^2} = \sqrt{2{,}25 + 0{,}75} = \sqrt{3}$$

and $$\varphi = \arctan\frac{\frac{\sqrt{3}}{2}}{\frac{3}{2}} = \arctan\frac{\sqrt{3}}{3} = 30°$$

the **phase-to-phase voltages results to**: $$\boxed{\underline{U}_{12} = \sqrt{3} \cdot U \cdot e^{j30°}}$$

Please note that the geometric addition according to Fig. 1.2 will lead to a more friendly solution.

Notification: All voltage values are rms values. The calculated phase shift of 30° is related to the real axis ! The network is characterized by the **phase-to-phase voltages**.

1.2 Tendency to higher Voltage Levels

In the early stage, in 1891 the first 3-phase power transmission was established, which connected the water plant in Lauffen with main city Frankfurt/M over a distance of 175 km. The operating voltage was 14 kV and the transferred energy was 140 kVA. Since that time, both parameters like voltage and energy has been increased steadily and the network structure was optimized such that in Europe a well cross-linked network is available called UCTE-Network. The following scheme is illustrating the functionality of a European network in principle.

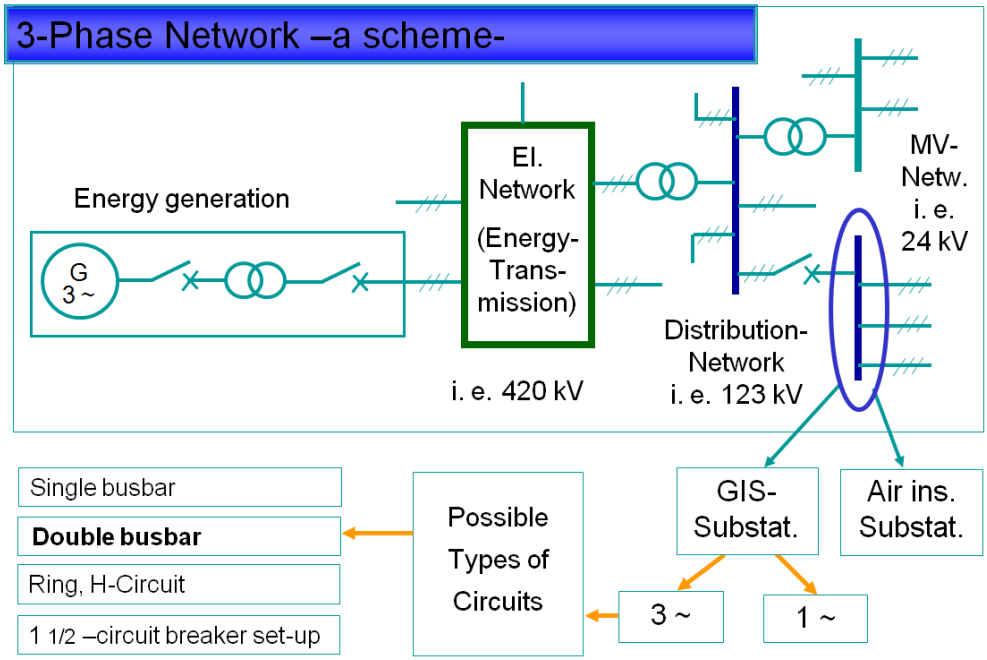


Fig. 1.3: 3-phase power system in Germany (scheme)

The energy is generated by 3-phase power generators with maximum operating voltages in the range of 12 to 36 kV. With aid of high power network Transformers, energies up to 1500 MVA are injected into the transmission network (420 kV), transferred over distances of some hundred kilometers and finally re-transformed to lower voltages such as U_m = 123 kV (for industry networks) or to U_m = 24 kV (for city networks). To serve private customers, the 400V-network is feeded by the 24 kV-network.

Usually, the networks are divided in
1. **Low** Voltage Networks (LV) up to 1000 V,
2. **Medium** Voltage Networks (MV) up to 40.5kV,
3. **High** Voltage Networks (HV) covering 52 kV up to 245 kV
4. **Ultra** High (UHV)/ **Extra** High (EHV) Voltage Networks covering the range of 300 kV up to 1200 kV.

In Europe, the HV is subdivided into Transmission Network (420 kV) and Distribution Network (123 kV).
The public energy consumption in Germany in the year 2004 at the rated power was 839100 MVA /Schlabbach,Metz/.

Why is it necessary to transfer the energy with high voltages ? The main reason is the quadratic dependency of the ohmic losses P_V on the transmision current ***I***.

Based on a 3-phase system with Line to Earth Voltage $\boldsymbol{U_{LE}}$ [Phase Voltage U_1, as explained before] and a Line to Line Voltage **U**, [Phase to Phase Voltage u_{12}] the following results are deduced:

Three phase ohmic losses are given by-

$$P_V = 3 \cdot R_L \cdot I^2$$

and

Three Phase transferred Power is given by -

$$S_{\ddot{U}} = 3 \cdot U_{LE} \cdot I = 3 \cdot \frac{U}{\sqrt{3}} \cdot I = \sqrt{3} \cdot U \cdot I$$

If the transferred power $\boldsymbol{S_{\ddot{U}}}$ is of high magnitude, the ohmic losses $\boldsymbol{P_V}$ can be reduced quadratical [$P_v \propto I^2$] by reducing the transmission current I. Assuming that the transferred power should be constant, the voltage level has to be increased:

$$P_V = 3 \cdot R_L \cdot \frac{S_{\ddot{U}}^2}{(3 \cdot U_{LE})^2} = \frac{1}{3} \cdot R_L \cdot \frac{S_{\ddot{U}}^2}{U_{LE}^2} \qquad \text{with} \qquad I = \frac{S_{\ddot{U}}}{3 \cdot U_{LE}}$$

The energy transmission with high efficiency, low losses and environmental friendly conditions can only be realized by using **high** voltages. Of course, there is an economic limit, which is determined by the increasing **insulating costs**.

Empirical Formula: It can be concluded that the voltage in **kV** should be chosen such that it represents the transmission distance in **km** (1km $\rightarrow$ 1kV).

Please note: The dimensioning of high voltage components is the most important issue. For that reason, the maximum operating voltage U_m is relevant, not the service voltage U_N (i.e.: Um = 123 kV, U_N = 110 kV /115 kV. Source: DIN IEC 60038, Table 3.3)

Actually, 3-phase power systems are developed with voltages above 1000 kV. Latest project is the 1000 kV-Transmission in China, started as a pilot project in 2009 and in the first step connecting a distance of 640 km.

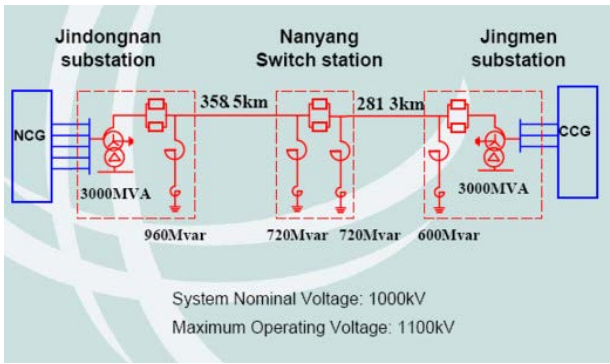


Fig. 1.4:
1000 kV-Pilot project China.
Source: ABB

As you can see from Fig. 1.4, feeding from both sides is possible with rated power up to 3000 MVA. The pilot character is visible by the fact that the substation in the middle does not provide a consumer connection. The high rate of capacitive reactive power is compensated at four locations (left side: 960 Mvar, 2x720 Mvar and 600 Mvar on the right side).

The next figure (source: ABB) shows different kind of UHV and EHV transmission lines:

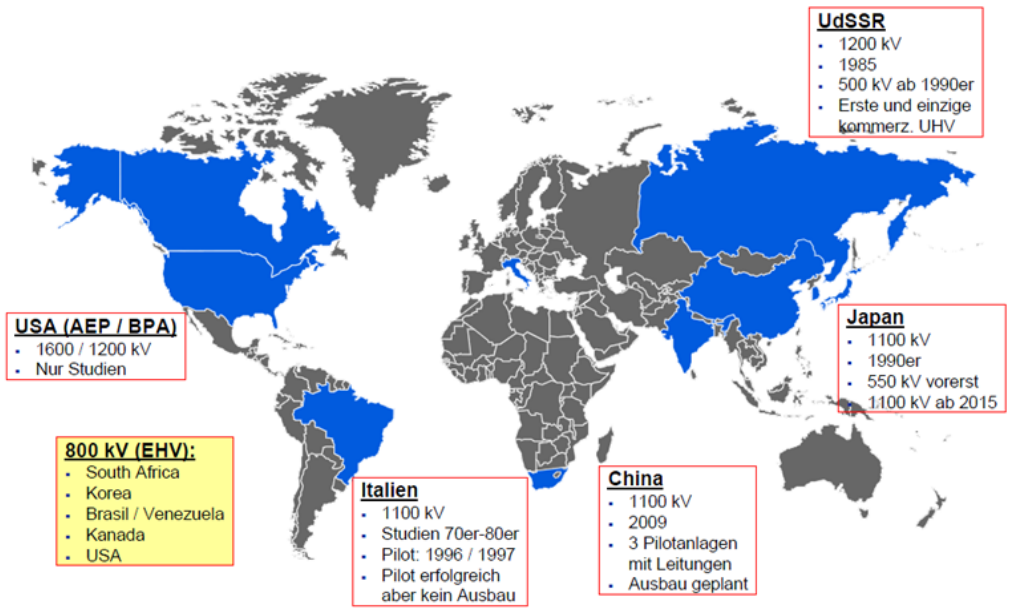


Fig. 1.5: EHV/UHV Transmission Lines Worldwide

The yellow marked 800 kV-transmission lines are commercially in service. Above 800 kV, actually all transmission lines have pilot character. It has to be mentioned, that in Russia the rated 1200 kV-Network is in commercial use only up to 500 kV. In China, the 1000 kV-Network is foreseen to be changed from pilot stage to extended commercial stage.

1.3 Application examples of high voltage equipment and components

To make the challenges of such ultra high voltages visible, the next figure shows a part of a transmission line for 1000 kV. The height of the pylons carrying the transmission lines starts with 77 meter and will be increased to 122 meter, when a river has to be crossed. This is necessary to ensure the needed distance between the conductors and the earth or an object to prevent a discharge and also to reduce the resulting field strength at earth.

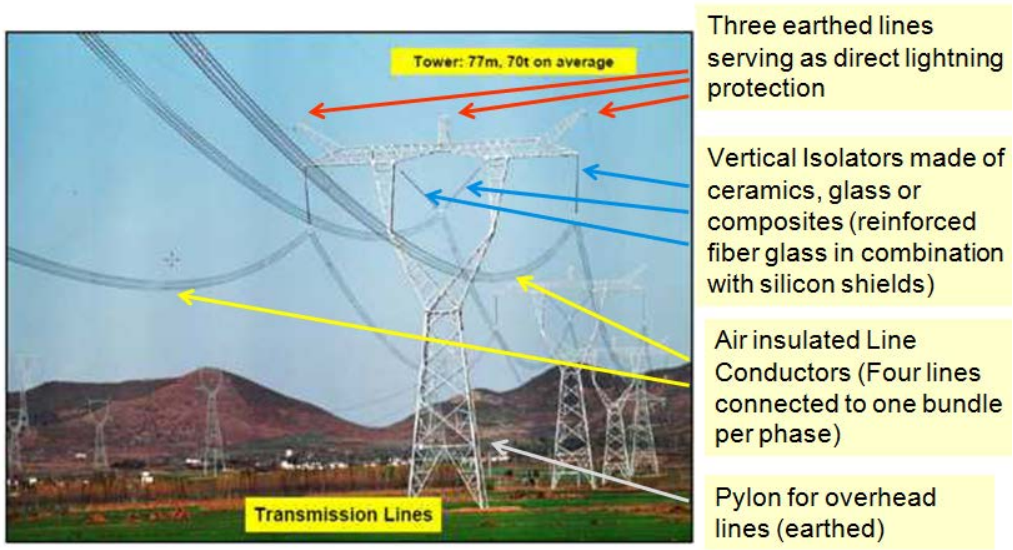


Fig. 1.6: Overhead Line 1000 kV: Pilot Transmission China

Lightning Protection

Usually, in case of voltage levels up to 420 kV, a **single** earthed line on top of the pylon is responsible to **protect** the transmission lines against **direct lightning**. In areas with a high probability of lightning, the amount of earthed lines is increased to two. As you can see from Fig. 1.6, in case of 1000 kV the needed distance from line to line increases the necessary protection area and results in **three** earthed lines.

For mechanical fixation of the transmission lines, several straight-line insulators are used. Nowadays, such insulators are compound insulators consisting of a core of glass fiber re-inforced plastic to ensure the mechanical stability (wind and short circuit forces) and many silicon rubber shields to ensure the electrical stability. The silicon rubber shields are not sensitive to ageing and can handle **climate and environmental stress** (pollution, UV-light).

Suspension Insulators

Conductor, conductor bundle

Reduction in field strength, Reduction of the corona losses by four bundled Conductors. The most important component is the current carrying line. The usage of four line conductors in a bundled form is the well proven method in the voltage range of 420 kV and also applied at 1000 kV. This has two advantages: first the resulting ohmic resistance is reduced down to a one-fourth, secondly the **electrical field strength** of the transmission line is **significantly reduced** (reason: the surface surrounding the four conductors is enlarged leading to reduced radii and therefore reduces or even prevents the corona losses.

As mechanical structure of the transmission line, the pylon has a height of 77 meter and a weight of 70 tons. This structure has to **withstand all kinds of forces** like ice, wind, lengthening/shortening and –in case of direct lightning onto the earthed lines- the pylon has to carry the short circuit currents to earth. Considering the last point, it is of high importance that the conductivity of the earth surface (on which the pylon is built) is high. Otherwise, it may happen that this earth resistance is so high, that the voltage along this earth resistance exceed the insulating level of the conducting line to the pylon. In this case, a backward flashover would be induced which has a high voltage steepness and could create big damage.

According to this practical application example, several aspects regarding High Voltage component can be derived from following factors:

- Dielectric stress (Deciding Factors-Line to Earth Voltage U_{LE}, Line to Line Voltage U, temporarily over voltages, switching over voltages, lightning over voltages)
- Discharge behavior of gaseous, solid and liquid insulating materials
- Generation of test voltages of high magnitude to prove the dielectric strength of HV-Equipment
- Measures to prevent direct lightning into high voltage components (earthed line, Earth resistance, lightning impulse withstand strength of straight-line insulators)
- Calculation of the dielectric field strength (Four-Line-Bundle), Reduction of the corona losses (small partial discharges from the line conductor to the near environment of the surrounding air, resulting in corona losses, but not leading to complete discharges)

Application example for DC-voltages up to 50 kV:

- Electron beam melting processes with following parameters:
  - Electrical power in the order of 100 kW to 600 kW
  - DC-Voltages from 25 kV to 45 kV
  - Electron gun working within a vacuum environment

Principle of electron beam melting: free electrons are generated by a DC source of some kilovolts. These electrons are highly accelerated by an electrical field generated by a DC-source in the range of up to 45 kV. When these electrons of high energy collides with the target (i.e material alloys) with typical power densities of 100 W/cm^2, the material (metal or non-metallic constituents) with vapor pressures higher than the base material will be melted and evaporated thus generating the desired high purity of the ingot material (i.e. tantalum, niobium, molybdenum, tungsten, titanium etc.). The exposure of the super-heated metal pool surface to a high vacuum environment (down to 0.001 Pa) results in excellent degassing of the molten material (ref-/ALD/).

The Electron Beam Melting Unit consists of the following main components:



- two electron beam guns
- Control of magnetic deflection in order to direct the electron beam to the shape of the target
- High voltage power transformer with integrated
- melting chamber within a vacuum environment

Fig. 1.7: DC-Application example: Electron Beam Melting [Source: ALD]

1.4 Overview over different types and shapes of voltages appearing in real power networks

Apart from the well known 50 and 60 Hz-alternating voltage there exist a lot of different types of voltages and voltage shapes. Mainly it can be divided into permanent and temporary voltages. The following figure describes different types of voltages appearing in real networks and which are arranged starting with zero frequency (DC) and leading to voltage types of very high frequency.

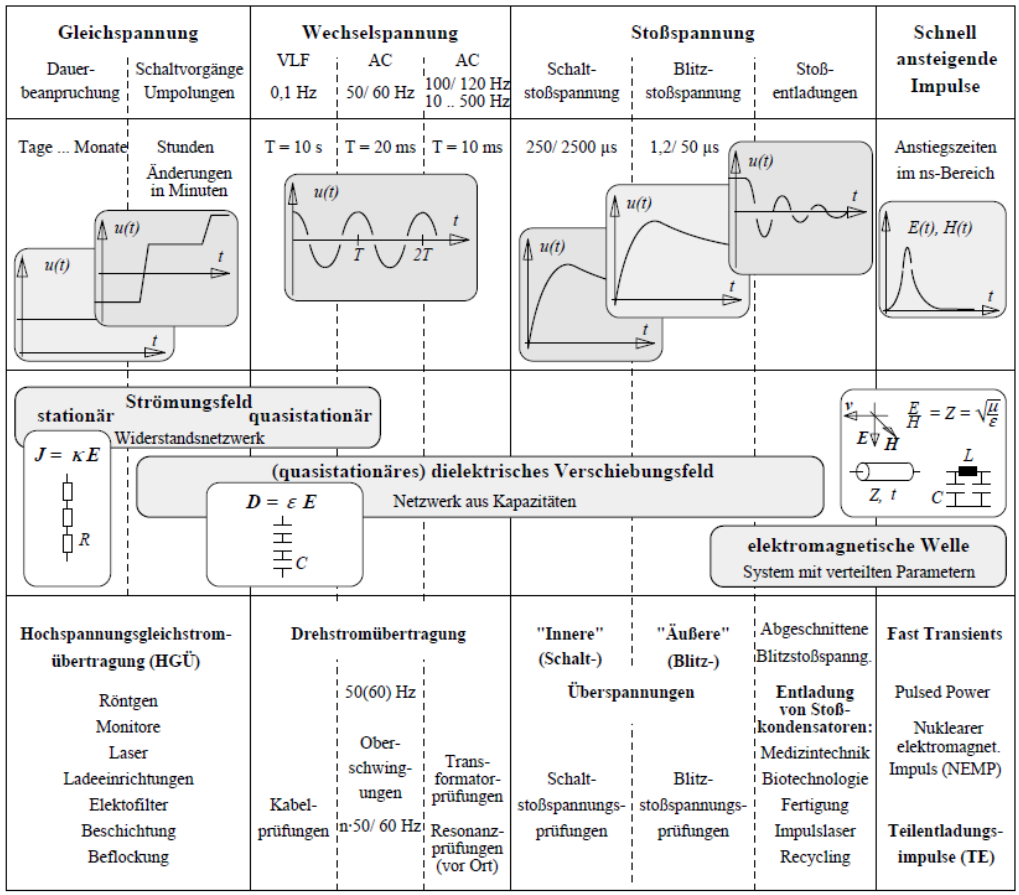


Fig. 1.8: Overview of characteristic types and shapes of voltage appearing in power networks (Source: Küchler)

An application example, which contains nearly all kinds of voltage types, is represented by the High Voltage Direct Current (HVDC) Transmission as illustrated in figure 1.8:

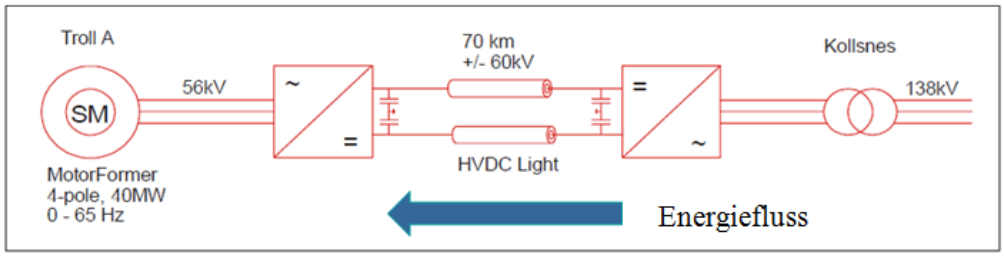


Fig. 1.9: HVDC-Connection between coast (Kollsnes) and oil platform (Troll A) using IGBT-Technology (HVDC "light").

Starting from a 138 kV / 50 Hz-3-phase power network, the conversion from AC to DC is realised by special rectifier (in reality a big converter hall), which leads to bipolar DC voltages of +/- 60 kV. With aid of XLPE-DC-Sea Cables the energy is transferred with high efficiency, low losses and without retroaction to the oil platform. At this point a second converter hall ensures the reversion of the energy into AC. This is realized with newest DCAC-converters using IGBT-Technology, which guarantee a frequency variable AC voltage (ref - /Hanson/). The advantage of such technology is that the DCAC-converters are able to control both the active and reactive power independently from the frequency. As a result the wind wheels can be driven in the most efficient speed depending on the wind force.
The voltage stress covers the range from **variable frequency voltages** (0-65 Hz), **DC** voltages (+/-60 kV), **mixed** AC and DC voltages at the converter stations towards rapid control **impulses** and **harmonics**. At the coast side there is a probability of direct lightning, which leads to **lightning impulse voltages** of high magnitude. **Switching over voltages** also have to be mentioned.

Please note: In 3-phase power systems up to 300 kV, **the lightning impulse voltage** determines the dielectric limit. In systems above 300 kV **the switching impulse voltage** is more critical.



Exercises

1.1 Please explain the difference between the service voltage U_S and the maximum operating voltage U_m ?

1.2 Please sort the following types of voltage with regard to increasing voltage level: Amplitude of Line to Line voltage, Line to Line Voltage, Phase to earth voltage, lightning impulse voltage, AC test voltage.

1.3 Application example:

Assuming a 123kV-Level (U_S = 110 kV) and a 36kV-level (U_S=34 kV). The maximum possible apparent power has to be calculated transmission line (cross section of steel 36 mm^2 and cross section of aluminum 210 mm^2, current density 0,8 A/mm^2) for a transmission distance of 100 km.

U_m	123		36		36	
U_S	**110**		**34**		**34**	
A_{Alu}	210		210		210	
A_{Stahl}	36		36		36	
S = 0,8 A/mm^2	0,8		0,8		0,8	
I	**168**		168		**?**	
$S_Ü$	32	MVA	?	MVA	32	MVA
Transmission distance	100	km				
Spec. Resistance	27mΩ*mm^2/m					
R=ρ*l/A	12.86	Ohm	12.86	Ohm	12.86	Ohm
$Pv=3*R_V*I^2$	1.1	MW	?	MW	?	MW

Which conclusion do you derive from the calculation above ?

- In the case of different voltages and identical currents
- In the case of different voltages and identical apparent power

2 Generation and Measuring of High Voltages

The aim of this chapter is to give you a basic understanding about the generation of high voltages and how to measure the different types of voltage.
Different concepts are described how to dimension a high voltage circuit, which kind of set-up is suited and how to calculate such circuits.
In the first chapter different types of voltages like DC, AC, Mixed voltage and Impulse voltages are mentioned. Nearly all of them need special measuring devices/principles. One of the exceptions is the direct measuring method of the **measuring spark gap** (sphere gap), which is suited for all kind of voltage types. The measuring result is the amplitude of the unknown voltage type. The measuring principle uses the discharge voltage as basis. Since the discharge voltage of the sphere gap is determined by the geometry (distance d and diameter of the spheres D), the voltage before discharge is known. The following figure shows a measuring spark gap with diameter D = 25 cm and gap distance of 5 cm:

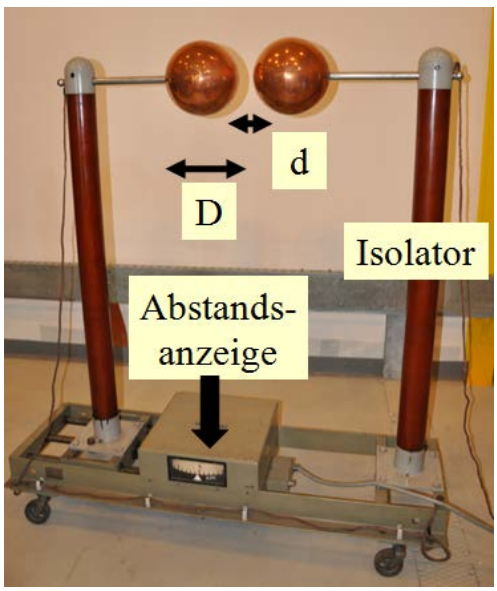


Fig. 2.1:
Measuring spark gap located in the HV-Laboratory of Darmstadt, University of Technology

Assuming, AC voltage is applied to the right sphere and the left sphere is earthed, it is possible to measure the _____ of the sinusoidal voltage curve (please decide: momentarily voltage value, peak value of rms-value of the voltage).
Example: we want to determine the exact discharge voltage based on special diagrams as a function of the geometry. As you can derive from Fig. 2.2, a voltage value of **137 kV** can be detected. In case of AC- (or Impulse-) Voltages, it has an **accuracy of +/- 3 %** and is valid under condition, that the

measuring voltage range in kV is smaller than the sphere diameter in mm (Prove: 137 (kV) < 250 (mm) => Condition fulfilled).

Please note: for DC-Voltages the accuracy is reduced to +/-5 %.

Tabelle 6.3.1-1: Scheitelwerte der 50 %-Durchschlagspannung von einpolig geerdeten Kugelfunkenstrecken in kV für verschiedene Kugeldurchmesser D und Schlagweiten d bei Gleich-, Wechsel- und negativer Stoßspannung (linke Spalte) sowie positiver Stoßspannung (rechte Spalte) unter atmosphärischen Normalbedingungen (T = 20 °C, p = 1013 mbar = 1013 hPa = 760 Torr).

Die Stoßspannungswerte gelten für Rückenzeiten größer oder gleich 50 µs, der Polaritätseffekt bei positiver Stoßspannung ist durch kursive Zahlen gekennzeichnet (vgl. Anmerkung im Text). Die fett gedruckten Zahlen entsprechen etwa den Werten des homogenen Feldes und gelten auch für größere Kugeldurchmesser D. Die Messunsicherheiten werden bei Wechsel- und Stoßspannung für d < D/2 mit ±3 % und bei Gleichspannung für d < 0,8·D/2 mit ±5 % angenommen, die in Klammern gesetzten Werte weisen größere, ungeklärte Streuungen auf.

d cm	D = 5 cm = ~	5 cm ⎍	D = 10 cm = ~	10 cm ⎍	D = 15 cm = ~	15 cm ⎍	D = 25 cm = ~	25 cm ⎍	D = 50 cm = ~	50 cm ⎍	D = 75 cm = ~	75 cm ⎍	D = 100 cm = ~	100 cm ⎍	d cm
0,5	17,4	17,4	**16,8**	**16,8**	→										0,5
1	32,0	32,0	**31,7**	**31,7**	→										1
1,5	45,5	*46,2*	**45,5**	**45,5**	→										1,5
2	57,5	*59,5*	**59,0**	**59,0**	→										2
2,4	65,5	*69,0*	69,5	*70,0*	**70,0**	**70,0**	→								2,4
3	(75,5)	*(81,0)*	84,0	*85,5*	85,5	85,5	**86,0**	**86,0**	→						3
4	(88,5)	*(97,5)*	105	*109*	110	*111*	**112**	**112**	→						4
5			123	*130*	133	*136*	137	*138*	**138**	**138**	→				5
6			(138)	*(148)*	152	*158*	161	*163*	**164**	**164**	→				6
7			(150)	*(163)*	169	*178*	184	*187*	189	189	**190**	**190**	→		7
8					(185)	*(196)*	206	*211*	214	214	**215**	**215**	→		8
9					(198)	*(212)*	226	*233*	239	239	240	240	**241**	**241**	9
10					(209)	*(226)*	244	*254*	263	263	265	265	**266**	**266**	10
12					(229)	*(249)*	275	*291*	309	*311*	315	315	**318**	**318**	12
14							(302)	*(323)*	353	*357*	363	363	**366**	**366**	14
16	Bereich mit						(326)	*(350)*	392	*402*	410	*411*	**414**	**414**	16
18	erheblichen Streuungen						(347)	*(374)*	429	*442*	453	*458*	**462**	**462**	18
20	und extern beeinflusstem Feld						(366)	*(395)*	460	*480*	492	*505*	**510**	**510**	20
24									515	*540*	565	*585*	595	*600*	24
30									(585)	*(620)*	665	*695*	710	*725*	30
40									(670)	*(715)*	(800)	*(835)*	875	*900*	40
50	*Ausführlichere Tabellen finden sich in*										(895)	*(940)*	1010	*1040*	50
60	*den einschlägigen Normen [142], [143]*										(970)	*(1020)*	(1110)	*(1150)*	60
70													(1200)	*(1240)*	70

Bereich mit nahezu homogenem Feld

$\hat{U}_{d50(0)}$ kV

Fig. 2.2: Breakdown voltage of sphere gaps [Source: Küchler]

Please consider that the resulting voltage of 137 kV is related to a temperature of ϑ_U = 20° C and an atmospheric pressure of p = 1013 mbar. In case of different conditions, a correction factor has to be considered:

$$\hat{u}_{d50} \approx \hat{u}_{d50(0)} \cdot \frac{p}{1013\,mbar} \cdot \frac{273K + 20\,°C}{273K + \vartheta_U\,(in\,°C)}$$

The discrepancy from the original value should be in the range of 0,9 to 1,1. The measuring sphere gap is highly suited for exploratory measurements, but not suited for standardized measuring.

In this context, it should be mentioned that there exist another measuring system which works as absolute voltage monitor for DC and AC-Voltages: the

electrostatic voltmeter. Catchwords: Field force, electrostatic deflection, homogeneous plate capacitor, rms voltage value. More details in /Schwab/.
Additional measuring methods which are related to the special type of voltages will be discussed in the following chapters.

2.1 Generation and Measuring of AC-Voltages of high magnitude

In case of power networks, the power transformers are constructed with 3-phases, whereas the test transformers for high voltage tests are always constructed in 1-phase set-ups. This has economic and test sequence reasons. To test a 3-phase test object, the 3 phases have to be tested sequentially.

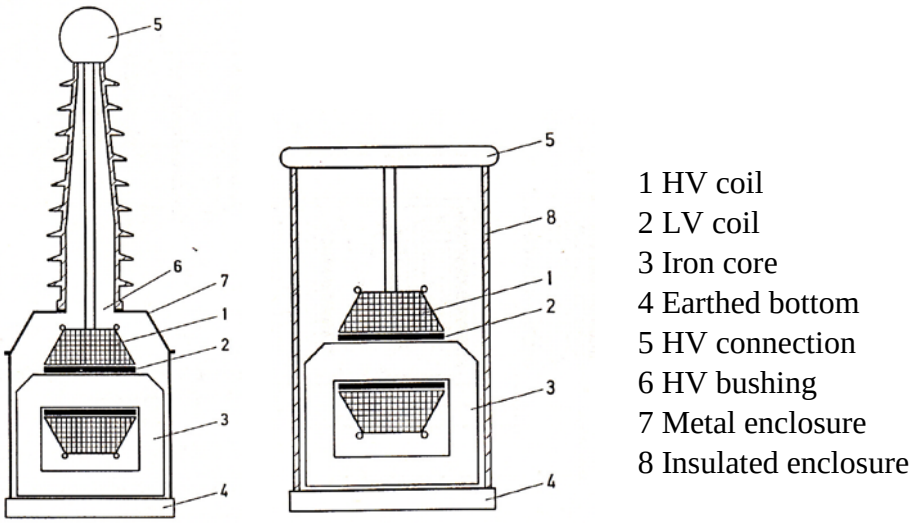


Fig 2.3: **Set-up of test transformers:**
a) Vessel set-up, b) Insulating sheath set-up

[Source: Kind/Kärner]

Fig. 2.3 shows two different setups: the vessel set-up is made from metal enclosure, which highly transfers the heat, but needs a expensive bushing.
The set-up with insulating sheaths is less expensive and can be equipped with large dielectric shielding.

Every technical equipment, which leaves the factory, must have been stressed successfully with AC-Test Voltages for a time period of 60 seconds.
For generation of such high AC-Voltages, usually a power testing transformer is used:

- Three test transformers in series resulting in 300 kV, 50 mA
- Single Test Transformer providing 200 kV, 125 mA
- Tesla-Transformer 300 kV, 1mA (constructed at HDA)

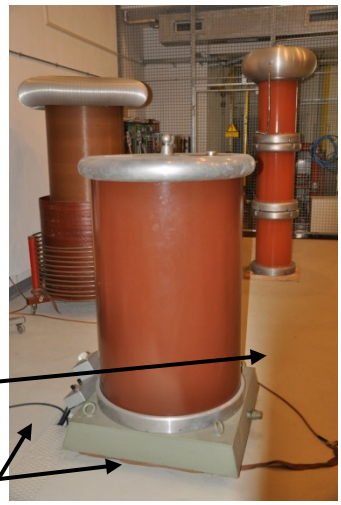


Fig. 2.4: Test Transformers

To provide cost effective AC test voltages of high magnitude, usually several modular test transformers are connected in series, so called „cascade“. Such a test set-up of AC-Cascade is visible in the next figure:



Fig. 2.5: Test Cascade

This AC-voltage cascade consists of two 400 kV test transformers connected in series. An aluminum bar is connecting the HV to one phase of the 3-phase GIS test object.

The related equivalent circuit of such a cascade is shown in the following figure:

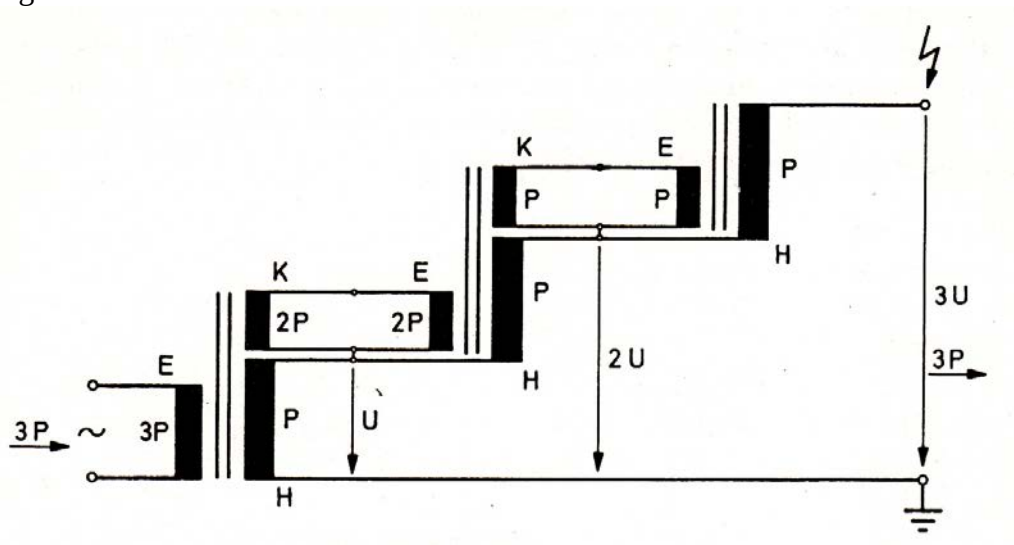


Fig. 2.6 Schematic set-up of a Cascade (E = Excitation coil, K = Coupling coil, H = HV Coil). Source: Kind/Feser

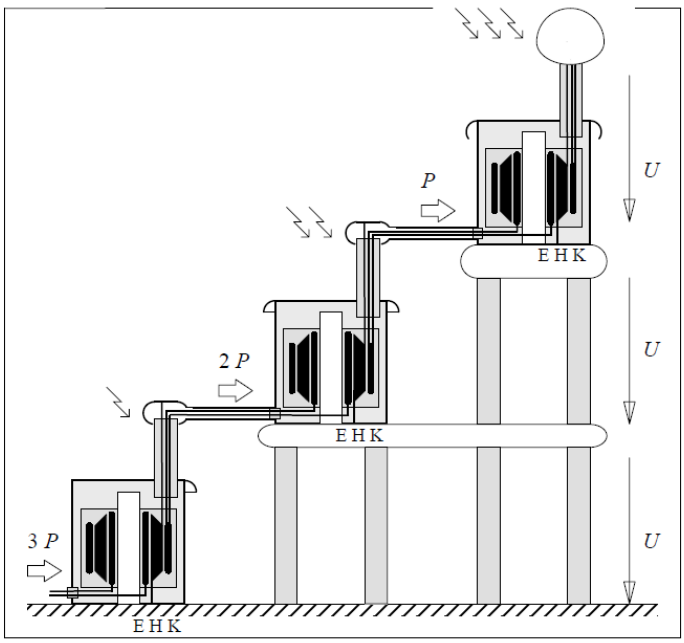


Fig. 2.7 Geometrical Set-up of a Cascade [Source: Küchler]

It has to be considered that the lower excitation coil has to feed the two upper excitation coils. Therefore the power of the lower coil has to manage three times the power of the upper coil. Consequently the short circuit voltage increases.

The following table is representing an extract of IEC 60694 :

Bemessungs-spannung Ur /kV rms/	Prüfwechselspannung Ud /kV rms/		Blitzstoßspannung Up /kV peak/	
	Phase-Phase, Phase-Erde	Über die Trennstrecke	Phase-Phase, Phase-Erde	Über die Trennstrecke
72,5 kV	140	160	325	375
123 kV	230	270	550	650
245 kV	360	415	850	950
	395	460	950	1050
	460	530	1050	1200

Table 2.1
Test Voltages

Which rated voltage could be tested with a Cascade of 2 x 200 kV ? Please consider that there are two possibilities. It should be assumed that the test object has no disconnecting gaps.

The following measuring systems are suited for AC-measuring /Kind, Feser/:

- Capacitive Divider (Earth capacitances to be considered ! Accuracy < 1 %)
- Peak voltage measuring devices according „Chubb und Fortescue“
- Peak voltage measuring with capacitive divider
- Electrostatic voltmeter
- Voltage transformer (inductive and capacitive voltage transformer, accuracies significantly smaller than 1 %, suited for calibration purposes)

2.2 Generation and Measuring of Impulse-Voltages of high magnitude

The limit of the voltage withstand capability of high voltage equipments up to 300 kV is given by the lightning impulse voltage.

As an example, such lightning impulse voltages are generated by so called

"Marx (Impulse-) Generators.
A simplified working principle is described now as follows: a DC-Voltage source will be connected to an arrangement, which contains a charging and a discharging unit with different time constants. The charging and discharging circuit consists of Resistors and Capacitances and will be connected to the DC-source via a switch. In the moment of closing the switch, both charging and discharging circuits are superposed resulting in two exponential voltage curves forming the lightning impulse voltage. The rising part is mainly determined by the charging circuit, whereas the depression part is mainly influenced by the discharging circuit.

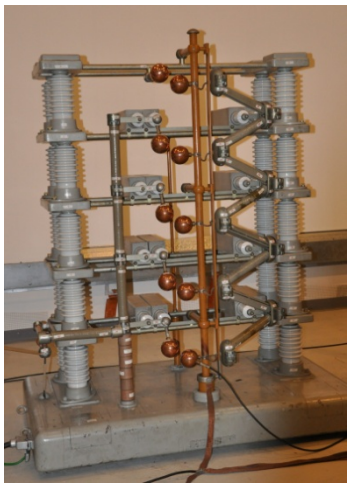


Fig. 2.8: Lightning Impulse Generator up to 400 kV

The figure above shows the impulse generator from Darmstadt, University of applied sciences (High Voltage Laboratory), which can deliver impulse voltages up to 400 kV. The generator consists of 4 units connected in series. The equivalent network of one unit is given in the following figure:

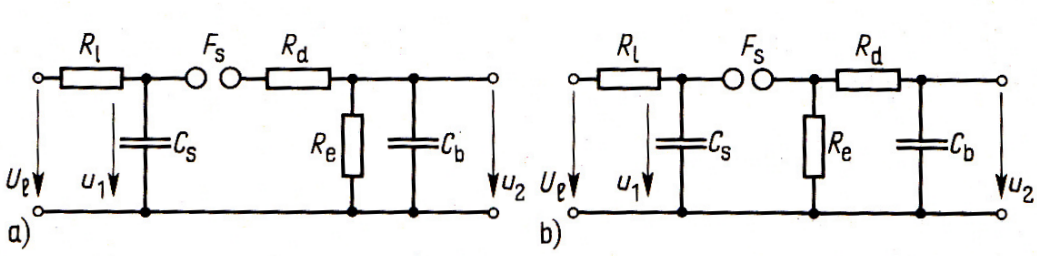


Fig. 2.9 Basic arrangements of single unit Impulse Generators
a) and b) both suited for lightning impulse voltages
b) advantageously in case of small load capacitances

U_l	DC-Voltage	C_S	Impulse Capacitance
R_l	Charging Resistance	F_S	Spark gap (switch)

R_E Discharging Resistance R_D (Charge-) Damping Resistance
C_B Load Capacitance

The principle of function is demonstrated in the next figure. Applying a DC-Voltage with a charging voltage U_l, this Source will be discharged down to the value $U_{2a\infty}$.
This value $U_{2a\infty}$ corresponds with the theoretical final voltage of the charging circuit (which will never be reached due to parallel discharging).
The curve U_{2a} shows the theoretical charging behavior, whereas curve U_{2e} shows the theoretical discharging behavior. As result, the impulse voltage curve U_2 is created by superposition of both circuits. It is also possible to calculate the curve progression of U_2 with aid of differential equations, which is done later.

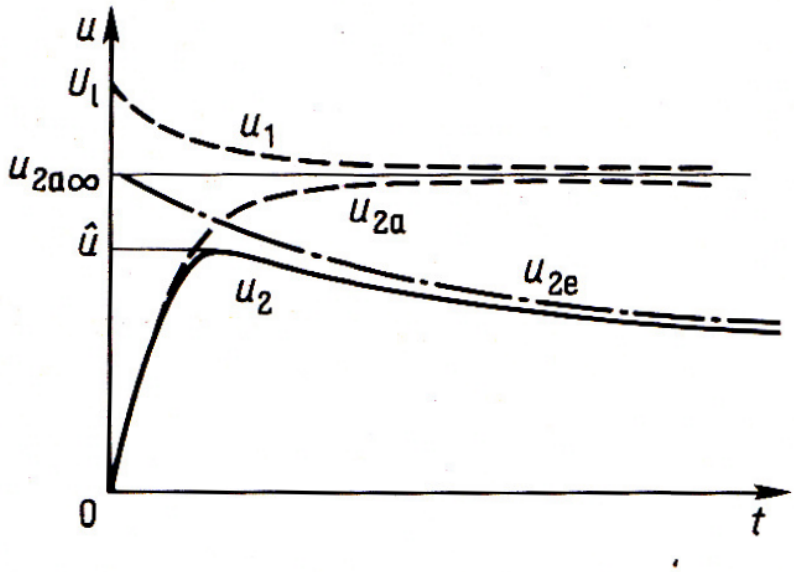


Fig. 2.10: Generation of Impulse Voltages by superposition of charging and discharging circuits

The characteristic of Impulse Voltages is defined by rise time T_1 and time to half-value T_2. In case of lightning impulse voltages T_1 corresponds with the linear part of the voltage rise with a time constant of 1,2 Microseconds (1,2 µs).

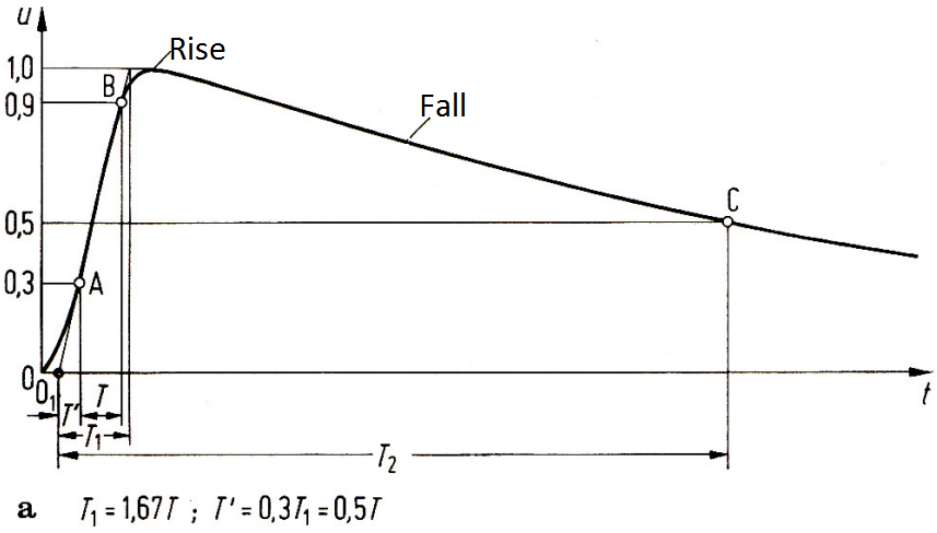


Fig. 2.11 Scheme of a standardized lightning impulse voltage

The time to half-value T_2 defines the time period from starting point of linear rise until the voltage is reduced to half-value of the peak voltage, in case of lightning impulse voltages it is defined to 50 Microseconds (50 µs). The overall short description of such impulse voltage is 1,2/50 µs.
A special variant of the impulse voltage is the cutting off in the tail of the impulse voltage. This test variant represents the effect of delayed breakdown of a insulating system. Especially for voltage transformers this cut off test applies a high stress.

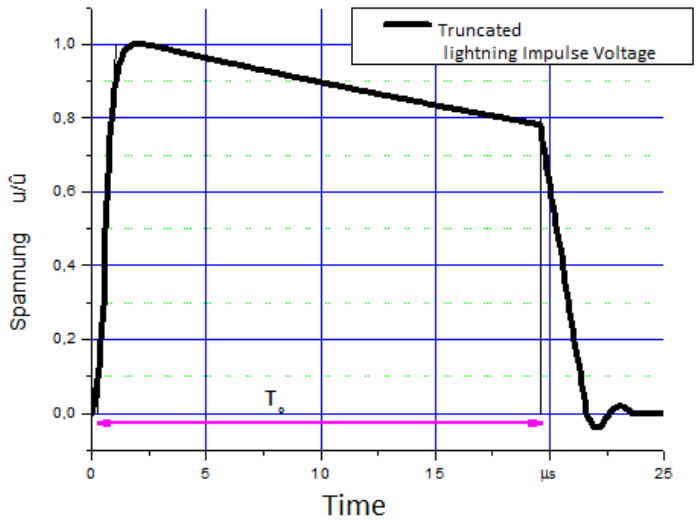


Fig. 2.12 Example of a impulse voltage with cutting-off after 20 µs

The time period T_C represents the breakdown or cutting-off-time, respectively. It can be derived from the figure above, that the dU/dt (falling slope) of the voltage breakdown is lower than the dU/dt (rising slope) of the voltage rise. In reality, this voltage breakdown can be much faster resulting in so called travelling waves. Such waves can stimulate high over voltages, at voltage transformer coils.

As a repetition of Fig. 2.9, the procedure of generation high impulse voltages is explained based on the following figure:

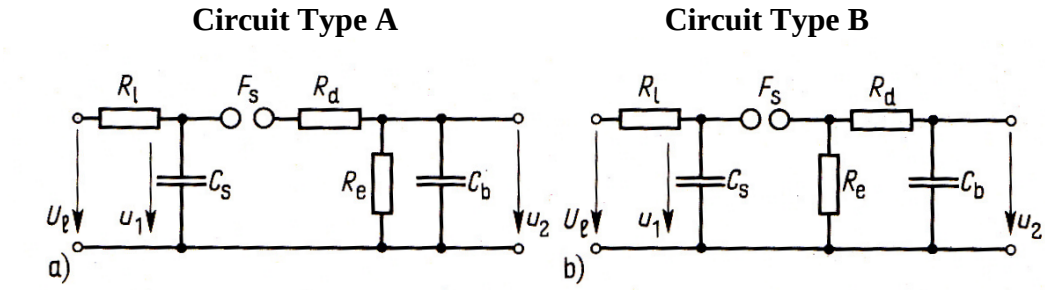


Fig. 2.13 Basic arrangements of single unit Impulse Generators

The impulse capacitor C_S is charged by a DC-source or a AC-source with rectifiers. C_S should be chosen 10 times higher in magnitude than the load capacitor C_b. Furthermore, the load capacitor should be 10 times higher than the capacitance of the test object C_{TO}. For exact ratio calculation, the measuring capacitances C_M and the cable capacitance C_C has to be considered. The resulting load capacitor is:

$$C_b = C_{b,Circuit} + C_{TO} + C_M + C_C$$

The switch is closed, when the sphere gap flashes over. Then a reloading appears from capacitor C_S to C_b following a differential equation of second order:

$$\frac{d^2u(t)}{dt^2} + B_1\frac{du(t)}{dt} + B_0u(t) = 0 \quad \Rightarrow u(t) = \frac{U_0}{K}\frac{1}{(\alpha_2 - \alpha_1)}\left(e^{-\alpha_1 t} - e^{-\alpha_2 t}\right)$$

Under condition that $R_E{*}C_S >> R_D{*}C_B$ the following factors are valid:

	B_1	B_0	K
Circuit A	$\frac{1}{R_d C_b} + \frac{1}{R_d C_s} + \frac{1}{R_e C_b}$	$\frac{1}{R_d C_b R_e C_s}$	$R_d C_b$
Circuit B	$\frac{1}{R_d C_b} + \frac{1}{R_d C_s} + \frac{1}{R_e C_s}$	$\frac{1}{R_d C_b R_e C_s}$	$R_d C_b$

To use the constants B_1, B_0 they have to be recalculated:

$$\alpha_{1,2} = \frac{1}{2} B_1 \mp \sqrt{\left(\frac{1}{2} B_1\right)^2 - B_0}$$

It results

$$u(t) = \frac{U_0}{K} \frac{1}{(\alpha_2 - \alpha_1)} \left(e^{-\alpha_1 t} - e^{-\alpha_2 t}\right)$$

with

$$K = \frac{1}{C_B \cdot R_D}$$

and

$$\tau_1 = \frac{1}{\alpha_1} \qquad \tau_2 = \frac{1}{\alpha_2}$$

the time dependent functionality of the basic elements of the basic circuit and the voltage:

$$u(t) = \frac{U_0}{R_D C_B} \frac{\tau_1 \cdot \tau_2}{(\tau_1 - \tau_2)} \left(e^{-t/\tau_1} - e^{-t/\tau_2}\right)$$

This dependency u(t) can be explained by superposition of two exponential curves:

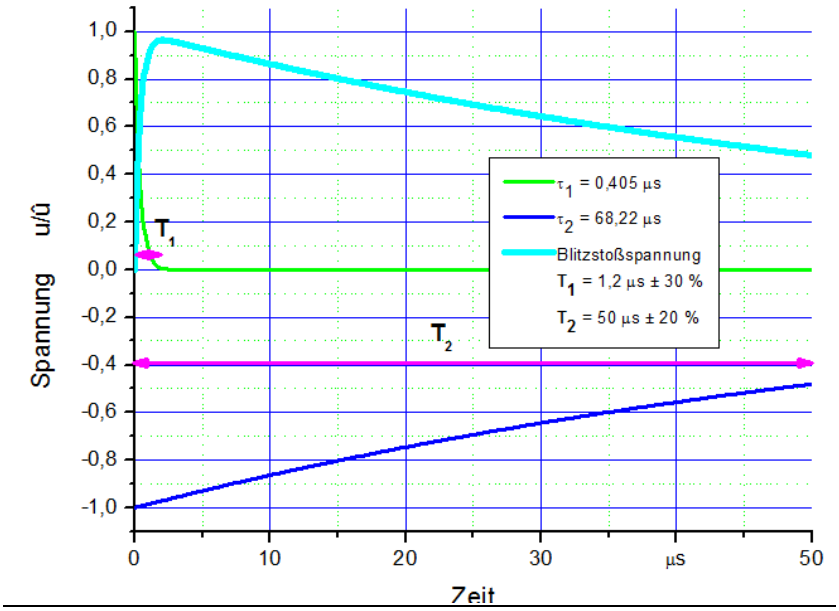


Fig. 2.14 Superposition of two exponential curves

The resulting factors for both basic circuits are:

Type of circuit	**A**	**B**
Constant of rise time	$\tau_1 = \frac{R_E \cdot R_D}{R_E + R_D} \cdot \frac{C_S \cdot C_B}{C_S + C_B}$	$\tau_1 = R_D \cdot \frac{C_S \cdot C_B}{C_S + C_B}$
Constant of tail time	$\tau_2 = (R_E + R_D) \cdot (C_S + C_B)$	$\tau_2 = R_E \cdot (C_S + C_B)$
Efficiency	$\eta = \frac{\hat{U}}{U_0} = \frac{R_E}{R_E + R_D} \cdot \frac{C_S}{C_S + C_B}$	$\eta = \frac{\hat{U}}{U_0} = \frac{C_S}{C_S + C_B}$
Advantages		Efficiency is higher

Table 2.2 Factors for lightning impulse test circuits

With aid of the factors above, the relationship between rise time and tail time can be determined:

$$T_1 = K_1 \cdot \tau_1 = 2{,}96 \cdot \tau_1 \qquad T_2 = K_2 \cdot \tau_2 = 0{,}73 \cdot \tau_2$$

Multi-stage Impulse Generators:

The already known relationships can now be extended to multi-stage circuits. The amount of stages is represented by „n“, leading to a series connection:

$$R_{D,multi-stage} = R_D \cdot n \qquad R_{E,multi-stage} = R_E \cdot n$$

$$C_{S,multi-stage} = \frac{C_S}{n} \qquad U_{multi-stage} = U_{1stage} \cdot n$$

The principle will be explained assuming the case of a 3-stage impulse generator. First the loading sequence will be demonstrated: in this sequence, the DC-source is connected via loading resistances R_L to the impulse capacitors C_S. The current backwards is driven through the damping resistances R_D and the discharging resistances R_E. The complete path is marked in green color.

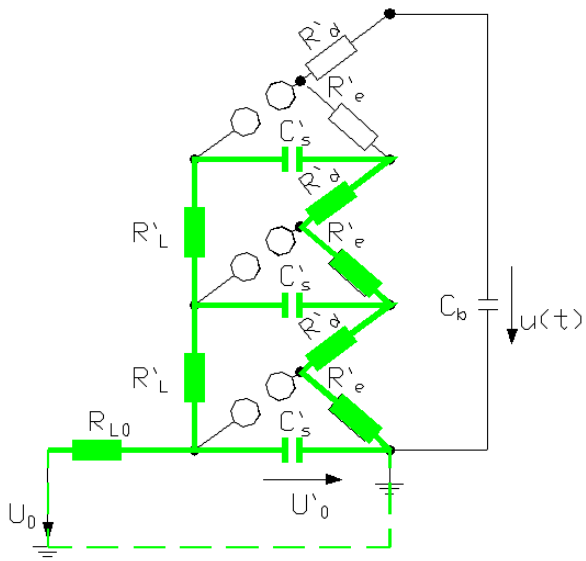


Fig. 2.15 Loading sequence of a 3-stage impulse generator circuit

Now the capacitors Cs are charged. When the first sphere gap is flashed, all sphere gaps will spark leading to another circuit equivalent: the impulse capacitors of the single stages now are connected in series feeding the load capacitances as can be seen in the next figure.

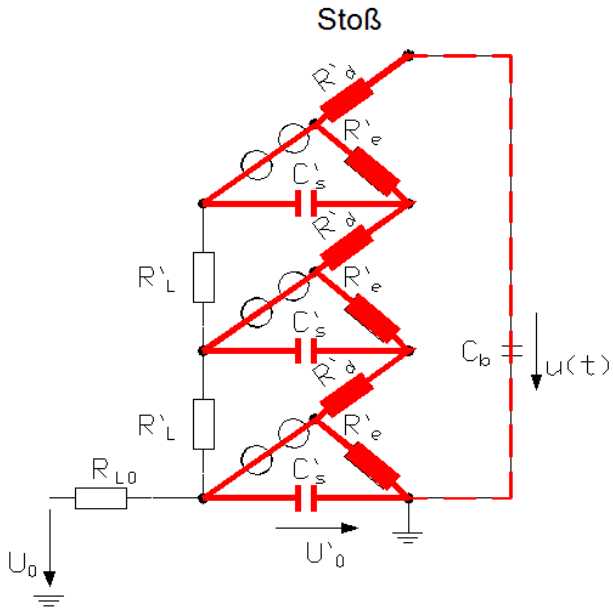


Fig. 2.16 Impulse generation sequence of a 3-stage impulse generator

It is visible, that the loading and discharging resistances are added, whereas the impulse capacitors of the single stages are divided by three due to the series connection. The impulse capacitance results to $C_S/3$. The Dc-voltages of the single stages are added to the total voltage of $3*U_{Stage}$. Figure 2.17 gives an overview over the complete test arrangement:

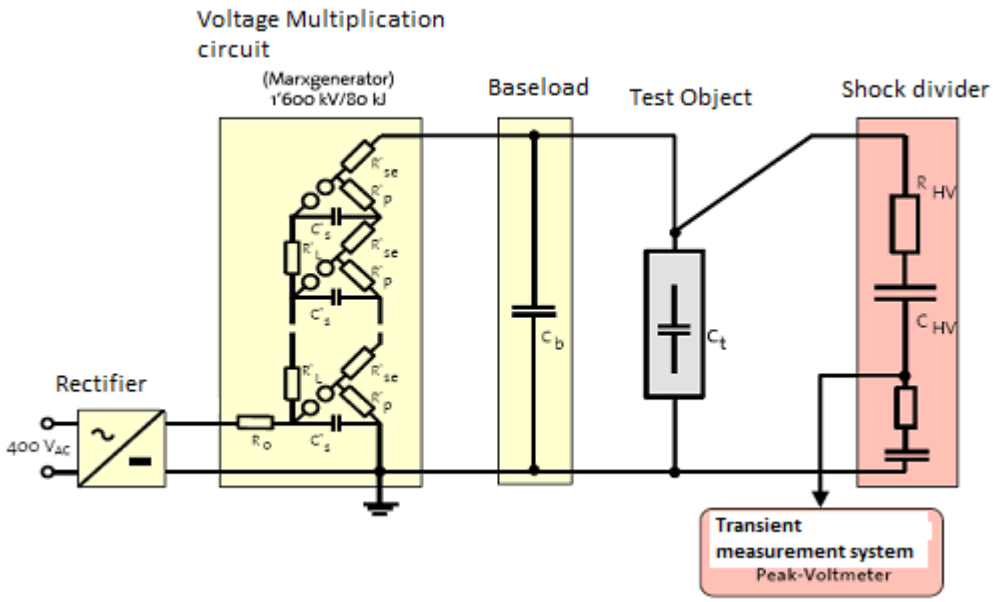


Fig. 2.17 Scheme of complete lightning impulse test arrangement

Which type of circuit was used in Fig. 2.17 ? _____

How can we prove the lightning impulse voltage strength ?

According to IEC 60694 the test object is stressed by 15 voltage impulses of positive polarity and 15 voltage impulses of negative polarity. In this context, only 2 breakdowns are allowed to occur. The magnitude of the test voltage is depending on the rated maximum operating voltage. For example, a 123 kV rated GIS-Module will be tested with 550 kV lightning impulse voltage.
The following figure shows a 3-phase gas encapsulated switchgear (GIS). According to the standards the circuit breaker has to be stressed by lightning impulse voltages with closed as well as with open contacts.



Fig 2.18 Impulse Voltage Test of 3-phase encapsulated circuit breaker

Since such tests can cause severe pre-damage of the insulation system due to very high test voltages, these types of tests are unique tests in the beginning of start of production, and so called as type tests of the product. It is not necessary to prove the lightning impulse voltage strength of every module which is produced. It is practical experience and shown in many publications, that possible production tolerances can be assessed by AC-Routine-Tests as well.



Exercises

2.1 Given is an unknown positive impulse voltage, which has to be measured by a measuring sphere gap. The sphere diameter is 10 cm and the gap distance results to 2 cm. The measuring starts at 9:00 a.m. The atmospheric condition is as follows: Air humidity 60 %, Room temperature 12°C, air pressure 1025 mbar. Please calculate the upper and the lower limit of the impulse voltage to be measured.

2.2 In the HV laboratory a 4-stage impulse generator is used providing the following technical data:

$$C_{S,Stage} = C_S^{'} = 10000\ pF$$

$$R_{E,Stage} = R_E^{'} = 68\,\Omega$$

$$R_{D,Stage} = R_D^{'} = 12\,\Omega$$

$$C_B = 2140\ pF$$

a. Please select any one type of impulse circuit and draw the scheme accordingly.
b. Calculate the rise time and the tail time of the given impulse test circuit.
c. Which maximum peak voltage is resulting, when a DC-voltage of 100 kV is applied at first stage ?
d. Please calculate the tolerance of the rise time and the tail time. Are such times within the standardized tolerances ?
e. Which values of the resistances should be chosen to have exact rise and tail times of 1.2/50 µs ?

2.3 Generation and Measuring of DC-Voltages of high magnitude

The most easy principle to generate DC of high magnitude is to rectify a given AC-voltage. Therefore transformers, rectifiers and capacitors are needed. In the following figure, the load is represented by the resistance R.

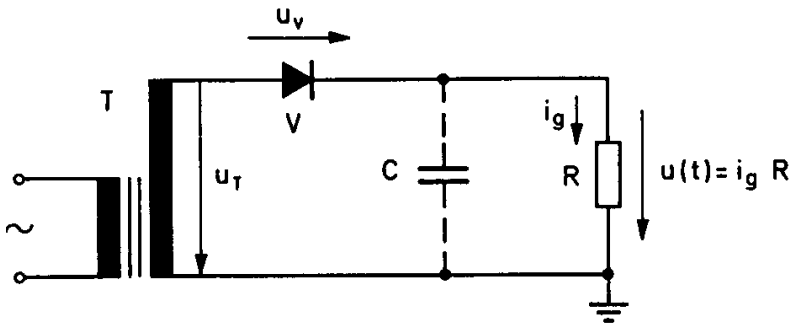


Fig. 2.19 DC Voltage generation: Half-cycle-Rectifier

The working principle is illustrated by Fig. 2.20:

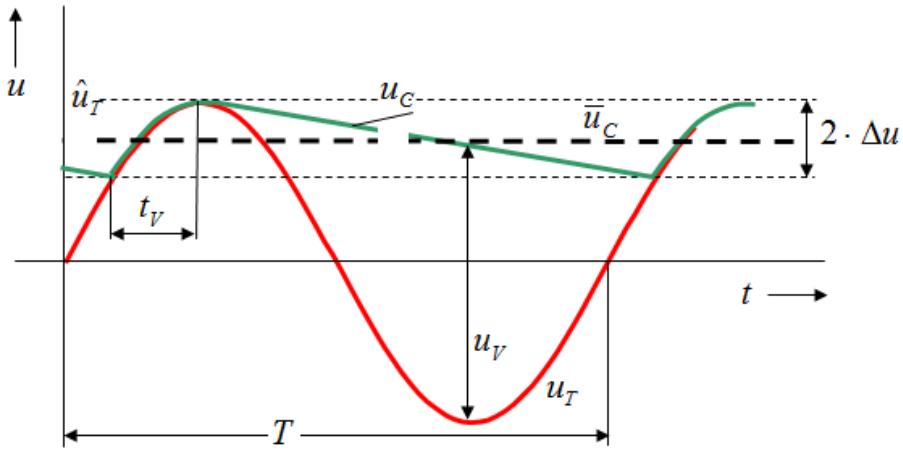


Fig. 2.20 Curve progression of Half-Cycle-Rectifier

The red curve represents the AC-voltage on the HV-side of the transformer u_T. During the positive half wave, the rectifier is conductive (forward biased) and thus charging the capacitor. In this figure the steady state is demonstrated: During the negative cycle the rectifier is reverse biased but the capacitor is already loaded and will be discharged continuously by the applied resistance R (see green curve). As soon as the green curve falls below the certain limit, the

capacitor will be loaded again. The related time period is defined as the Rectifier-time t_V. As consequence, the voltage appearing at the capacitor is periodically charged and discharged resulting in a time dependent voltage U_C, an average DC-voltage $\bar{U}_C$ and a variation of ΔU. It has to be mentioned that the rectifier has to withstand the voltage U_V.

The mathematical equation for the time dependent voltage U_C is given by:

$$u_C = \hat{u} \cdot e^{-\frac{t}{\tau}} \approx \hat{u}\left(1 - \frac{t}{\tau}\right) \qquad \tau = R\,C$$

The variation can be calculated by:

$$2 \cdot \Delta u \approx \hat{u} \cdot \frac{T}{RC} = \frac{\hat{u}}{RC \cdot f}$$

For 50Hz and a load resistance of 20 MΩ and C = 10 nF a so called ripple (variation related to average voltage) will appear:

$$\frac{\Delta u}{\bar{u}} \approx \frac{1}{2 \cdot 20 \cdot 10^6 \cdot 10^{-8} \cdot 50} = 5\%$$

To reduce the ripple, the full-wave-rectifier or a bridge rectifier circuit can be used. Especially the bridge-rectifier needs a lot of expensive HV-components, therefore we concentrate only on the full-wave-rectifier. It follows:

$$\frac{\Delta u}{\bar{u}} \approx \frac{1}{2 \cdot 20 \cdot 10^6 \cdot 10^{-8} \cdot 100} = 2{,}5\%$$

The generation of very high DC-voltages is done usually with multi-stage circuits comparable to the examples in the last chapter.

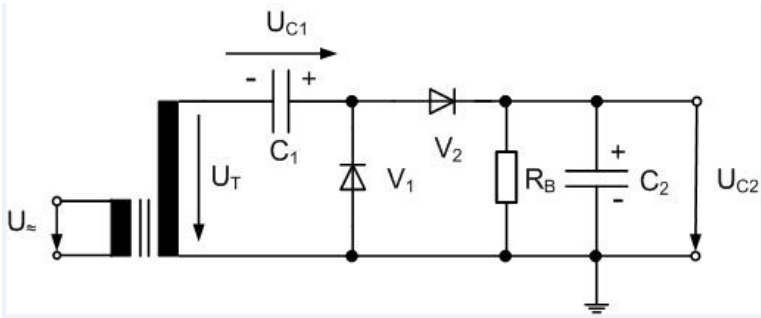


Fig. 2.21 Voltage duplication circuit

The principle of the “Greinacher-Cascade“ is based on the half-wave rectifier. The capacitor C_1 is charged to the peak value of the AC voltage source: U_{C1} =

$\hat{U}_T$. The voltage potential of this capacitor is now frozen, whereas the polarity of the transformer voltage changes to the negative peak value of the AC-voltage and charges the other capacitor in that stage. This leads to the effect that the capacitor can be charged up to $U_{C2} = 2*\hat{U}_T$.
With aid of such multiplication circuits, several MV can be reached as you can see in the next figure:

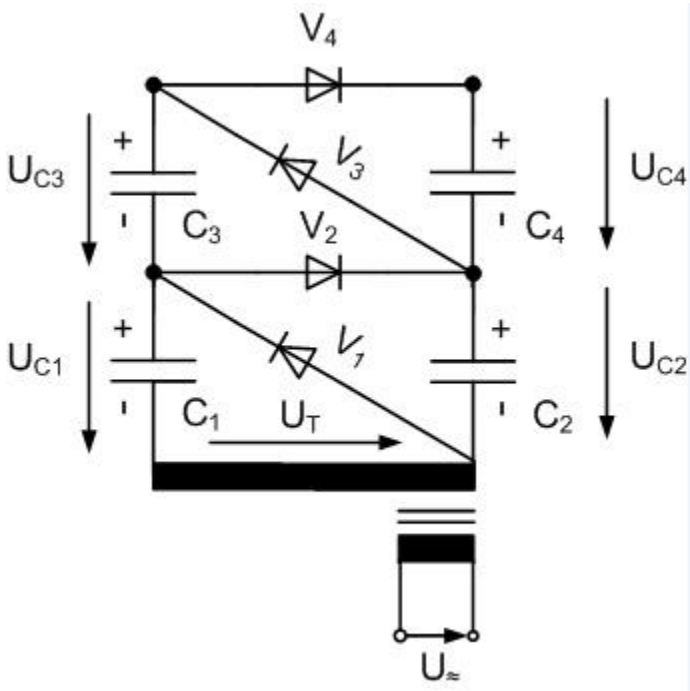


Fig. 2.22 Voltage multiplication by Greinacher-Cascade

It is visible from the 4-stage-circuit that many components are identical resulting to economy of scale effects. It is easy to extend the circuit to higher voltages. In this case the distance to the top and distance to the side have to be considered.

How to measure such high DC-voltages?
In case of low ripple, an ohmic-divider is suited. In case of high loads with increasing ripples and increasing AC-ratio, stray capacitances would disturb the measuring. To prevent this effect, capacitive controlled ohmic-dividers are used. An overview of different measuring dividers is given by Fig. 2.23.

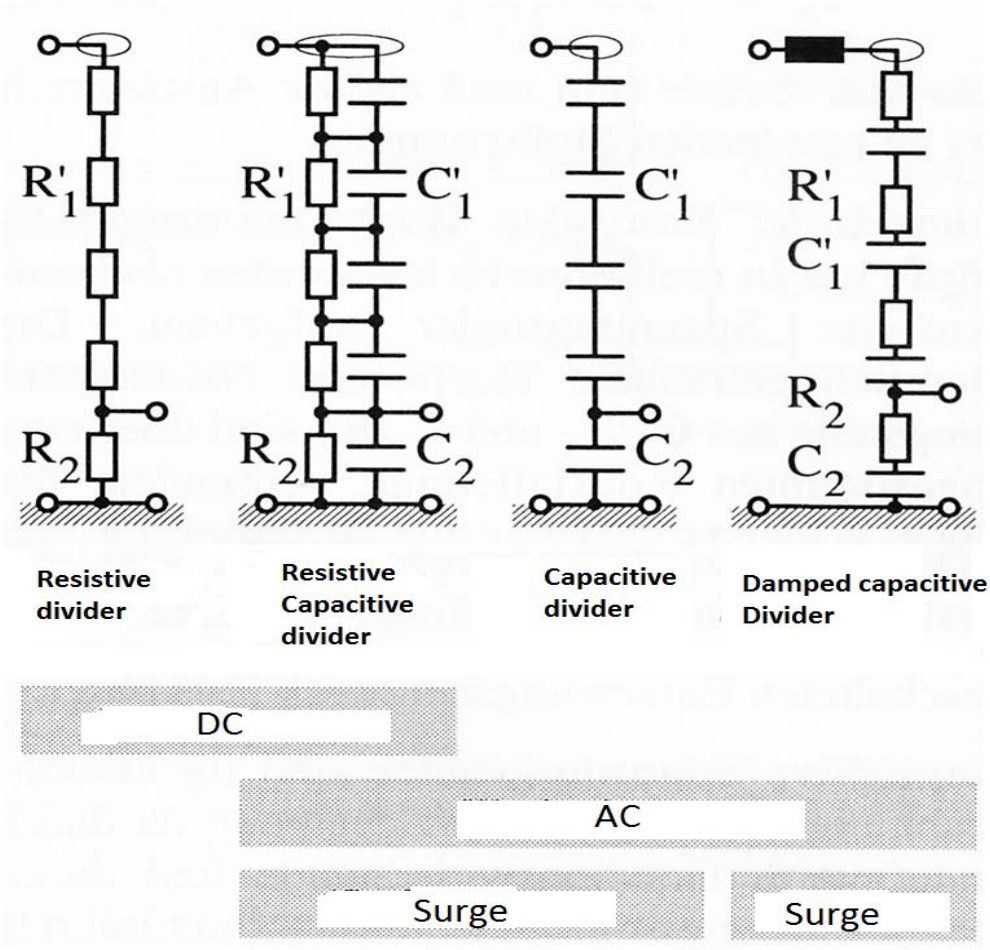


Fig. 2.23 Overview of different voltage dividers [Source: Küchler]

The ratio „ü“ of a pure ohmic or a pure capacitive divider results to:

$$\ddot{u} = \frac{R_2}{R_1 + R_2} = \frac{C_1}{C_1 + C_2}$$

If mixed ohmic-capacitive or damped-capacitive dividers are used, the different time constants have to be adapted to each other:

$$R_1' \cdot C_1' = n \cdot R_1' \cdot \frac{C_1'}{n} = R_1 \cdot C_1 = R_2 \cdot C_2$$

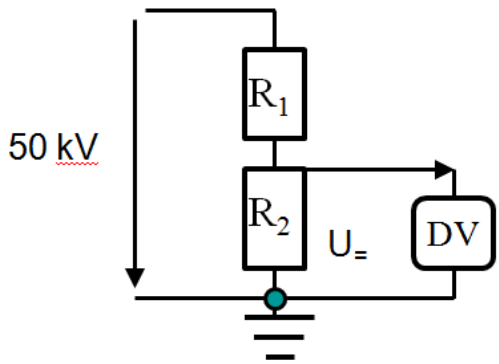
Please note: The higher the frequency of the measured signal, the more adapted and controlled the divider has to be dimensioned.

Exercises

2.3 A HV-transformer is given with outgoing HV-voltage of 100 kV. You want to generate a DC-voltage up to 1000 kV (1MV). How many stages are necessary? Please make a scheme of the circuit set-up and summarize the amount of components.(Neglect the efficiency factor of the circuit)

2.4 Given is a DC-voltage of 50 kV. This DC-voltage shall be measured with an ohmic divider. For this purpose, several components are available:

- Digital voltmeter DV with internal resistance of > 10 MΩ and a maximum voltage display of 50 V
- Several resistances are available R1: 280 MΩ, 500 kΩ, 10 kΩ
- The secondary resistance R_2 can be chosen free



The measuring set-up is shown above. Please calculate the suited resistances R_1 und R_2 considering losses in the circuit.

3 Electrical Fields

The knowledge of electrical phenomena goes back to the ancient world. If two materials are electrically charged, a force between such materials results, which was formerly interpreted as far force. Far force means that there exists a force from one material to the other material without influence of the space between them, similar to the gravitational force. According to Faraday the electrical forces are described as effect of an electrical field applied to charged material. This electrical field is not necessarily connected to a material, since the effect is present even in vacuum. The strength or magnitude of the field at a given point is defined as the force that would be exerted on a positive test charge of 1 coulomb placed at that point; the direction of the field is given by the direction of that force.
The field can only be confirmed by the force impact due to charged material. The electrical field is therefore a result of charges. In the following we investigate *electrostatic fields*, which are generated by *non moving charge distributions*.
In this case, neither an electrical current is flowing nor time dependent functions will appear.

3.1 The effects of Electrical Fields

Every electrical field generates a force at every space point on a charged material. Every point of such a force is vectors resulting to the effect that the electrical field $\vec{E}$ is a vector field as well.
This implies that the field can be described at every space point by „absolute value“ and by „direction“, illustrated by an arrow. Since such an arrangement of arrows is not very clear to interpret, the field line figure is used instead. The density of such field lines represents the “absolute value”, whereas the direction of the field line is given by the tangential line applied to such a field line.
We divide into homogeneous and inhomogeneous electrical fields. Homogeneous fields are characterized by parallel and equidistant field lines, the vectors are identical directed and have same length. The electrical field is influencing a probe charge at every space point with the same force.
In case of inhomogeneous fields the electrical field strength vectors are changing with regard to “absolute value” and “direction”. Fig. 3.1 shows the

electrical field of a plate capacitor, which is homogeneous at the inner side and tends to inhomogeneous vector field at the edge of the plate capacitor.

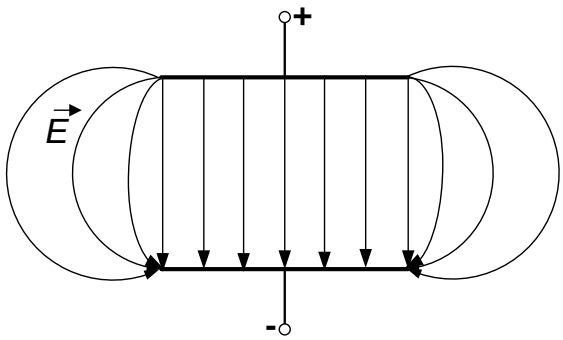


Fig. 3.1 Cross section of a plate capacitor. Inside, the electrical field is homogeneous, at the edge inhomogeneous. The field lines are leaving the positive electrodes at right angle and end at right angle at the negative electrode.

Please note: Field lines cannot cross each other, because in this case the direction of the force applied to a point charge would not be defined. This would not fit to the fact that movements of macroscopic structures are determined by the field and can therefore be explicitly calculated.

When the electric field provides the electrical field strength $\vec{E}(\vec{r})$ at the space point $\vec{r}$ to a charge, then the force is given by

$$\vec{F} = Q \cdot \vec{E} \tag{3.1}$$

Inside a conductor an electrical field cannot be present, since such an electrical field would shift charges and therefore would create a current.

Every vector field is related to a potential field. Such potential fields describe lines of identical potential. Easy example: the potential of the earthed electrode is zero over the complete electrode. In the same way you can imagine a 50% potential, which would be in case of homogeneous structures in the middle of the structure. Inhomogeneous structures are more complicated.

Potential fields are scalar fields and for this reason every space point can be related to a defined potential value (no vector):

The electrical field $\vec{E}(\vec{r})$ is forming a potential field $\varphi(\vec{r})$ given by:

$$\vec{E}(\vec{r}) = -grad\,\varphi(\vec{r}) = -\left\{ \vec{e}_x \frac{\partial \varphi}{\partial x} + \vec{e}_y \frac{\partial \varphi}{\partial y} + \vec{e}_z \frac{\partial \varphi}{\partial z} \right\}. \tag{3.2}$$

This operation is called „gradient equation“, showing that the potential is

always present, where an electrical field exists. The potential can be calculated by differentiation leading to a **steady function** (no steps !).
φ(r) is representing the physical gain of energy or the order of energy needed for moving a charge along or against the electrical field, respectively. The electrical field then creates working energy at this structure. In case of movement, potential energy is created. The potential field can be illustrated by areas with identical potential value (equi-potential areas, see lines marked in red in the following figure).

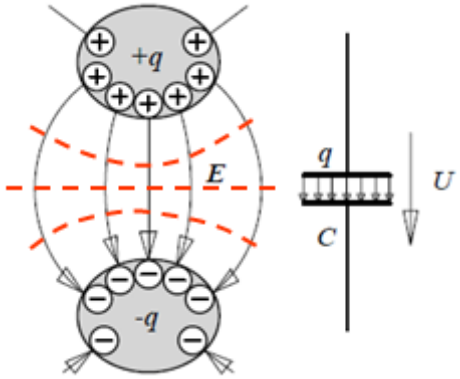


Fig. 3.2 Illustration of field lines and equi-potential lines

Electric potential is a location-dependent quantity that expresses the amount of potential energy per unit of charge at a specified location. This context is comparable to the mass in a gravity field. Movements of charges are creating potential energy: assuming that the charge is moving from A to B: when turning back from B to A the charge must have the same potential energy. The same work would have to be done on the charge by an external force.
This behavior can mathematically be formulated by:

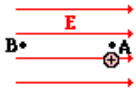
$$W = \oint \vec{F} d\vec{s} = \oint q \cdot \vec{E} d\vec{s} = 0 \qquad (3.3)$$

The work *W* as integral over force and distance will become zero, when a force is applied on a charge „q" along the complete distance. The closed loop is represented by distance segments of a infinite length $d\vec{s}$.
By definition, the electric potential difference is the difference in electric potential (V) between the final and the initial location when work is done upon a charge to change its potential energy. The result is voltage U_{AB}:

$$U_{AB} = \int_A^B \vec{E} d\vec{s} \equiv \varphi(A) - \varphi(B) \Rightarrow \varphi(B) = \varphi(A) - \int_A^B \vec{E} d\vec{s} \,. \qquad (3.4)$$

This equation is interesting from three points of view:

1. Instead of expected φ -values resulting from the rules of defined integrals, the sequence of $\varphi(A)$ and $\varphi(B)$ is reversed. But this is demanding from reasons of constant energy. Positive charges are moving always in direction of potential difference (Ohmic Law). Thus Positive charges which are accelerated on their way from A to B,



are gaining kinetic energy and have to reduce their potential energy due to constant energy law. Consequently, $\varphi(B)$ must always be smaller than $\varphi(A)$. In other words: along a field line, the voltage increases, whereas the potential decreases!
2. Obviously the potential at point B is only dependent on a constant and on the potential at the reference point $\varphi(A)$. This point A can be chosen arbitrarily, i.e. for simplification reasons the point is chosen to at infinity => potential is zero:

$$\varphi(B) = -\int_{\infty}^{B} \vec{E} d\vec{s} \qquad (3.5)$$

3. Between two points which belongs to an equi-potential area no voltage results (U = 0). When charges move along such equi-potential areas, they do not gain energy. It follows that no forces are applied on charges and this leads to the effect that all field lines will leave such equi-potential areas **at right angle**.

3.2 The Excitation of Electrical Fields

The causes of electrical fields are charges or materials which are charged. But how are such fields be excited and which shape is related to electrical fields?
Since every material structures are electrically neutral (except plasma effects in fluorescent lamp or the inside of stars), such charged materials can only be created by separation of charges, since the sum of all charges is Zero in a material. The electrical field is a source field, meaning that all field lines starts from positive charged structures and end at negative charged structures. The field lines do have a defined starting point and a defined end point (the end point could be in the infinity).
Coulomb's law states that: "The magnitude of the Electrostatics force of

interaction between two point charges Q_1 and Q_2 is directly proportional to the scalar multiplication of the magnitudes of charges and inversely proportional to the square of the distances between them:

$$\vec{F}_{12} = \frac{Q_1 \cdot Q_2}{4\pi\varepsilon r^2}\frac{\vec{r}}{r} \qquad (3.6)$$

Where ε represents the dielectric constant of the space between both charges resulting to ε = 8,854·10^{-12} As/Vm for vacuum and $\frac{\vec{r}}{r}$ as unit vector along the connecting axis of both charges. If we compare the equation (3.1) with equation (3.5), it can be stated that the force applied i.e. to charge Q_1 can be interpreted as influence of the electrical field excited by charge Q_2. The electrical field of charge Q_2 results to:

$$\vec{E}_2 = \frac{Q_2}{4\pi\varepsilon r^2}\frac{\vec{r}}{r}.$$

Without the index „2“, this equation has general meaning for a point charge Q with distance r:

$$\vec{E} = \frac{Q}{4\pi\varepsilon r^2}\frac{\vec{r}}{r} \qquad (3.7)$$

Such a field is directed radial and is inhomogeneous (Fig. 3.3).

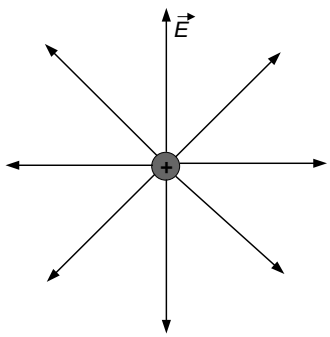


Fig. 3.3 **Electrical field of a positive point charge.** The field lines end at infinity, which is assumed as negative reference point.

It follows from equation (3.7) that increasing charges Q increases electrical field strength and therefore increasing density of field lines. Vice versa, if the density of the field lines is high, then the magnitude of the charge must be high. The density of the local charge determines the distance of neighbored field lines, according to the findings of Faraday in 1831. The amount of field

lines, which starts at one electrode, is a value for the amount of positive charges, which are present at the electrode.
The sum of all field lines is called „*displacement flux*“. This displacement flux can be compared with the current of the electrical drift field, resulting from all current wires linked to the conductor cross section. In case of this current, the charges are moving , whereas the **displacement flux** is related to a **static field.**
The current density $\vec{j}$ in case of electrical drift field is corresponding to the *displacement flux* $\vec{D}$ in the static field. $\vec{D}$ is a vector, which is defined at a special point in the electrical field. The value of this vector is identical to the displacement flux of the field line of *dQ*, running through a small volume *dA*. To obtain the density, the ratio of displacement flux and related value of the area *dA* has to be calculated:

$$\left|\vec{D}\right| = D = \frac{dQ}{dA}. \qquad (3.8)$$

The displacement flux, flowing through a given area, can be divided into a normal and a tangential component of the flux vector. The tangential component does not have an effect.
If we introduce now a normal vector $d\vec{A}$, which is at right angle to the selected area and which absolute value is given by the area dimensions, then the related charge can be determined:

$$\Delta Q = \int \vec{D} \cdot d\vec{A} \qquad (3.9)$$

When the area *A* is representing a closed envelope, then the area integral of the displacement flux over the envelope corresponds to the **total charge being present** within the envelope:

$$Q = \oint \vec{D} \cdot d\vec{A}. \qquad (3.10)$$

This is the **first basic law of the electrostatic field**.

How is the relation between electrical field $\vec{E}$ and the displacement flux $\vec{D}$? Since the displacement flux is correlated to the charges, the ratio of electrical field strength and displacement flux can be measured very easy.
If we charge a capacitor, the charge *Q* can be determined by measuring and integration of the current. If the plate distance d is small and the plate area is of high magnitude, then the displacement flux is practically moving completely through the volume represented by both plates and the small distance. During charging, the voltage will increase. The electrical field is homogeneous. The field lines are leaving one plate and are ending in the other plate at right angle.

The electrical field strength is given by:

$$\left|\vec{E}\right| = \frac{U}{d}.$$

The total charge Q is distributed equally over the complete plate area. The absolute value of the displacement flux results to:

$$\left|\vec{D}\right| = \frac{Q}{A}.$$

The electrical field strength is resulting from the voltage measurement, whereas the displacement flux follows the current measurement. If those measurements are done at different voltage levels and on different materials, you will find out that for usual materials the displacement flux is linearly depending on the electrical field strength. It results:

$$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}. \qquad (3.11)$$

This equation is called the **second basic law of the electrical field**.

The constant $\varepsilon_0\varepsilon_r$ is representing the dielectric constant, based on the multiplication of the dielectric constant of the vacuum, ε_0 = 8.85419 pF/m, and the dimension free dielectric constant, ε_r, which usually is > 1 in a material filled space. In case of linear dielectric material ε_r is constant, in case of non linear dielectrics ε_r is a function of the electrical field.
The relative dielectrics of usual (linear) materials are given in tables. Typical values are varying between 2 (Oil of transformers) and 8 (mica, glimmer).

3.3 Simulation Methods to Simulate the Field Strength of Electrical Fields

Several methods are known from literature how to calculate electrical fields. In general, it has to be stated that only very few geometries can be calculated analytically. Usually general field problems can only be calculated numerically. Some special examples are presented which could be solved by easy equations.

3.3.1 Method of Superposition

The direct method to calculate electrical field strength is the superposition of fields generated i.e. by single point charges. This superposition is done according equation 3.7 by superposition of E-field-vectors.
As example three point charges are considered with $Q_1 = 2.5 \cdot 10^{-8}$ As, $Q_2 = 1.5 \cdot 10^{-8}$ As and $Q_3 = -2.0 \cdot 10^{-8}$ As. These point charges are placed at the edges of an equilateral triangle with side length of a = 10 cm, see Fig. 3.4. The envelope of the charges is air with dielectric constant of $\varepsilon_0 = 8.85 \cdot 10^{-12}$ As/Vm. Considering that the center of the triangle is representing the origin of the Cartesian coordinate system, it results:

1. The resulting electrical field generated in the center of the triangle can be calculated by three superposing electrical fields.
2. The absolute value of that resulting electrical field appearing in the center of the triangle can be determined.

At first, we will calculate the distance *b*. The distance from the edges to the center is identical for all point charges:

$$b = \frac{\frac{a}{2}}{\cos 30^{\circ}} = 5,77 \cdot 10^{-2} m \,. \tag{3.12}$$

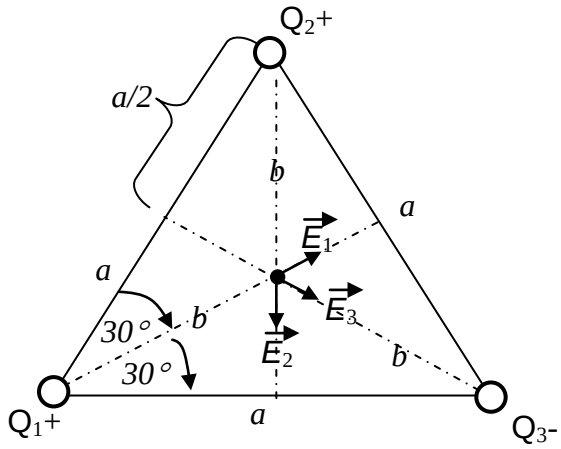


Fig. 3.4 **Three point charges placed in a equilateral triangle.** The distance b is representing the distance to the center.

The electrical field strength is given by:

$$\left|\vec{E}\right| = \frac{Q}{4\pi\varepsilon_0 b^2}$$

When applying the related charges at the edges of the triangle, the absolute values of the three electrical field strength results to
$\left|\vec{E_1}\right| = 6,57 \cdot 10^4 V/m$, $\left|\vec{E_2}\right| = 4,05 \cdot 10^4 V/m$ and $\left|\vec{E_3}\right| = 5,40 \cdot 10^4 V/m$
To add the electrical field strength vectorially, the x- and y-components have to be evaluated. It will be explained based on $\left|\vec{E_1}\right|$.

$$E_{1x} = 6,57 \cdot 10^4 V/m \cdot \cos 30^\circ = 5,85 \cdot 10^4 V/m, E_{2x} = 0, E_{3x} = 4,58 \cdot 10^4 V/m$$
$$E_{1y} = 6,57 \cdot 10^4 V/m \cdot \cos 60^\circ = 3,38 \cdot 10^4 V/m, E_{2y} = -4,05 \cdot 10^4 V/m, E_{3y} = -2,7 \cdot 10^4 V/m$$

For vectorial addition, we change to a vector matrix:

$$\vec{E} = \left\{ \begin{pmatrix} 5,85 \\ 3,38 \end{pmatrix} + \begin{pmatrix} 0 \\ -4,05 \end{pmatrix} + \begin{pmatrix} 4,68 \\ -2,7 \end{pmatrix} \right\} \cdot 10^4 \frac{V}{m} = \begin{pmatrix} 10,52 \\ -3,38 \end{pmatrix} \cdot 10^4 \frac{V}{m} \quad (3.13)$$

Now we know the electrical field strength exactly which is appearing at the center of the triangle. To obtain the complete field distribution, this calculation should be repeated for every point within the 3D-volume. This is usually done with special software programs. The problem of such an example is the low grade of symmetry. The result is only as good as you define the right density of the calculation mesh.

3.3.2 Gauss´ Formula

When special charge distributions like cylinder symmetry (equilateral charged thin wire) or sphere symmetry (equilateral charged sphere), then it is very helpful to use the **Law of Gauss** (equation 3.9).
First the electrical displacement flux $\vec{D}$ is calculated. In this respect, an integral over an envelope has to be solved, but the choice of the envelope, which covers the complete charge, is free. Therefore it is recommended to choose an envelope, where the multiplication of the scalars $\vec{D} \cdot d\vec{A}$ is very easy. This is the case, when $\vec{D}$ **is within** the envelope, since then D is at right angle to the normal area vector and the scalar product is zero at all points. Or $\vec{D}$ is at right angle to the envelope, that means parallel to the normal area vector: then the scalar product is a usual algebraic product.
The more **symmetrical** the geometry is, the less complicated is the calculation.

The equilateral charged sphere
If we consider a metal sphere with a radius R and a charge of +*Q*, then the

charge is distributed equilaterally over the sphere surface. The field lines are leaving the sphere surface at right angle. The sphere surface is representing the equi-potential area in case of an electrostatic field. The field lines end at infinity, where a virtual sphere is considered as negative electrode. Since the sphere is sphere symmetry, the electrical field must be sphere symmetry as well. This implies that the field strength and the displacement flux is constant at the concentric sphere surface. Inside the sphere, the electrical field must be zero, because all possible envelopes do not cover charges.
To calculate the displacement flux outside the sphere, a concentric sphere envelope is assumed with distance $r > R$ from sphere center. Since field and the normal area vector is in every case parallel to the envelope and the displacement value is constant at the concentric envelope, the result is simplified to:

$$Q = \oint \vec{D} \cdot d\vec{A} = D \oint dA = D \cdot 4\pi \cdot r^2 \tag{3.14}$$

This leads to the displacement flux and the electrical field:

$$\vec{D} = \frac{Q}{4\pi \cdot r^2} \vec{e_r},\ \vec{E} = \frac{Q}{4\pi\varepsilon_0 \cdot r^2} \vec{e_r} \tag{3.15}$$

The potential of the electrical field at point P with distance $r > R$ from sphere center can be evaluated by integration along a field line. At the field line, the field is always parallel to the distance element (in this case r is directed radial): $\vec{E} \parallel d\vec{s} \Rightarrow \vec{E} \cdot d\vec{s} = E \cdot ds$.

$$\varphi_P = \varphi_\infty - \int_\infty^P \vec{E} \cdot d\vec{s} = 0 - \int_\infty^P \frac{Q}{4\pi\varepsilon_0 \cdot r^2} \vec{e_r} \cdot dr = \frac{Q}{4\pi\varepsilon_0 \cdot r} \tag{3.16}$$

Since the potential of every point with distance r has to be equal from symmetry reasons, the potential follows:

$$\varphi(r) = \frac{Q}{4\pi\varepsilon_0 \cdot r}\ . \tag{3.17}$$

If the radius of the sphere tends to zero, we get the field and displacement of a point charge: for $r \to 0$ the field and displacement diverges. Within the sphere with real diameter > 0, the electrical field is zero, but the potential is staying on the value given by the sphere surface (due to equi-potential behavior)

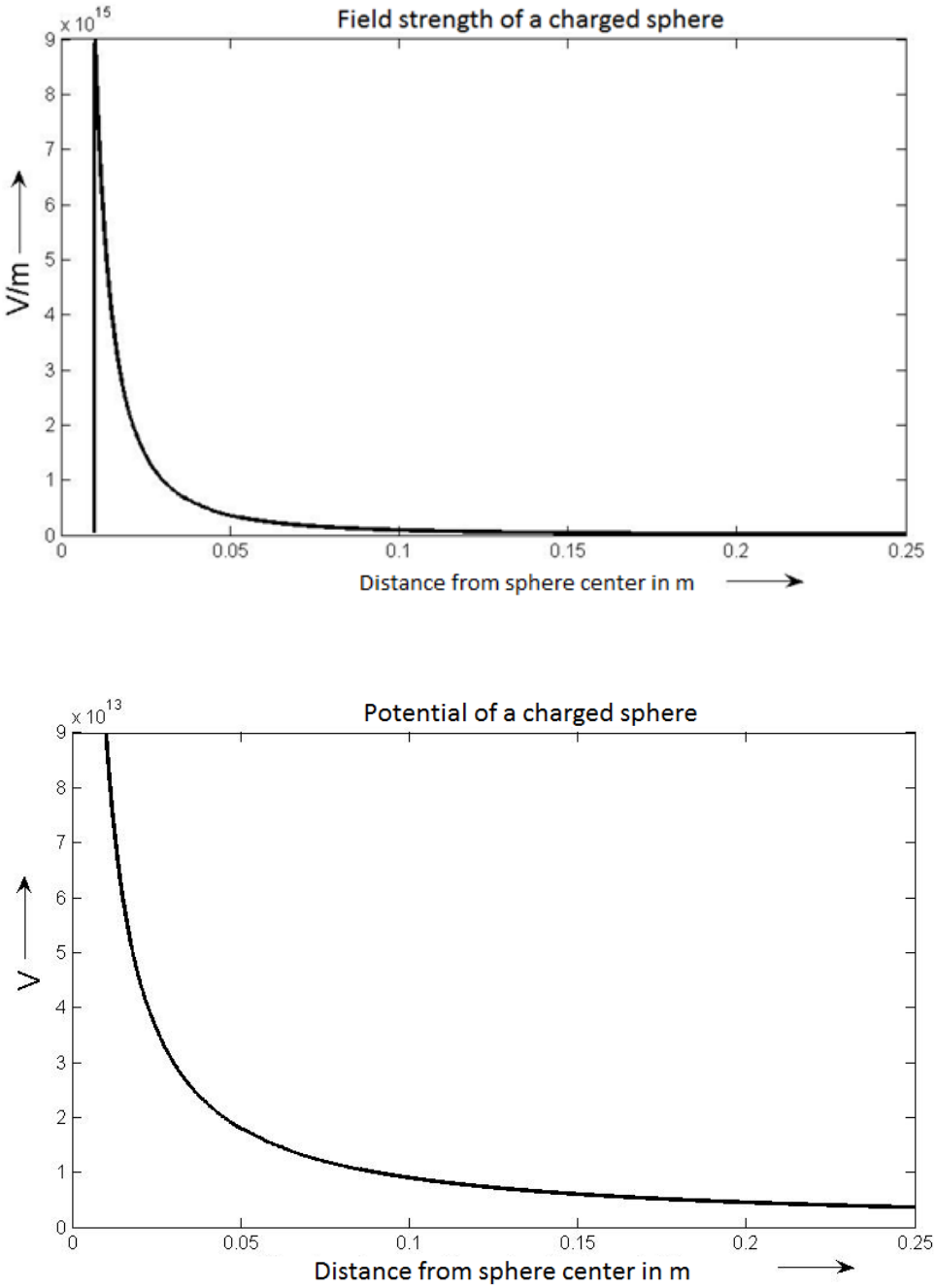


Fig. 3.5 **Electrical field strength and potential of a sphere.** The charge is 100 As and the diameter of the sphere is 1 cm.

The coaxial cable

This arrangement is cylindrical symmetrically. We assume that the inner conductor with radius R_i is charged with Q and that the outer conductor with radius R_a is carrying the negative charge $-Q$. Between both conductors, the dielectric material is present, the outside of the cable is insulated. The length of the cable is sufficient long to fulfill the condition $l >> R_a$ to avoid boundary effects at both sides of the cable.

We solve the displacement calculation with law of Gauss. Afterwards we calculate the field strength and the potential. The whole arrangement is cylinder symmetrical around the inner conductor as axis. At every cylindrical envelope, the electrical field has the same absolute value and is at right angle to the cylindrical envelope. Therefore $\vec{D}$ and $d\vec{A}$ are parallel. The displacement flux through such an cylindrical envelope can be easily calculated:

$$Q = \oint \vec{D} \cdot d\vec{A} = D \cdot \oint dA = D \cdot 2 \cdot \pi \cdot r \cdot l \qquad (3.18)$$

The crossing sections at the end of the cable do not add a value, since the scalar product is zero at these points (field along the cylindrical envelope). The electrical field strength results to:

$$\vec{D} = \frac{Q}{2\pi \cdot r \cdot l}\vec{e_r},\ \vec{E} = \frac{Q}{2\pi\varepsilon_0\varepsilon_r \cdot r \cdot l}\vec{e_r} \qquad (3.19)$$

The potential between R_i and R_a is given by integration along a field line. The reference potential is the potential at the inner conductor, which was chosen to be zero ($\varphi(R_i) = 0$):

$$\varphi(r) = \frac{Q}{2\pi\varepsilon_0\varepsilon_r \cdot \ln\frac{r}{R_i}},\ R_i < r < R_a \qquad (3.20)$$

Outside of the cable, the field is zero. This results from the fact that the enclosed charges (Q and –Q) outside the cable are summarized to zero. The potential of the area outside the cable is staying on the value of the outer conductor. Inside the inner conductor, the field is zero as well (please see sphere as well).

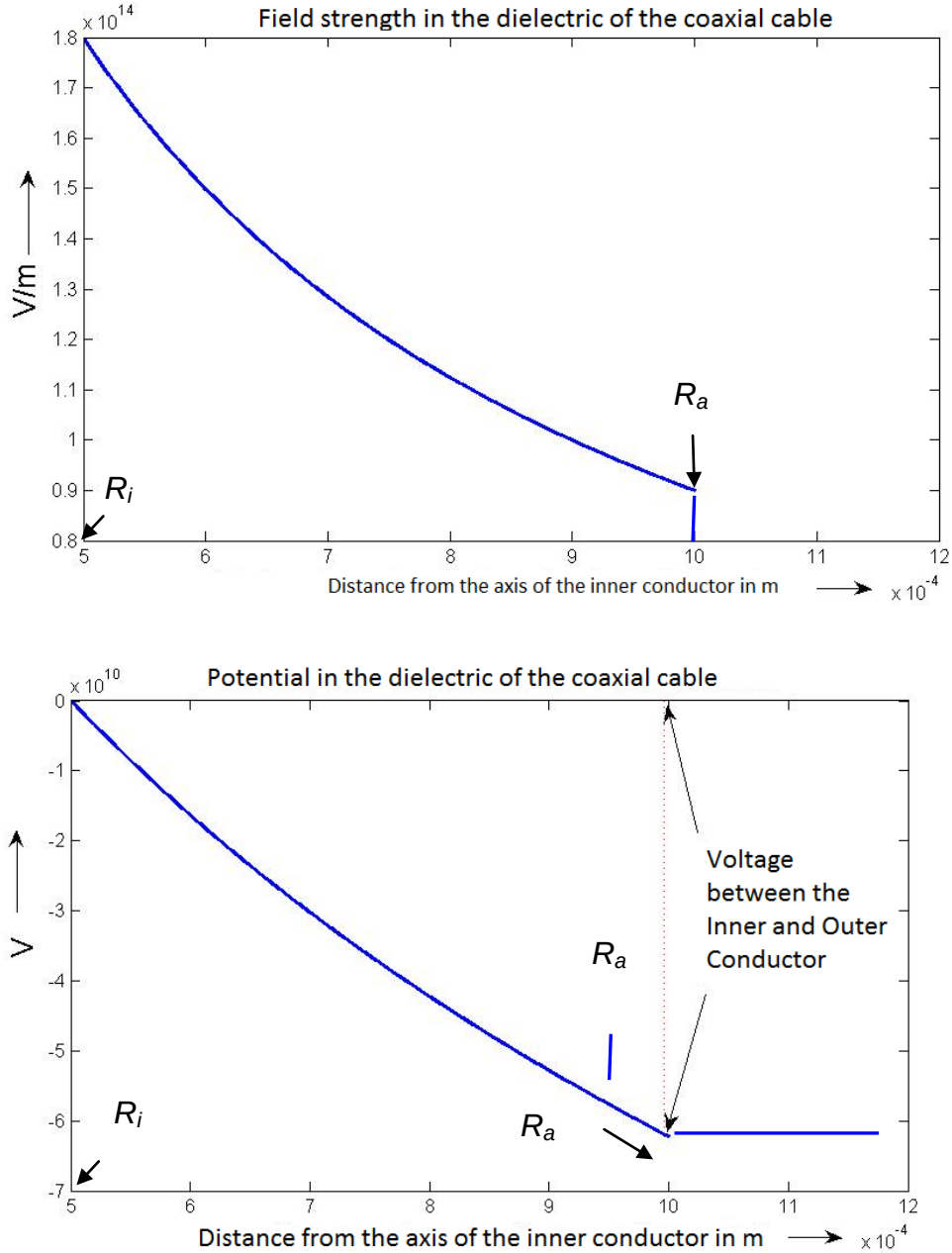


Fig. 3.6 **Electrical field strength and potential of a coaxial cable.** The charge is 100 As and -100 As, respectively. The diameter of the inner conductor is 1 mm and 2 mm of the outer conductor. The length of the cable is 10 m.

In case of a coaxial cable, the maximum electrical field strength (and therefore the maximum stress for the dielectric) is appearing at the surface of the inner conductor. In this example the maximum E-field is $1.8 \cdot 10^{14}$ V/m!

3.3.2 Differential Methods for Electrical Field Calculation

Charges are in reality not only single point charges, but quasi-continuously distributed charges over 3D-space or 2D-areas. If we consider the charge *dQ* in a small volume element *dV* located at the point $\vec{r}$, then $\rho(\vec{r}) = \frac{dQ}{dV}$ is representing the charge density at this location.

If we know the amount of charges, defined by the charge density, the law of Gauss changes to:

$$\iiint_{V} \rho \cdot dV = \oiint_{HülleV} \vec{D} \cdot d\vec{A} \tag{3.21}$$

The volume integral is written as triple integral and the surface integral as double integral over an envelope V.
The law of Gauss is valid for every volume segment and for every charge distribution. Assuming a field, which directs only to x-direction, V is defined as a small volume element with 3D-border *dx*, *dy* and *dz*. Only the areas of the cuboid are adding values to the displacement flux:

$$\begin{aligned} \rho \cdot dV &= \vec{D} \cdot d\vec{A} = -D(x)dydz + D(x+dx)dydz = \frac{dD}{dx}dxdydz \\ \rho \cdot dV &= \rho dxdydz \\ \frac{dD}{dx} &= \rho \\ \frac{dE}{dx} &= \frac{\rho}{\varepsilon_0 \varepsilon_r} \end{aligned} \tag{3.22}$$

If the field does not direct to *x*-direction, the local derivation has to be replaced by the divergency:

$$\begin{aligned} div\vec{D} &= \frac{dD_x}{dx} + \frac{dD_y}{dy} + \frac{dD_z}{dz} = \rho \\ div\vec{E} &= \frac{dE_x}{dx} + \frac{dE_y}{dy} + \frac{dE_z}{dz} = \frac{\rho}{\varepsilon_0 \varepsilon_r} \end{aligned} \tag{3.23}$$

These partial differential equations are known as **Poisson-Equations**, which describes the charge density in every space point linked the related field at this point. These equations can be solved exactly only in special cases, as known from Gauss law. The advantage of the differential formula is the direct relation between charge and field at same location, which allows a numerical calculation with aid of computers.

3.4 Application Examples for High Voltage Components

As typical example of a HV-component, the principle and the peculiarities of a HV cable is considered, which is cylindrical symmetrical. Nowadays, the insulating material used for voltages above 72.5 kV is the cross-linked Polyethylene (XLPE or VPE), which has a high reliability and does not tend to initiate the so called **water trees** (**Treeing** is an electrical pre-breakdown phenomenon. It is a damaging process due to partial discharges and progresses through the stressed dielectric insulation, whose path resembles the form of a tree.).
High voltage cables are produced up to voltage levels of 500 kV and conductor cross sections of roughly 2500 mm^2. In special cases, oil filled cables are used up to voltage levels of 400 kV, which are not described here.
The general design of XLPE-HV-Cables is shown in the next figure.

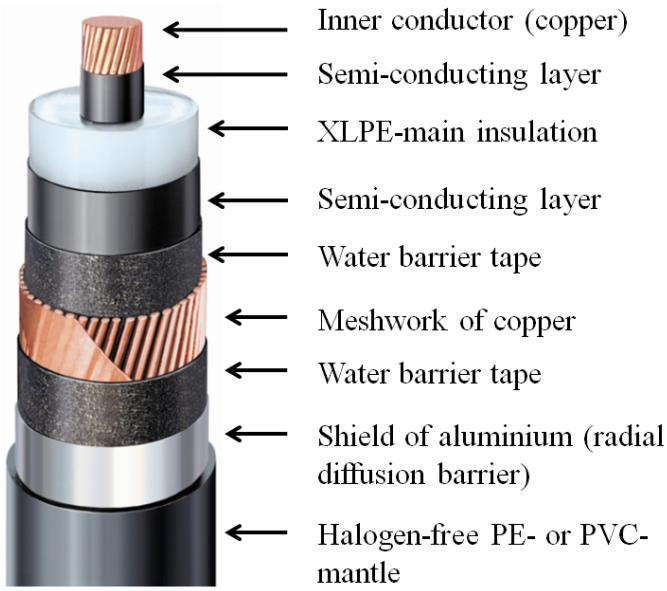


Fig. 3.7 Design of a XLPE-HV-Cable [Source: Brugg Cable]

The inner conductor is carrying the current and is wrapped with a semi-conductive layer to avoid field enhancement at the protrusions. The main insulation is cross-linked PE, which is wrapped by a semi-conductive layer as well. All three insulating layers are produced in one extrusion step. Therefore trapped air is prevented. Together with the inner conductor, these layers are forming the core of the cable. Further components are two semi-conductive hygroscopic tapes providing a water barrier in longitudinal direction. These tapes are wrapped upon the metal shield (meshwork of copper).
As optional element a radial diffusion barrier can be selected consisting of an aluminum mantle.

The outer protection (inclusive corrosion) of the cable is ensured by an extruded PE-mantle or a PVC-mantle.
Now we want to apply the learned steps from the last chapters to calculate the maximum electrical field strength of a 220 kV-3-phase power cable. We assume that the cross section of the inner conductor is 2000 mm^2 and the outer diameter of the insulation is given by 113 mm.
We evaluate the inner radius r_1, the outer radius r_2 and the applied voltage U_B as follows:

$$r_1 = \sqrt{\frac{2000\,mm^2}{\pi}} = 25{,}2\,mm \qquad r_2 = 56{,}5\,mm \qquad U_B = 127\,kV$$

In case of a homogeneous field, the electrical field strength E_0 would result:

$$E_0 = 5{,}74\,\frac{kV}{mm}$$

Since this arrangement is a quasi-homogeneous arrangement with resulting field enhancement, the following procedure is recommended and applied:

Step 1: Evaluation of the displacement flux D

$$Q = \oint_A \vec{D} \cdot d\vec{A} = \oint_A D(r) \cdot dA \cdot \cos\alpha = D(r) \cdot \oint_A dA \cdot \cos 0° =$$
$$= D(r) \cdot 2\pi \cdot r \cdot l$$

$$\boxed{\Rightarrow D(r) = \frac{Q}{2\pi \cdot r \cdot l}} \qquad (3.24)$$

Step 2: Pre-Evaluation of the electrical field strength E

$$D = \varepsilon \cdot E$$
$$\Rightarrow E = \frac{D}{\varepsilon}$$

$$\boxed{E(r) = \frac{Q}{2\pi \cdot r \cdot l \cdot \varepsilon}} \qquad (3.25)$$

Step 3: Calculation of the voltage U_B

$$U_B = \int_{r_1}^{r_2} \vec{E}(r) \cdot d\vec{r} = \frac{Q}{2\pi \cdot l \cdot \varepsilon} \cdot \int_{r_1}^{r_2} \frac{1}{r} \cdot dr = \frac{Q}{2\pi \cdot l \cdot \varepsilon} \cdot \ln\frac{r_2}{r_1} \qquad (3.26)$$

Step 4: Evaluation of the electrical field strength as function of the voltage

$$\Rightarrow Q = \frac{2\pi \cdot l \cdot \varepsilon}{\ln \frac{r_2}{r_1}} \cdot U_B$$ Changing of (3.24)

$$\boxed{with\ E(r) = \frac{Q}{2\pi \cdot r \cdot l \cdot \varepsilon}}$$ Result from (3.25)

The following relation will result:

$$E(r) = \frac{Q}{2\pi \cdot r \cdot l \cdot \varepsilon} = \frac{\frac{2\pi \cdot l \cdot \varepsilon}{\ln \frac{r_2}{r_1}} \cdot U_B}{2\pi \cdot r \cdot l \cdot \varepsilon} = \frac{1}{r} \cdot \frac{U_B}{\ln \frac{r_2}{r_1}} \qquad (3.27)$$

It results from the equation above that the electrical field strength is only depending on the radius r (in case of given voltage and given geometry r_1, r_2). To obtain the maximum electrical field strength, the smallest radius ($r = r_1$) hast to be chosen. This leads to the resulting field strength of:

$$E_{\max}(r = r_1) = \frac{1}{r_1} \cdot \frac{U_B}{\ln \frac{r_2}{r_1}} = 8{,}8 \frac{kV}{mm}$$

In the same manner, the minimal field strength can be evaluated to 3.9 kV/mm. You can see the functionality of the electrical field strength in the following figure.

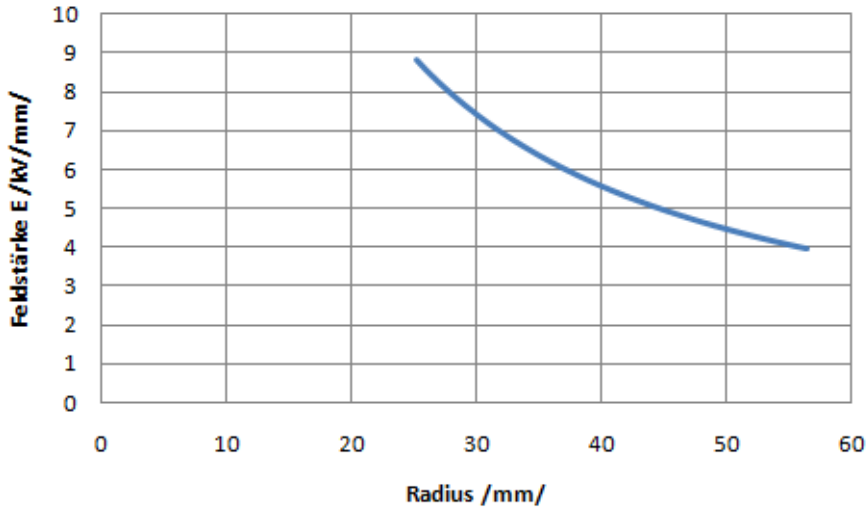


Fig. 3.8 Electrical field strength of a HV-Cable

Please note: the maximum electrical field strength is always appearing at the inner conductor. In general, the electrical field strength is increasing with increasing curvature or decreasing radius. The dependency of the electrical field strength from the curvature is valid for all other geometries as well.

3.5 Inhomogeneous Electrical Fields and Calculation Examples

Homogeneous and nearly homogeneous fields (quasi-homogeneous) are in most cases easy to calculate, as shown in the last unit. In case of inhomogeneous arrangements it is recommended to use special diagrams, which are derived from experimental investigations of nearly every type of geometry. We will now look at different types of geometries:

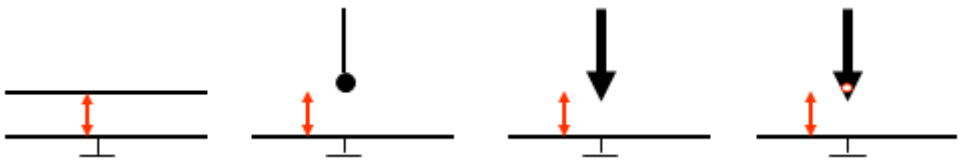
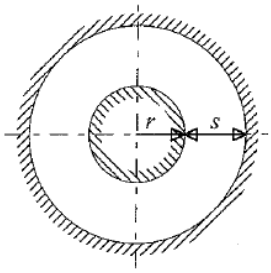


Fig. 3.9 Different arrangements with different grade of homogeneity
(from homogeneous to strongly inhomogeneous)

We apply 100 kV to all arrangements, assuming a gap distance of 10 cm. The homogeneous field can be calculated very easy:

$$E_{Max} = E_{Mittel} = \frac{100\,kV}{10\,cm} = 10\,\frac{kV}{cm}$$

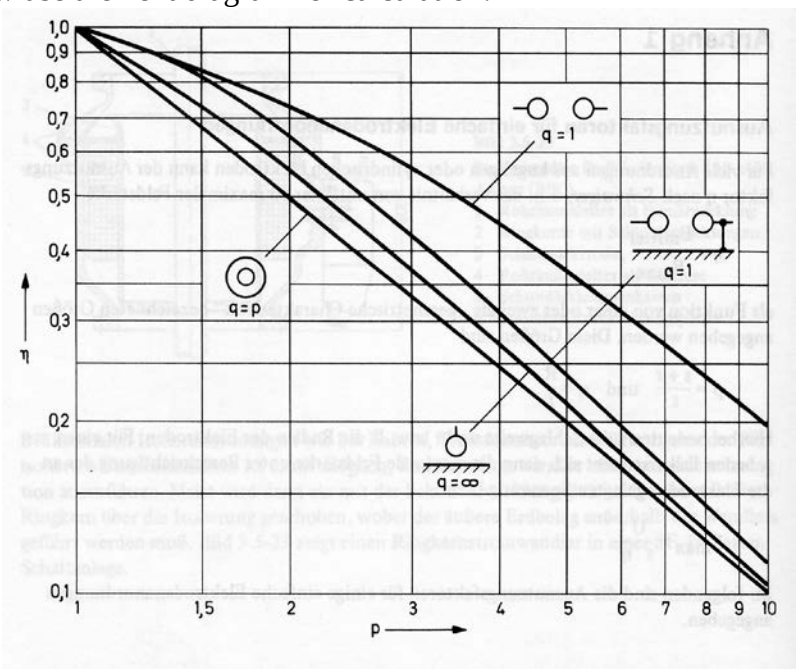
In case of the sphere-plate-arrangement with a radius of r_{Sphere} = 1.2 cm, the following „Schwaiger-Model“ is used. Every geometry has a special geometry-factor „p“. This factor p only considers the gap distance s and the radius with higher curvature:



$$p = \frac{s + r}{r}$$

$$\Rightarrow \quad p = \frac{10cm + 1{,}2cm}{1{,}2cm} = 9{,}3$$

We now use the next diagram for calculation:

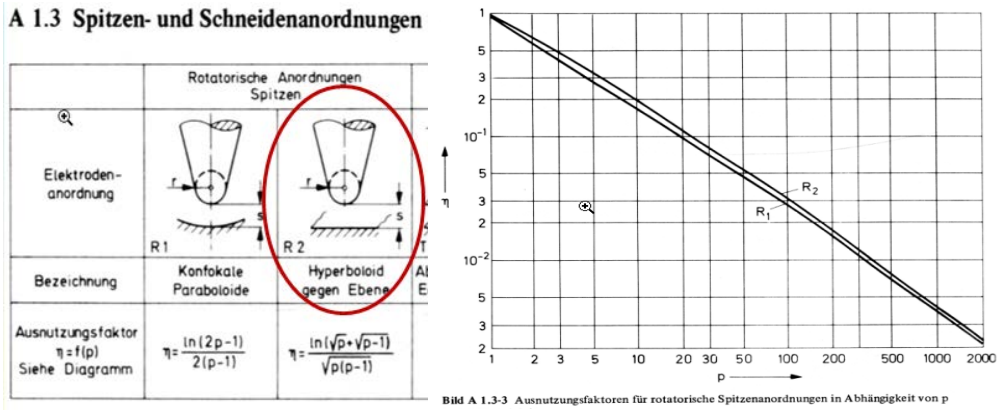


It results: $\eta = 0{,}12$ $\qquad \eta(p) = \frac{E_{mittel}}{E_{\max}}$ $\qquad E_{Max} = \frac{E_{Mittel}}{\eta} = 83\,\frac{kV}{cm}$

You can see the effect of the field enhancement of inhomogeneous arrangements: the former 10 kV/cm is now increased to 83 kV/cm.

In case of a needle-plate-arrangement, it is not possible to get a direct result, since such a needle has theoretically a infinite high field strength. For practical

considerations, an approximation is used. We use the field distribution of a cone-plate and assume that the radius of the cone is very small. If we estimate a radius of 0.01 cm, the needle-plate-arrangement can be calculated:



With these assumptions, it results:

$$\eta = 0{,}004 \qquad p = \frac{(10 + 0{,}01)cm}{0{,}01cm} = 1001 \qquad E_{Max} = \frac{E_{Mittel}}{\eta} = 2500\ \frac{kV}{cm}$$

The field enhancement factor is very high, which is caused by such sharp needles or protrusions. In practice, such protrusions could generate partial discharges and –under extreme conditions- could initiate complete voltage breakdowns. It should be mentioned that for special purposes such field enhancements are used i.e. in electro-filter, whereas in usual cases such high electrical fields are not wanted.

Please note: you will find other diagrams within the script of formularies for calculation of field strength in different arrangements.

With aid of the described methods, you can check the results of field simulation programs, especially at zones with critical dielectrics.

The following procedure is applied when HV-components are dimensioned with aid of 3D-field calculation programs:

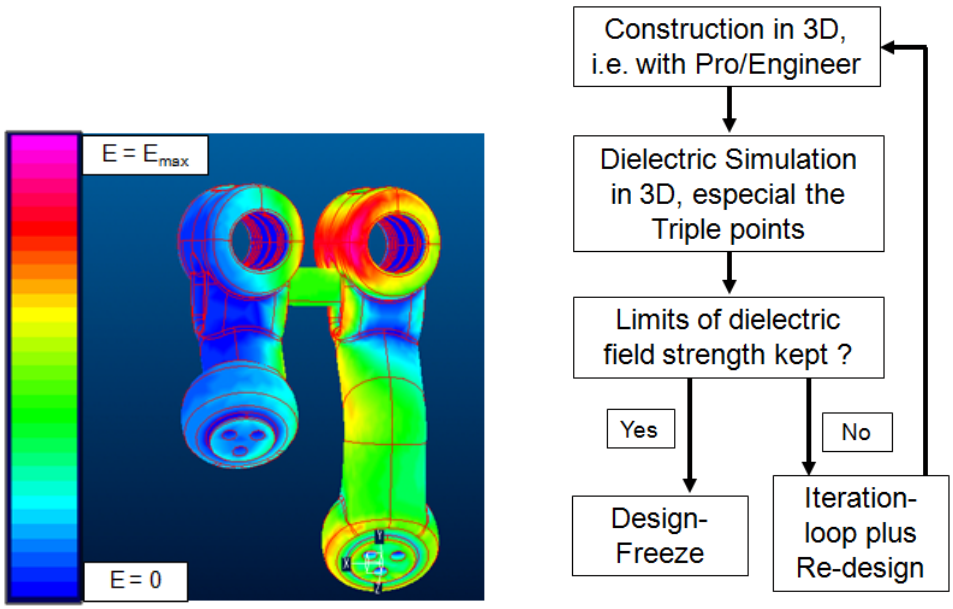


Fig. 3.10 Procedure to use field calculation programs

It is obvious that this procedure has iterative character and needs experience with regard to the resolution of the mesh and to judge the effect of critical zones.

4 Discharge Behavior of Insulating Materials

High Voltage components need insulating media to ensure their main functionality. The most efficient insulating materials are gaseous media like air insulation. Based on atmospheric conditions (20°C, 1013 mbar air pressure), the theoretical breakdown field strength results to roughly **30 kV/cm**. You can find this range within the next table, which shows the practical breakdown voltages as function of different insulating media. The practical breakdown value is reduced from 30 kV/cm to 25 kV/cm. Fig. 4.1 illustrates the linear functionality between breakdown voltage and gap distance, leading to i.e. 75 kV breakdown voltage at 3 cm gap distance.

When we compare the breakdown voltage of the insulating and arc quenching media “oil”, it results that even at short gap distances of 1 cm the breakdown voltage already reaches values of 110 kV! In case of “vacuum” as insulating and arc quenching media, the breakdown voltage at 1 cm is even higher and results to 170 kV.
Considering increasing gap distances, it has to be mentioned that both insulating media tend to follow a kind of saturation and are leaving the linearity.
The main application of „oil“-insulation medium is the power transformer, where oil ensures additionally the heat transfer to the outside.

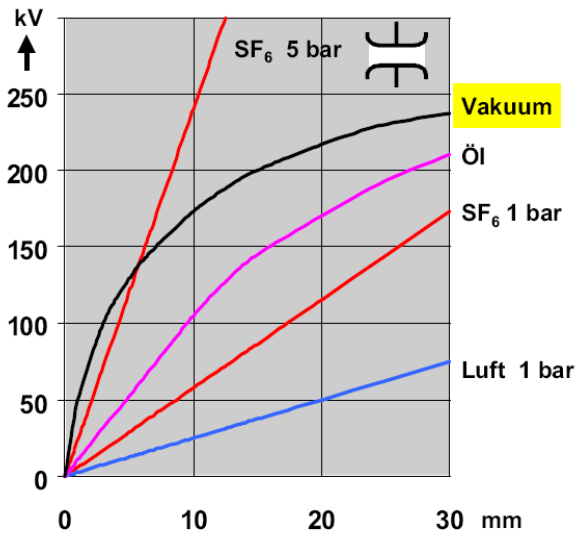


Fig. 4.1 Comparison of different insulating and arc quenching media [Source: Siemens]

The main application field of vacuum insulation is the circuit breaker for

medium voltages. In the range up to 52 kV, the “Vacuum Circuit Breaker” (VCB) is dominating. The related contact distances are varying between 10 mm and 30 mm and are covering rated voltages up to 36/52 kV. Based on these small contact distances, the needed drive energy is comparable low.
Please note: in special applications and special countries like Japan and nowadays China and Korea, VCBs are applied at voltage levels up to 145 kV, but the cost are high and there is actually no application field in Europe.

In former times „Air Switches“ were working with very high air pressures and were applied in voltage levels >= 72.5 kV. The arc quenching principle was such that the arc was exposed to very high air pressure beams, which leads to efficient cooling and quenching of the arc. Then a time period with “Oil Switches” followed.
Nowadays the transmission and distribution voltage levels are dominated by the synthetic gas „sulfur hexa fluoride“ SF_6 applied as insulating medium and additionally as excellent arc quenching medium in SF_6-Circuit Breakers.
SF6-Circuit Breakers are used up to the highest voltage levels and they guarantee highest arc quenching power and therefore are without competition.
The excellent dielectric characteristics can be derived from Fig. 4.1.
If the SF_6-gas is exposed to pressures of i.e. 5 bar, the breakdown voltage at 1 cm reaches 250 kV!
But is has to be considered that SF_6 belongs to the category of gases with a very high global warming effect. The related GPM-factor results to 22.300 compared to the reference gas CO_2. Those gases affecting the global warming are listed in the so called “Kyoto Protocol”. Due to the excellent and actually unbeaten advantages of SF_6, there is a special approval for use of SF_6 in High Voltage Levels and the manufacturer and electrical power supplier has agreed to a self control procedure to prevent the loss of SF_6 into the atmosphere.

4.1 Insulating Material Gas (Air, SF_6, ...)

Main components of gaseous media are neutral molecules, positive charged atoms and negative charged electrons, which are compensating each other from point of view of electrical charge. Furthermore, free electrons can occur under special conditions, which are charged with negatively charged and can be influenced by the electric field. When these free electrons are accelerated by the electrical field they can hit neutral molecules and can create positive ions. The most important charge carriers are:

- Electrons: Charge: $q_e = - e$
  - Elementary charge $e = 1.602 \cdot 10^{-19}$ C
  - Mass: $m_e = 9.11 \cdot 10^{-28}$ g

- Ions: Charge in case of single grade of ionization : $q_I = +/- e$
  - Mass: molecular weight M • mass of protons: $m_I \approx M \cdot 1836\ m_e$

- Big Ions: Such ions are created by accumulation of electrons at i.e. dust particles, water drops or comparable macroscopic parts

Even without applying an external electrical field, there is an amount of free electrons within gases. Their density depends on the equivalent between regeneration and loss of charge carriers by recombination and migration.
A source for external ionization is the natural radiation. Beneath corpuscular-radiation (α- / ß-radiation) there often exist high frequency electromagnetic radiation (γ-radiation and ultraviolet light).
In atmospheric air the cosmic and radioactive radiation creates a density of ions of roughly 10^3 cm $^{-3}$, without applying an electrical field. The average life duration of single ions is in the order of 18 seconds.

One special characteristic is given by electronegative gases: they provide a high affinity to electrons: that means, they tend to adsorb electrons. This leads to the effect of generation of negative ions.
Electronegative gases are oxygen, air, halogens and hybrids of halogen and other gases like SF_6. The probability to adsorb electrons is high in case of electrons with low kinetic energy. In case of electrons with high kinetic energy, the probability to generate negative ions is more likely than being adsorbed.
Other gases like nitrogen, hydrogen and nobel metals do not tend to generate negative ions.

4.1.1 **Dependent Discharge**

A gas discharge needs moving charges which represent a kind of electrical conductivity. Such discharges can be kept alive by external influences like radiation or heating or by the discharge itself.
In case of external influences the discharge is dependent, in case of effects resulting from the discharge itself, it is called independent discharge.

We consider now the case of a point charge q, which is influenced by a electrical (vector-) field E and lead to the Coulomb-force F:

$$\vec{F} = q \cdot \vec{E}$$

Assuming a particle with a charge q is travelling through the distance x without collision and accelerated by the voltage ΔU, then the increase of kinetic energy is given by:

$$\Delta W = \int_0^x F(x) \cdot dx = q \cdot \int_0^x E(x) \cdot dx = q \cdot \Delta U, \qquad \Delta W = \frac{m_q \cdot (v_0 + \Delta v)^2}{2}$$

In gases with high density, the electron movement is affected by collisions with other particles, such that we can assume an average free distance λ before it collides with another particle. In case of a collision between heavy ions and gas molecules, a main part of the kinetic energy may be transferred to the collision partner. Considering an elastic collision between an electron and a gas molecule the electron will be only deflected and the complete energy is kept.
Only in case of many elastic collisions or single non-elastic collisions (ionization) the electron will lose the main part of its energy. If a constant electrical field E is given and collisions are present, then a constant drift of the charge carriers is initiated. In homogeneous applications the average drift $\vec{v}$ is constant.

The drift velocity is –assuming a small collision-free distance- linear depending on the field strength E and the polarity is given by the polarity of the charge carrier:

$$\vec{v} = \pm b \cdot \vec{E}$$

The mobility factor b results in case of atmospheric air at normal condition (1013 mbar, 20 °C) to:

for electrons $$b_e \approx 500 \frac{cm/s}{V/cm}$$

for positive ions $$b_e \approx 1 \ \frac{cm/s}{V/cm}$$

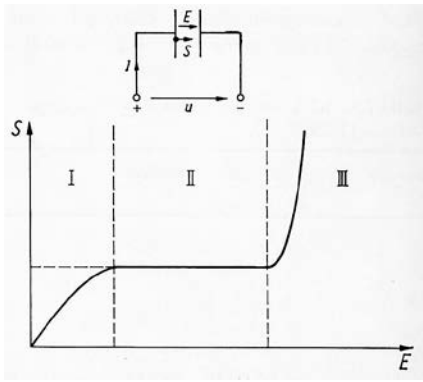
for large ions $$b_e \approx 10^{-4} \cdots 10^{-1} \ \frac{cm/s}{V/cm}$$

The maximum electrical field strength in air is 30 kV/cm. Please calculate the drift velocity for such electrical field strength of the ions and electrons:

$$v_I = b_I \cdot E = 1 \ \frac{cm/s}{V/cm} \cdot$$

$$v_e = b_e \cdot E = 500 \ \frac{cm/s}{V/cm} \cdot$$

It follows that for short term voltage stresses in the range of some µs the ions and especially the large ions do not affect the breakdown process. Maybe the presence of big charges could influence the voltage distribution indirectly. When applying a homogeneous electrical field to a gas with the electron density n_e and the density of positive ions n_I, a carrier flow is resulting with the current density S and the conductivity κ:



$$\vec{S} = \kappa \cdot \vec{E}$$

Fig. 4.2 Flow of Charge Carriers [Source: Hilgarth]

The current density of the flow of charge carriers is defined to:

$$\vec{S} = n_i \cdot \vec{v}_i \cdot e - n_e \cdot \vec{v}_e \cdot e =$$
$$= (n_i \cdot b_i + n_e \cdot b_e) \cdot e \cdot \vec{E} = \kappa \cdot \vec{E}$$

The curvature of the current density can be separated into three different areas: area I is representing the initiation of new charge carriers by external ionization, area II shows the saturation zone of a low current discharge (dark discharge or dependent discharge), in area III an independent discharge is generated by collisions which lead to ionization and charge multiplication. This area III is described now more in detail.

4.1.2 Ionization by electron collision

If the collision-free distance λ is long enough and an electrical field E is applied, then the electrons can reach such high kinetic energies that in case of collisions with neutral molecules new electrons could be generated and that the molecule changes into ion. The collision is successful, when a minimum ionization energy W_I is reached:

$$\Delta W = \lambda \cdot e \cdot E \geq W_i = e \cdot U_i$$

The average collision-free distance λ for atmospheric air results to 10^{-4} cm. The condition for ionization is realized at the following voltages:

Gas type	N_2	0_2	Hg	Cs	SF_6->SF_5 +
U_I in V	15,8	12,8	10,4	3,9	15,9

If this condition of ionization is fulfilled, then a self-dependent multiplication process of the electrons will start. Along a defined length dx a related amount of new electrons dn are generated, depending on the value of the electrical field:

$$d\,n = \alpha(E) \cdot n(x) \cdot dx$$

The ionization coefficient α = α (E) and the adsorption coefficient $\eta_e = \eta_e$ (E) of the electrons are influencing the electron multiplication. In case of electronegative gases like SF_6 the adsorption of electrons to molecules creates negative ions. This leads to the effect that the new electrons generated by collisions are reduced by adsorption, along the distance dx:

$$d\,n = (\alpha - \eta_e) \cdot n(x) \cdot dx \quad => \quad d\,n = \alpha_{eff} \cdot n(x) \cdot dx \quad \text{with } \alpha_{eff} = \alpha - \eta_e$$

For homogeneous electrical fields the integration based on initial amount of electrons n_0 results to:

$$n(x) = n_0 \cdot e^{\alpha(E) \cdot x}$$

and for inhomogeneous electrical fields:

$$n(x) = n_0 \cdot e^{\int_0^x \alpha(E)\,dx}$$

This charge multiplication leads to the following electron avalanche:

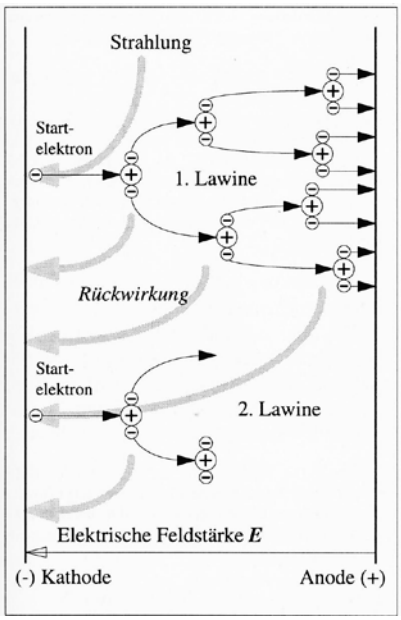


Fig. 4.3 Initiation of an electron avalanche [Source: Küchler]

Every discharge starts with an initial electron. This initial electron is placed at the cathode as you can see in Fig. 4.3. Due to the applied electrical field this electron is accelerated towards the anode. If it has gained enough kinetic energy and then it collides with a gas molecule, then the collision creates an additional electron (negatively charged) and leaves back a positively charged ion. Both electrons will now be accelerated by the electrical field, they can cause collisions and this leads to the multiplication effect. This multiplication can be interpreted as an electron avalanche according to the following scheme:

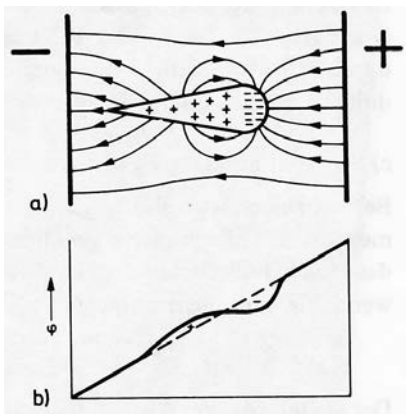


Fig. 4.4 Electron avalanche in case of homogeneous electrical field
a) Field Distribution b) Potential Distribution

At the front of the avalanche the amount of electrons is very high which leads

to high charge densities. Behind this zone of electrons the positively charged ions are located. Such positively charged ions move slowly towards the cathode. When they hit the cathode, they can release new so called "secondary" electrons.

4.1.3 Law of Analogy

Ionization only starts, when

$$\lambda_e \geq \lambda_i = \frac{U_i}{E}$$

For the following consideration, we assume a gap distance x and within this distance an average amount of x/λ collisions. Furthermore we assume that the real ionizations are reduced by factor "exp (λ_I / λ)", because only electrons with collision-free distances in the range of λ_I will cause ionization. This results to an ionization coefficient α as follows:

$$\alpha_{eff}(E) = \frac{1}{x} \cdot \left[\frac{x}{\lambda} \cdot e^{(-\lambda_i / \lambda)} \right] = \frac{1}{\lambda} \cdot e^{(-U_i / E\lambda)}$$

The average collision-free distance λ is anti proportional to the pressure (in case of constant temperature) and therefore leads to:

$$\frac{\alpha_{eff}(E)}{p} = A \cdot e^{(\frac{-B \cdot p}{E})}$$

This dependency is called law of analogy. Fig. 4.5 shows the relation between the effective ionization coefficient and E/p for air and SF_6:

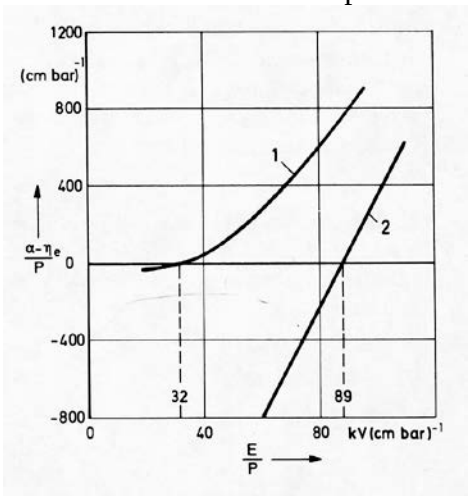


Fig. 4.5 Effective ionization coefficient of Air (1) and SF_6 (2)

It can be derived from the diagram above that the effective ionization coefficient in case of Air (η_e is low) follows an exponential curve, whereas in case of SF_6 the strong electronegative behavior leads to a linear relation.

The diagram shows furthermore that a_{eff} of Air gets > 0 at the value E/p = 32 kV per cm per bar. This means that above this value the charges are increasing rapidly and therefore enables the start of an electron avalanche.
The pass through zero is present at 89 kV per cm per bar in case of SF_6. This is the reason why technical SF_6-arrangements do provide nearly three times higher dielectric strength than compared to Air at same pressure.

4.1.4 Independent Discharge (Townsend-Mechanism)

The Townsend-mechanism describes a self-depending discharge. In this model it is assumed that electron avalanches are already created. As already mentioned the positively charged ions are moving slowly towards the cathode and can release secondary electrons from the cathode (or secondary electrons are released by photons). It is expected that the amount of new secondary electrons is a function of the amount of positively charged ions.
The related factor is called the secondary emission coefficient γ and varies between 10^{-8} and 10^{-1}.

Considering $\mathbf{n_o}$ primary electrons initiated in the near of the cathode, an exponential increase of electrons n_0 (e^{as} - 1) is expected during crossing the gap distance. The same amount of ions is generated. When these ions collide with the cathode, a secondary emission of electrons follows:

$$\gamma \cdot n_0 \cdot \left(e^{\alpha \cdot s} - 1\right)$$

These new secondary electrons cause new avalanches and so on …

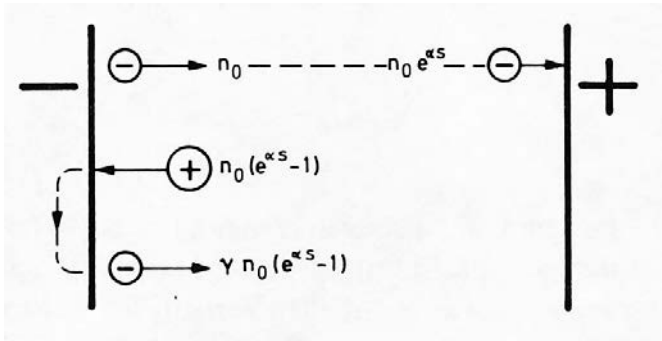


Fig. 4.6 Breakdown model according to Townsend Mechanism

When the amount of secondary electrons is higher than the original amount of electrons n_o, then the discharge is getting self-dependent and leading to a complete breakdown. We can derive the spark condition:

$$\gamma \cdot \left(e^{\alpha \cdot s} - 1\right) \geq 1 \qquad \text{or} \qquad \alpha \cdot s \geq \ln\left(\frac{1}{\gamma} + 1\right)$$

In case of homogeneous arrangements the condition can be simplified to *[J.S. Townsend* 1901]:

$$\alpha \cdot s \geq k$$

Practical values for **k** are varying between **k = 2,5 and 18**.

Analogue to that a law for inhomogeneous arrangements can be determined *[W. 0. Schumann* 1922]:

$$\int_0^s \alpha(E)\, dx \geq k$$

In both cases α is depending on the electrical field strength E. At the moment of reaching the spark condition, this field strength is called breakdown field strength E_d.
If the spark condition is already fulfilled at gap distances x < s, then partial discharges can occur. These partial discharges are local and do not cross the complete gap.

For **homogeneous fields** it can be concluded:

$$\alpha \cdot s \geq k$$

The Law of analogy $\qquad \frac{\alpha_{eff}(E)}{p} = A \cdot e^{\left(\frac{-B \cdot p}{E}\right)}$

leads together with the spark condition to:

$$\alpha_{eff}(E) \cdot s = A \cdot p \cdot s \cdot e^{\left(\frac{-B \cdot p \cdot s}{U_d}\right)} \geq k$$

By changing this equation to U_d, the general **„Law of Paschen“** follows:

$$U_d = B \cdot \frac{p \cdot s}{\ln\left(\frac{A}{k} \cdot p \cdot s\right)}$$

The most important outcome from this equation is that the breakdown voltage is only depending on the product p*s. This relation was already found experimentally in 1889 by *F. Paschen*. The basic curvature of the curve of Paschen is illustrated in Fig. 4.7:

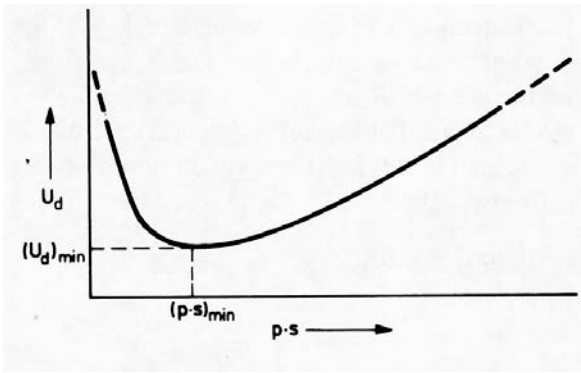


Tabelle 3.2-3: Paschen-Minima für verschiedene Gase [16], [46].

Gas	$U_{d\,min}$ V	$(pd)_{min}$ bar·µm
SF_6	507	3,5
O_2	450	9,3
CO_2	420	6,8
Luft	330 350	7,3
N_2	240 250	8,6
H_2	230 270	14
Ne	129 245	53,2
Ar	94 265	-
He	155	53,2

Abb. 4.7 Curve of Paschen with minimum voltages = f(type of gas) [Source: Kind, Kärner]

Looking at the **left** side of the curve of Paschen, the breakdown process is determined by the spark condition (collision-free distance is increasing, but less probability to hit a molecule). Below 10^{-6} bar, the breakdown behavior changes to the law of vacuum breakdown.

The **right** side of the curve of Paschen represents the „far"-discharge and is influenced by the condition of ionization. If we assume that the pressure increase and the gap is constant, than the average collision-free distance will decrease and therefore the necessary energy for starting an electron avalanche might not be reached. Therefore the breakdown voltage increases.

The law of Paschen is suited mainly for static breakdown phenomenon and therefore valid in the range of

$$p^*s = 10 \text{ bar mm}$$

Even in case of over voltages up to 5 % above the static value this limit is valid.
But when we consider technical arrangements with homogeneous field distributions, usually the validity of the law of Paschen is no more given. In this case, you can use the experimental relationship of SF_6-Insulations:

$$E_{d,\,technical} = 65\frac{kV}{cm} \cdot (10 \cdot p_{20}[MPa])^{0,73}$$

This dependency of the gas density is valid in the range of 1 .. 10 bar [Mosch, Hauschild 1978].

The right side of the curve of Paschen is characterized by an increase in scattering of voltage withstand, since either the pressure is high or the distance is high. The right zone of the curve of Paschen is therefore determined by other physical models like the **Streamer-Discharge** and –under condition of gap distances in the range around 1 m and above- the **Leader-Discharge**.

4.1.5 Streamer-Mechanism

If either the gap distance or the pressure or the rate of rise of the voltage is increased, then the experimental results differ from the curve of Paschen. It was found that the length required for an electron avalanche could not be increased according to the gap distance, because the avalanche tend to get instable at a critical length $\mathbf{s_k}$. For atmospheric air and homogeneous field distributions, the following equation results

$$(\alpha - \eta) \cdot s \geq \ln N_{krit} = k_{st} = 14 \ldots 18$$

Explanation: with increasing amount of ionization processes, the original electrical field will be continuously disturbed by the related space charges.

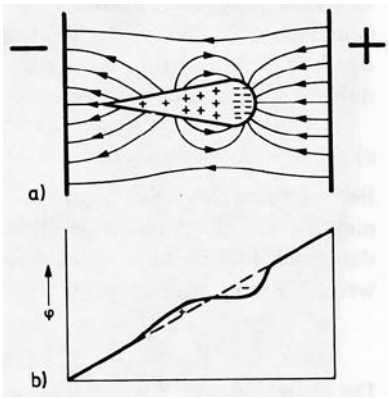


Fig. 4.8 Space charge of the electron avalanche (a) and related potential distortion (b) [Source: Küchler]

Especially in the front of the electron avalanche, local field enhancement is appearing, which accelerates the electrons such that further ionization

processes are initiated. Furthermore, atoms and molecules are stimulated to send out photon radiation. These short-wave light quantum particles can generate new starting electrons within the complete volume of the gap. As a result, new electron avalanches are initiated, which could be bundled together and linked by photon radiation. Consequently the gap could be crossed much faster than compared to the model of Townsend. As you can see in Fig. 4.9, photon radiation is directed mainly towards the anode, but some photon beam is directed backwards and therefore connects the avalanches to each other.
The quick growing of the Streamer-avalanches is accelerated by the light velocity of the photons and leads to growing velocities of 1 .. 10 m / µs. As soon as those channels are connecting the electrodes, the resulting current is heating up the plasma channel and initiates the breakdown:

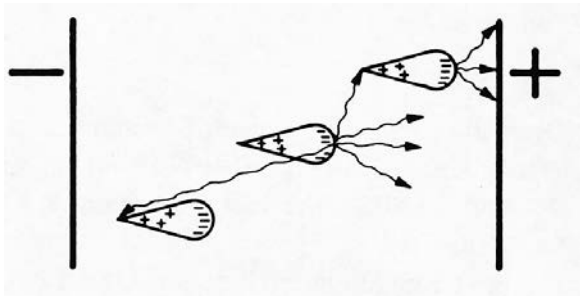


Abb. 4.9 Generation of plasma channels acc. Streamer-Mechanism

Inhomogeneous electrical fields: the critical length of the avalanche s_K can be derived from:

$$\int_0^s (\alpha - \eta) \cdot dx \geq \ln N_{krit} = k_{st} = 14 \ldots 18$$

Streamer discharge only appears for $s > s_k$. Due to the field dependency of α , the critical length s_k decreases for higher voltages. This explains why impulse voltage stresses lead to very short breakdown times down to some nano seconds. The Streamer-Mechanism can be described only limited from quantitative point of view.

4.1.6 Leader-Mechanism

Very large gap distances like s > 1 m, appearing in air insulated switchgear, are often related to inhomogeneous and unsymmetrical arrangements. Such arrangements are affected by the **Streamer-Leader-Mechanism**, especially

when electrodes with strong curvatures are exposed to positive switching impulse voltages. This Streamer-Leader-Mechanism can cross large gap distances with low average electrical field strength.

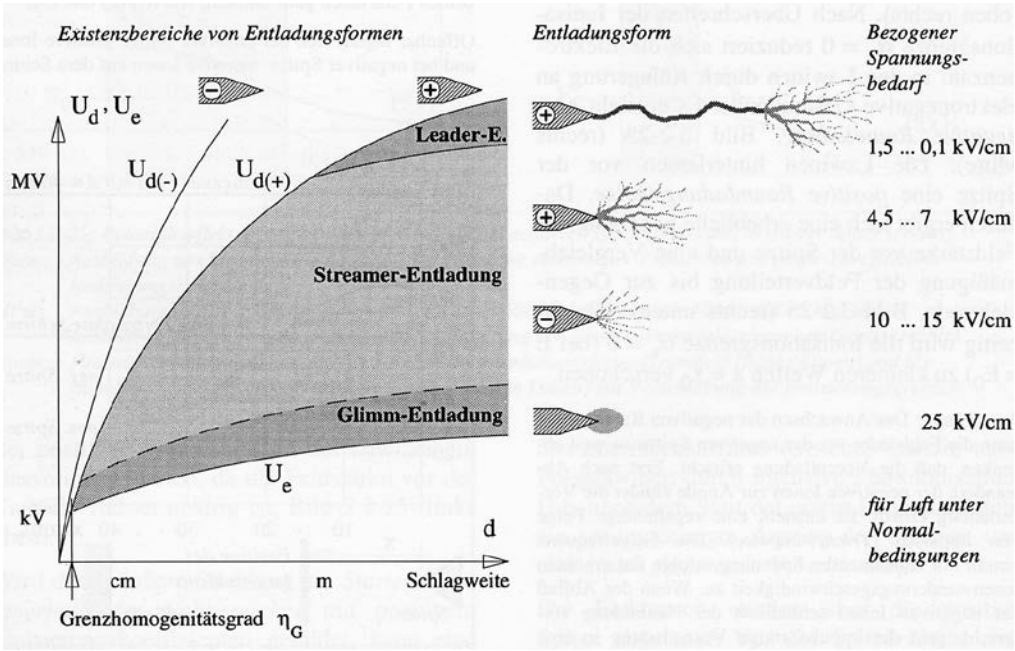


Abb. 4.10 Application of Glow- / Streamer-/ Leader-Discharge [Source: Küchler]

Additionally to the mentioned influencing effects, the surface roughness determines the breakdown voltage significantly. Especially local field enhancements induced by spines, scratches, protrusions etc. can lead to high field values above the field limits. Usually, the field enhancement is locally limited that the electrical field distribution is drastically reduced with the distance from such zones.

4.1.7 Dependency of discharge voltage on boundary condition

Air insulation under atmospheric condition is determined by the following parameters like *geometry, pressure, temperature, humidity, type of gas* and is influenced by *Field distortions*:

The influence of **pressure, temperature and humidity** is represented by an air density correction factor k_d and by an humidity correction factor k_h. The real breakdown voltage is based on the basic breakdown voltage U_{d0} *under*

$$U_d = U_{d0} \cdot \delta$$

atmospheric conditions

The correction factor for changes in pressure or temperatures is resulting to:

$$\delta = \left(\frac{p}{p_0}\right) \cdot \left(\frac{T_0}{T}\right) \qquad \delta = \left(\frac{p}{1013 mbar}\right) \cdot \left(\frac{293°K}{273°K + T}\right)$$

This correction is based on a linear extension of the curve of Paschen and is valid in the range of:

$$0{,}9 < \delta < 1{,}1$$

The influence of the humidity is negligible in case of homogeneous arrangements and in case of negative Streamer-Discharges. Only in case of positive Streamer-Discharges the real breakdown voltage is increasing with the absolute value of the humidity.

4.1.8 Partial Discharge (PD)

We want to discuss now a thought experiment: we apply 100 kV DC-voltage to an air-insulated-arrangement. The resulting average field strength is determined by the voltage and the contact distance of 10 cm, leading to a electrical field strength of 10 kV/cm in average.

> We know from the earlier chapter that this electrical field strength is uncritical in the first stage, since the breakdown field strength is in the order of 30 kV/cm. A breakdown will not occur.

Furthermore we imagine that at a local limited zone a sharp protrusion or a curvature of high magnitude is present leading to a field enhancement of factor 5. This would mean that at that local zone the field strength will be increase to 50 kV/cm.

> At this stage we can conclude that the arrangement is strongly inhomogeneous (factor 5 results to a grade of inhomogeneity of 0.2. Additionally we know that the breakdown field strength is exceeded.

What will happen at this zone of E > 30 kV/cm ? In every case an electron avalanche will start due to fulfilled spark condition.

> Such charge carriers resulting from this zone with fulfilled spark condition will be sent out into the contact volume. It can be summarized that a small part of the contact gap is conductive, meaning a locally present partial discharge.

This behavior is called imperfect discharge or a partial discharge (PD) in air. How far do the electron avalanches move ?

> The length of the zone with electron avalanches depends on the value

of the locally enhanced electrical field and from the reduced electrical field in the rest of the contact gap. The field strength in the rest of the gap decides about acceleration of the electrons and the ability to ionize neutral molecules.

Mainly we can state about the value of the electrical field strength in the rest of the gap. We simplify such that the field enhancement is present in 1/10th of the complete gap distance and afterwards will be reduced in one step to an constantly low electrical field strength. It follows:

$$0{,}1 \cdot 50\frac{kV}{cm} + 0{,}9 \cdot x\frac{kV}{cm} = 10\frac{kV}{cm} \qquad x = 5{,}55\frac{kV}{cm}$$

This leads to a constantly low electrical field strength of 5.55 kV/cm along 9/10th of the contact gap. The probability for a complete breakdown is low. Please check the calculation above graphically.

This thought experiment hast o be extended now by the effect of charged ions and the effect of different voltage polarities.

Effect of Polarity:

We discuss the example of a needle-plate-arrangement, where the needle is exposed to positive polarity. The electron avalanches starts around the needle due to the high electrical field strength. The electrons are such versatile that they are adsorbed to the anode very quick. As a result, positive ions are positioned around the needle. The effect of these positively charged ions is an reduced field strength around the needle.

In case of needles with positive polarity the electron avalanches must start within the gas volume, since the field strength is too low to start at the cathode. When the amount of electrons exceeds the critical number of electrons, the Streamer-Mechanism is incepted leading to many electron avalanches which are interconnected by photon beams.

The electrons are absorbed by the positive needle. The positive space charge is caused by the positive ions in the near of the needle. These space charges are acting like a sphere around the needle and therefore are reducing the resulting field strength around the needle. Consequently the field strength in the rest of the contact gap is increased. This increased field strength in the low-field range positively affects the growth and move of the Streamer-Discharges towards the anode. As a result, the breakdown voltage is reduced as compared to the situation without space charges.

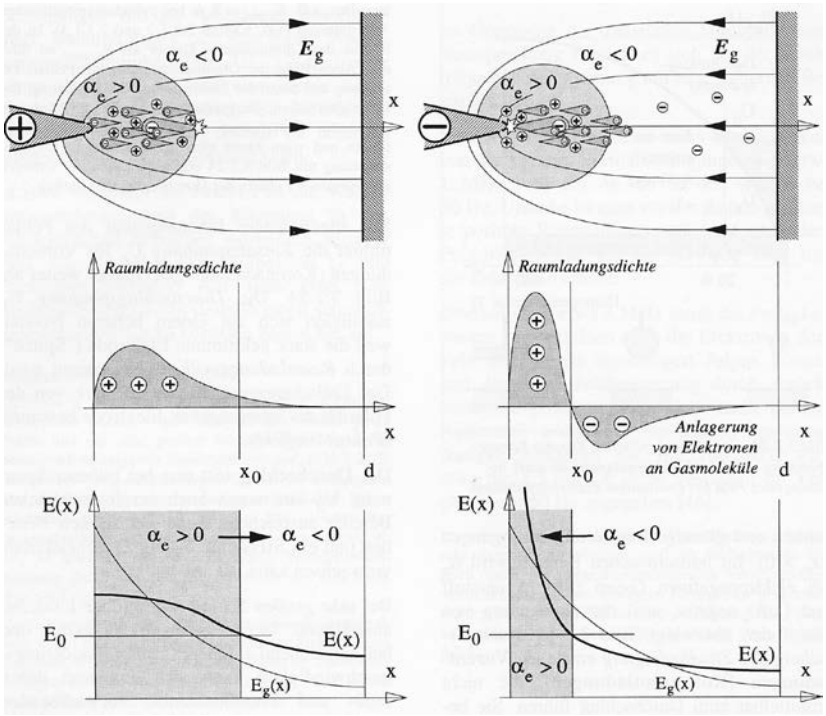


Fig. 4.11 Effect of Polarity [Source: Küchler]

If the voltage is increased continuously, first the „Cluster"-Discharges appear and are getting longer and more frequent. Finally by reaching U_d the breakdown follows.

In case of **negative needle** and earthed plate, a nearly similar process starts but leading to other consequences. When the spark condition is fulfilled, the electron avalanches are starting from the negative needle towards the anode. While the electrons are moving very quickly, the slow ions are staying around the negatively charged needle thus generating a very high electrical field (short distance of positively charged ions and negatively charged needle).
The initiated Streamer-Beams are moving into a zone of very low electrical field strength. When the ionization condition is not more fulfilled ($\alpha_e <= 0$) then the amount of electrons will be drastically reduced by absorption to the electronegative molecules.
In the vicinity of the anode, a negative space charge results. Both effects (positive space charge around the needle and the negative space charge near the plate) lead to the effect that in the zone between the field strength is too small to keep the discharge alive.
The discharge starts again, when the negative space charge has declined. This alternating process activates small impulses which can be detected in the outer current loop. Current pulses with durations of some 10 ns are resulting, firstly

detected in 1938 by *G. W. Trichel*.
If the voltage is increased continuously, then such „Cluster"-Discharges will start as well and finally a complete breakdown will appear.

It results:

$$U_{d,\,Positive\;\;Polarity} < U_{d,\,Negative\;\;Polarity}$$

The above relationship is illustrated by the following figure. You can see the inception voltage U_e, which represents the start of partial discharges.

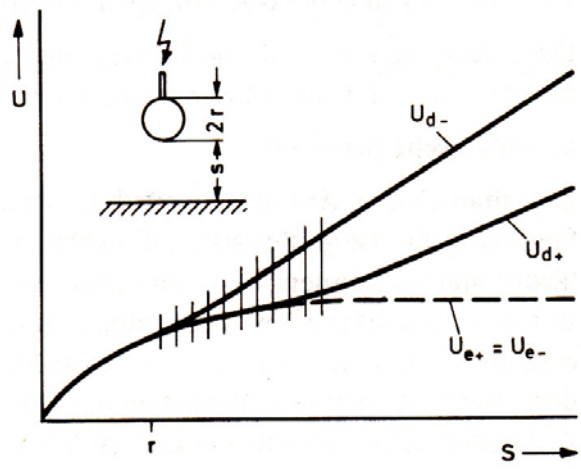


Fig. 4.12 Effect of Polarity at inhomogeneous arrangements [Source: Kind, Kärner]

Please note: practical problems of partial discharges arises at air insulated overhead lines: so called corona discharges (continuous or impulse discharges) leads to ohmic losses. Such losses strongly depend on the weather conditions. The losses vary between 1...10 kW/km. To prevent the inception of corona discharges, either the diameter of the conducting line has to be increased or the height of the overhead line has to be increased.
For voltage levels above 110 kV a bundle of lines is used (i.e. two-line-bundle or 4-line bundle) instead of single lines. Power lines are dimensioned in the range of 15 kV/cm average field strength.

4.1.9 Discharge Voltages

The breakdown voltages for **atmospheric air** are given by the following

relationships:

- **Nearly homogeneous electrical fields:**
  - Grade of homogeneity η = 1 to 0.8

$$U_d = E_d \cdot d$$

In this case, the equations of homogeneous field are valid. This covers the voltage values for DC, AC, switching impulse voltages and lightning impulse voltages, since the growing velocities of the streamer is very high in homogeneous arrangements.

- **Weakly inhomogeneous fields**
  - Grade of homogeneity η = 0.8 to 0.3

$$U_d = U_e = E_e \cdot \eta \cdot d$$

In these cases, the field efficiency factor η has to be considered.

- **Strongly inhomogeneous fields**
  - Grade of homogeneity η < 0.3

Long before the discharge appears, stable partial discharges are present. The breakdown voltage is estimated by the length of the pre-discharge comparable to the complete contact distance "d". Depending on the potential distribution and considering a small contact distance, the following relation is resulting: (with $E_{Glow} = E_G = 25$ kV/cm):

$$U_{d,\,Glow} = E_G \cdot d$$

For increasing contact distances, a streamer will be present in any case (with $E_{S,\,-}$= 10-15 kV/cm and $E_{S,\,+}$= 4,5 - 7 kV/cm):

$$U_{d,\,Streamer} = E_S \cdot d$$

In case of contact distances in the range of > 1 m, the discharge is depending on the streamer and as well depending on the leader effect (with E_L= 0,1 -1,5 kV/cm):

$$U_{d,Leader} = U_L + U_S$$

Based on the low voltage drop of the leader mechanism, the breakdown voltage will not be significantly increased by increasing the contact distance. The leader mechanism is initiated only in case of AC and positive switching impulse voltage and is the dimensioning factor for voltages above 300 kV.

4.2 Liquid Insulating Materials

Liquid insulating materials are used in many applications, such as power transformers, voltage and current transformers, in switchgear and in oil cables. They are of high importance due to the following aspects:

- **Impregnating and filling materials:**

Liquid materials benefit from the effect that they can cover voids completely, i.e. laminated dielectric in oil/paper-Cables or Capacitors. Even the expansion of the length of active parts of power transformers or measuring transformers is compensated without creating mechanical forces to the equipment. Liquids prevent from having voids between the active parts and the insulating material. Without voids, no partial discharge is generated, which increases the life time of an equipment drastically.

- **Cooling aspect:**

Liquids provide a large thermal capacity (in contrary to gases). Liquid flows are able to transfer heat energy (due to losses) to the outside. The transition resistance is very small, since the liquid always perfectly adapt to the contour of the active parts.

- **Insulating- and arc quenching media:**

In standard conditions, liquids act like a dielectric insulation. In case of oil circuit breakers and short circuit conditions, the oil is vaporized by the arc leading to cooling effects, whereas the liquid additionally acts as dielectric insulation.

Technical application of insulating materials

The following table gives an overview over the technical data of various liquids:

Mineral Oil	**$\varepsilon_r \approx 2$**	Refining of natural oil. Mixture of hydrocarbons. Problems: chemical ageing, water absorption (Gas-in-oil-analysis). Power transformers: it is recommended to check the humidity in defined time steps	200 - 350 kV/cm
Chlorinated hydrocarbon fluids (Ascarels, PCB)	**$\varepsilon_r \approx 4..6$**	High chemical reliability and low ageing effect. Forbidden due to environmental and health reasons (allowed for oil-paper-cables due to high ε_r)	200 - 300 kV/cm

Silicon oils	**$\varepsilon_r \approx$ 2,3..2,7**	Synthetic oils. Long structures with silicon oxygen connections + CH_3. Nearly no ageing, robust against firing. Expensive. Low ε_r fits to dielectrics of foils	150 - 250 kV/cm
Herbal Oils	**$\varepsilon_r \approx$ 4..6**	Natural resource, tendency to ageing. High loss factor	200 - 300 kV/cm

Mineral insulation oils are generated by Refining from natural oil. They exist as a mixture of different hydrocarbons (Paraffin, Cyclohexane, Aromatic Hydrocarbons, Olefin). A reference value of the dielectric strength is 350 kV/cm, the relative dielectric constant is given by $e_r \approx 2.2$ (at 50 Hz). The **chemical ageing** and the affinity to absorb water can significantly reduce the dielectric strength. The breakdown voltage of oil insulations has to be checked regularly. For that reason, a small amount of oil is taken from the equipment and checked by a special oil test. Such a gas-in-oil-analysis is very important to judge about possible partial discharges.

Chlorinated hydrocarbons, so called ascarels or polychlorinated Biphenyls (PCB), are very stable from chemical point of view, resistant against ageing and difficult to fire. PCB benefits from its high permittivity of $e_r = 4... 6$ (at 50 Hz) and therefore is used for capacitors. In Germany, it is forbidden to produce or use PCB-insulation.

Silicon oils or liquid silicon is consisting of large chemical structures with a central silicon-oxygen-connection. Silicon oil is resistant against ageing and difficult to fire. It is comparable to mineral oils with respect to the insulating behavior. But the price is high and the production effort is limiting the use of such insulating liquids.

Herbal Oils (Raps oil, **Rizinus oil)** are suited as insulating liquids in general. The disadvantages are the accelerated ageing and high loss factors. „Rizinus“-oil is used in DC- and Impulse Voltage Capacitors due to high permittivity factor $\varepsilon_r = 4.5 ... 5.7$.

Discharge behavior of liquids:

For liquids, no closed solution is available for the discharge behavior. When applying very high electrical field strength (some 100 up to some 1000 kV/cm), the liquid is transferred into a „frozen“-status. Depending on the

application time of the dielectric strength, there are different theories present. The following diagram shows the time dependent behavior of liquids with regard to the voltage strength. Four different zones can be matched with 4 ageing effects:

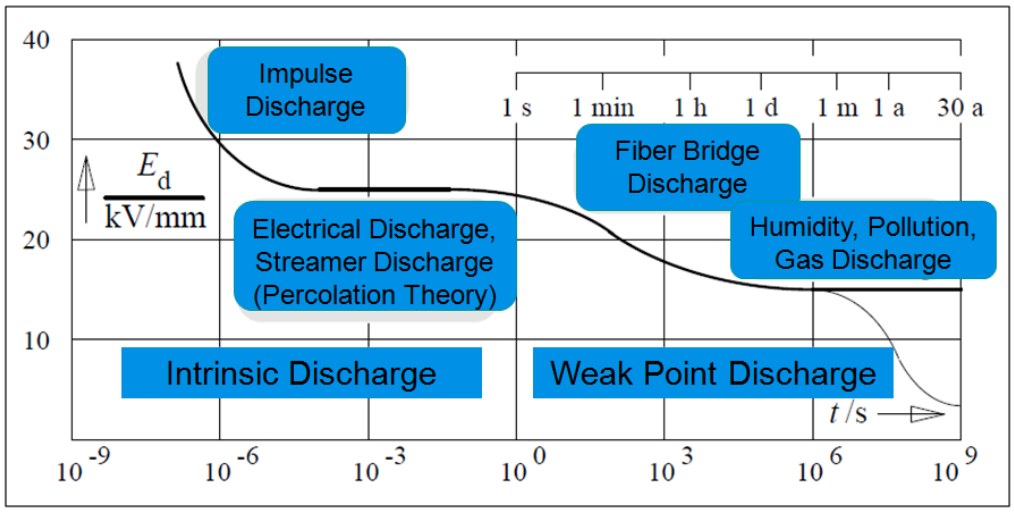


Fig. 4.13 Principle time dependent curve of the breakdown electrical field strength illustrated for mineral oils [Source: Küchler]

In the beginning, the „intrinsic electrical"-discharge is dominating and related to highest electrical field strength. After some short-time-ageing, the „weak zone"-discharge is following.

Electrical Discharge (virtual gas discharge, „micro drops"- theory):

According to the theory of the gas discharge, little micro-drops are generated within an oil vapor. The resulting free-distance of electrons is comparable to gases. At high electrical field strength, an ionization and electron-multiplication-effect is started comparable to gas discharges. Such discharges are typical for short voltage stresses (10^{-6} ... 10^{-3} s) superposed with high electrical field strength (> 200 kV/mm), i.e. at sharp edges or needles (field enhancement).

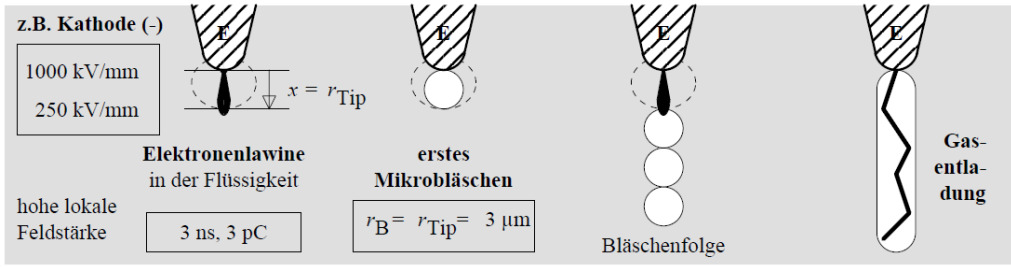


Fig. 4.14 Discharge development initiated by micro drops [Source: Küchler]

Percolation theory: According to that theory, the non-uniform structure of the molecules can be transferred (under a high electrical field strength) to quasi-crystalline structures, which gives a uniform direction for charge carriers.
This effect is shown in Fig. 4.14, where the electrode (E) is connected with a zone of unstructured conductive areas (A). Due to the high electrical field strength, such conductive areas are concentrating around an origin point (U), which is in connection with the electrode. As a matter of time, the conductive areas are connecting successively and tend to form a uniform direction for conductive channels. In these channels, the current is increasing. With increasing current density, local areas of high conducting oil are vaporized. Electron multiplication effects within such vaporized oil initiate a gas discharge.

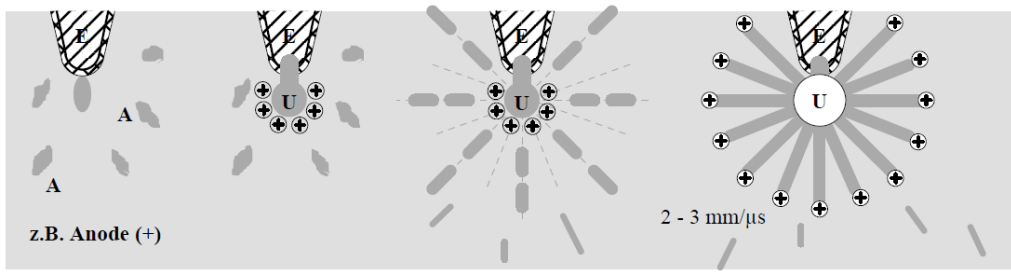


Fig. 4.15 Percolation theory [Source: Küchler]

Based on experimental results, the drift velocity of such conductive channels is estimated to 2-3 mm per microsecond.

Discharge of „Fiber-Bridges“:

The oil-paper-insulation of power transformers always contains fibers of

cellulose“, which are movable in oil. Under the influence of the electrical field, such fiber can connect to a channel between the electrodes within some minutes. Such cellulose-fiber can absorb much more humidity than pure oil, therefore the current density is high within such fiber-bridges leading to evaporation of dust and oil and resulting to initiation of a discharge.
To avoid complete bridging of such fibers, there are insulating barriers inserted within oil-insulated-arrangements.

Discharge of weak points (humidity, pollution, gas content):

Main factors for reduced dielectric strength of liquid insulations are:

- **Pollution within the liquid:**
  - Result: strong deviation of breakdown voltage (+/- 20%)
  - Non-regular pre-discharges (which must not lead to breakdown, even not in homogeneous arrangements)

- **Voltage-Time-Stress:**
  - In case of short voltage stress durations (i. e. at impulse voltages in the range of microseconds), the voltage strength is increasing by factor 2-3 compared to voltage stresses with alternating voltages (up to 200 kV/cm)

- **Humidity, Temperature, Gas Content, Loss Factor, Emulsion**

The self recovery of liquids is limited. For this reason, the liquid materials can be ranged between the behavior of gases and solid insulating materials.

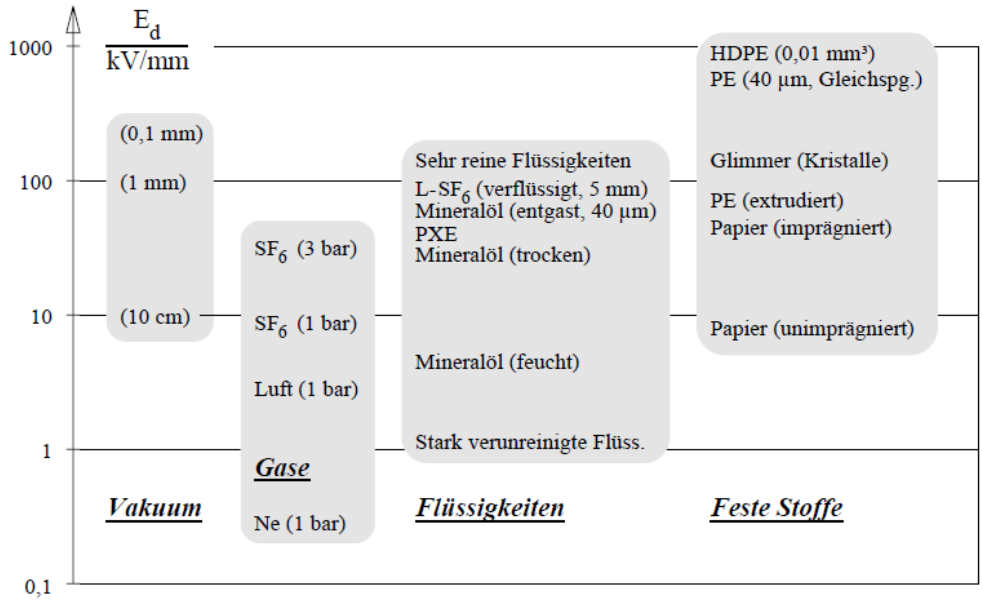


Fig. 4.16 Comparison of the breakdown field strength for vacuum, gases, liquid- and solid-insulations [Source: Küchler]

A special result derived from above is that the breakdown-strength of liquids is one order higher in the initial state as compared to the aged state.
The ageing is influenced by increasing humidity and increasing pollution.

A short overview about the influencing factors of liquid and solid insulating materials is given for the example of power transformers:

Non-recovery factors:

- **Solid-isolation based on Cellulose:**
  - Isolation-paper
  - Pressboard
  - Wood
  - Laminate

Recovery factors or exchangeable insulations:

- **Liquid Dielectric**
  - Mineral oil
  - Ester

The following factors are negligible in general:

- **Metals, Metal Components**
  - Copper, Aluminum
  - Iron, Electro Sheet

A power transformer is one of the most expensive high voltage components and therefore should be monitored very carefully. This monitoring is done with various methods, which are repeated in defined time steps (i.e. every year, every 5 years, depending on manufacturer recommendations or depending on the maintenance strategy of the utility).

Monitoring Measure	Physical Result of the Measure
Release/check of insulating oil	Content of humidity, Grade of pollution
Analysis „Gas in oil“-Content	Type of Failure, Intensity of Failure
Measuring of Resistance	Contact Resistance, Hot Spots
Isolation Resistance	Content of Humidity
Measuring of voltage ratio	Failure of windings (based on initial measurement)
Polarization currents	Humidity
Tanδ and Capacitance	Ageing, Humidity
Measuring of Partial Discharges	Partial Discharges, Ageing Online or Cycle-Monitoring
Analysis of „Flur“	Thermal status of the paper-set-up
Measuring of due-point	Thermal status of paper set-up
Thermograph	Thermal anomalies, Hot Spots

A power transformer consists of different insulating materials, which are affected by different physical influencing factors. Some of these factors are coupled. For this reason, it is worth to apply many measures as possible to detect a failure as soon as possible. An early warning can protect the power transformer effectively.
The ageing process can be reduced depending on minimization of possible failure durations by factor 10 or more.
In case of liquids, ageing effects can be recovered based on a high effort. In contrary, ageing effects on **solids cannot be recovered**.

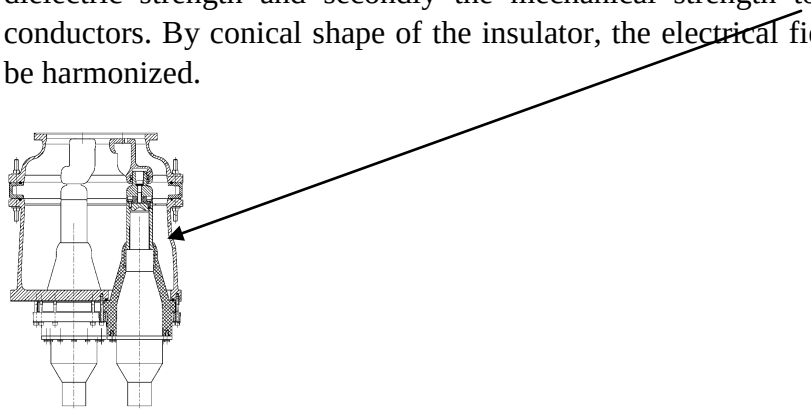
4.3 Solid Insulating Materials

Solid insulating materials do not transfer heat effectively. They do not recover after breakdown and therefore are not self-healing.
Especially under thermal influence, they tend to age very fast. The dimensioning factor is therefore depending on the ageing effect.
The following tasks are suited for solid insulations:

- **Material for special constructions used for** support or fixation of active parts, i.e. isolators of overhead lines

- **Filling materials** to fill voids, i.e. epoxy resins for cable terminations

The following example shows a cable termination of an gas-insulated switchgear. The cable plug-in has to provide two functionalities: first the dielectric strength and secondly the mechanical strength to fix the active conductors. By conical shape of the insulator, the electrical field strength will be harmonized.



In case of a cable termination, the solid cones have to withstand the overpressure, which results from the difference of the high pressure enclosure to the outside atmospheric pressure. For security reasons, the bursting pressure of such cable end terminations has to be demonstrated by experimental tests. Such values are defined as above 3-time of the construction pressure.

- **Enveloping Materials** to envelope active parts, i.e. PVC-mantling of a low voltage cable

- **Laminating Materials** for Impregnation of liquid or gaseous insulating materials, i. e. „Pressspan“ is used in oil-paper power transformers.

Solid insulating materials and insulating systems are developed in a high variety. They can be classified by:

Chemical compound

Solid insulating materials are organic or inorganic materials.

Important **organic Insulation Materials are:**

- **Thermoplastics,** i.e. Polyvinyl chloride (PVC), Polyethylene (PE)
- **Curable Plastics,** z. B. Polyester-, Silicon-, Epoxy-, Phenyl-Resins
- **Elastomere, i.e.** Silicon rubber

Important **inorganic Insulating Materials are:**

- Glass
- Glimmer
- Quartz
- Electro porcelain

Structure of Insulating Materials: Solid Insulating Materials provide:

- amorphous Structures (i.e. natural resins)
- crystalline Structures (i.e. Quartz, Glimmer)
- amorphous Structures with crystalline areas (i.e. Polyethylene)

Insulating systems

Insulating Systems based on organic and inorganic insulating materials provide excellent electrical and mechanical characteristics like mechanical strength and low shrinking.
Insulating Systems can provide an electrical strength which is not depending on the direction, a so called **isotropic** behavior**:**

- i.e. epoxy resin filled with quartz filler; pure non crystalline insulating materials

In case of laminated insulating systems like epoxy-glimmer-insulation used in electrical drives, the electrical strength is depending on the direction, so called anisotropic behavior. The same behavior is given by fiber reinforced insulating systems, i.e. fiber reinforced polyester resin.
The reason for the dependency on the direction of the electrical field rises from the excellent growing conditions of discharges along the interfaces.

4.3.1 Dielectric Loss Factor

If a dielectric is stressed with AC-voltages, the phase shift between current I and voltage U is given by $\varphi \leq 90°$. The polarization and conductivity losses are generating a small loss angular φ, which is slightly differing from 90°. The current component I_δ ("active current") is in phase with the voltage U and results to a real power generated within the dielectrics: that means the dielectric loss energy Pg. The current component I_C is advanced against the voltage U by 90° and is representing the capacitive reactive power Q_c.

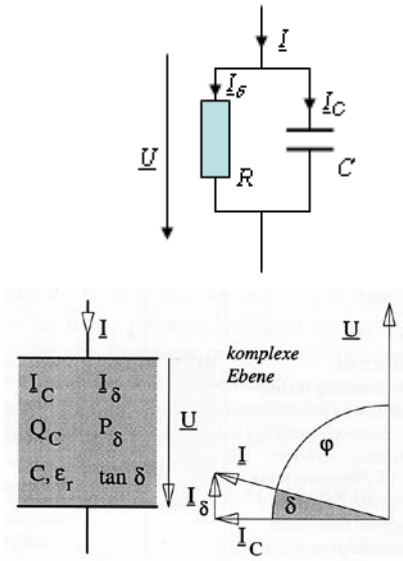


Fig. 4.17 Definition of the dielectric loss factor

The dielectric loss factor δ is represented by:

$$\tan\delta = \frac{I_\delta}{I_C} = \frac{P_\delta}{Q_C}$$

The loss factor tan δ is defined by the relation between the dielectric loss power P_δ and the capacitive reactive power Q_c in a dielectric. If the capacitive reactive power is known, then the dielectric loss power can be calculated directly from the loss factor:

$$P_\delta = Q_C \cdot \tan\delta$$

The loss factor tan δ is a material constant, which is determined by the polarization and the conductivity losses. The loss factor is usually greater than expected from the DC-conductivity. This can be explained by additional AC-conductivity and by the fact that polarization losses are generated by 50Hz-change of the electrical field direction.

High Voltage Insulating Systems usually uses materials with very small loss factors below 10^{-2} (1 %). Loss factors above that range can lead to thermal problems in service.

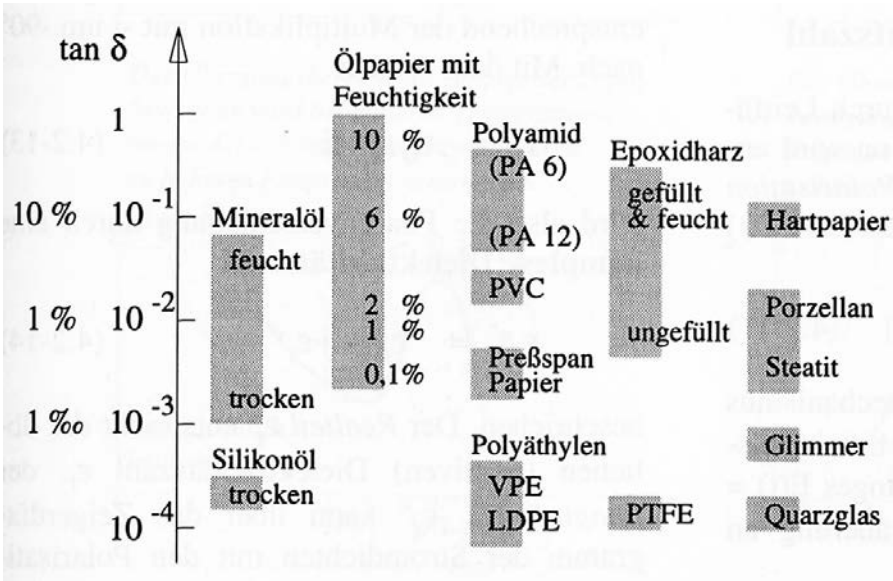


Fig. 4.18 Loss factors for different insulating materials

Materials which are sensitive against orientation-polarization, tend to high dielectric coefficients (i.e. PVC, Polyamide, Epoxy resin, Cellulose, Laminated Paper) and therefore show significant polarization losses.

Humidity increases the losses due to the excellent polarization effect of water and due to the increase of the conductivity. This is especially critical in case of humidity-sensitive materials like paper, lamination, Polyamide and fiber reinforced or filled plastics.

The loss factor is sensitive to parameters like **Frequency** and **Temperature. The loss factor increases with conductivity** and therefore with increased electrical field as well. In case of partial discharges, the loss factor will increase immediately *("PD-Step").*

4.3.2 Discharge effects

Pure solid insulating materials, which provide homogeneous structures, can withstand higher electrical field strength as compared to liquid insulating materials. But due to production processes, irregularities in the structure of the

material and inhomogeneous areas within the insulating material, the theoretical electrical field strength are reduced by roughly one order.
Practical values for poly ethylene cable are in the range of 30 to 70 kV/cm.
Depending on the time of the applied voltage, there are different discharge effects known. Such discharge effects are influenced by the material volume and the grade of homogeneity.

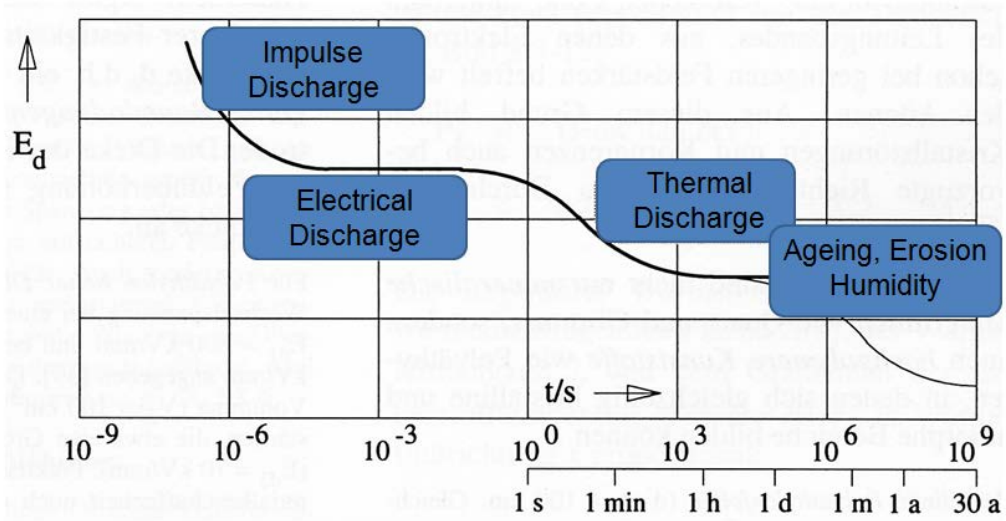


Fig. 4.19 Time depending ageing of solid insulating materials

Electrical Discharge:

The electrical discharge is based on a mechanism, which appears especially at high electrical field strength and in case of thin insulating layers. This effect appears without having significant thermal heating or without having partial discharges. The discharge is initiated after short voltage durations of some nanoseconds, when electrical field of some MV/cm are applied. The origin of that discharge is expected in the effect of free electrons, which are multiplied by the electrical field resulting in a voltage breakdown. Where are these electrons coming from and which mechanism leads to the multiplication effect?

- Initiation of electron avalanches is created by impulse-ionization. That means that the solid material acts like a gas with high density. With increasing temperature, the creation of electron avalanches is limited by the oscillations of the crystal structure => the limit of the electrical field leading to a discharge is increasing with temperature => **Low Temperature Breakdown (see curve 1: Quartz crystal)**

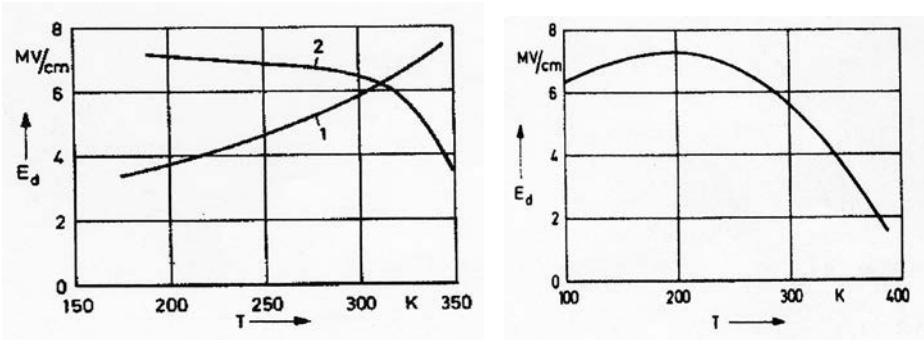


Fig. 4.20 Breakdown field strength as function of the temperature ϑ [Mosch, Hausschild]

- The electrons are released by inner field emission (electrons "jump" from the valence band to the conductive band). This effect grows with increasing temperature => **High Temperature Breakdown** (see curve 2 in left diagram representing quartz glass or PE-foil in right diagram). The field emission appears more likely at amorphous structures like glass (amorphous ≡ non crystalline). Mainly all other solid insulating materials are crystalline.

Thermal Discharge:

Heat Losses created by conductivity and polarization, are given in every volume unit of an insulating material. Such heat losses can be transferred easily in the regions near to the surface, whereas the heat transfer from the inner part of the insulating material is less efficient. This means that in the inner part of an insulating material the temperature tend to be higher as compared to the outer areas. In a region with high temperature, the resistance will decrease. As a consequence, the current density increases, which lead to higher temperatures. In this case, a "hot spot" is generated.
This hot spot represents an energy source, which will be driven by the voltage. At a certain voltage level, the thermal equivalent between heat transfer and heat generation will be disturbed such, that a breakdown will be initiated. This voltage is called "Thermal Breakdown Voltage" or "Forward Breakover Voltage U_K".

Fig. 4.21 is showing the heat loss generation as a function of the voltage of a solid insulating material, considering a constant temperature of the environment. When the voltage is below the critical "Thermal Breakdown Voltage", then the linear heat dissipation is dominant. When the voltage is above the "Thermal Breakdown Voltage", then the heat loss energy is higher than the dissipation energy, which results to a thermal breakdown. The

"Forward Breakover Voltage U_K" is representing exactly the limit, where heat loss generation is equal to the heat dissipation.

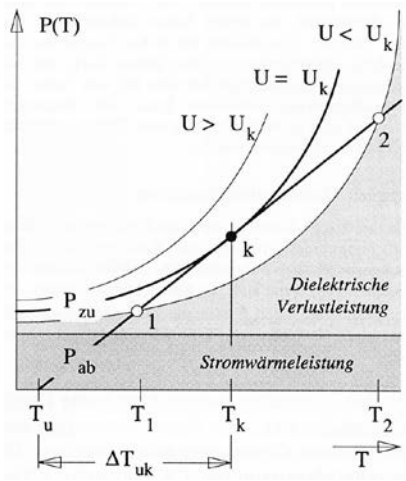


Fig. 4.21 Heat loss generation ⇔ dissipation as a function of U [Source: Küchler]

Point 1 represents a stable condition, point k is exactly the equivalent of heat generation and heat dissipation and point 2 is unstable (which means that a breakdown will occur).
Depending on the geometry of the arrangement consisting of solid insulating materials, there are two possible discharge variants:

Global Thermal Discharge
This global thermal discharge is applicable, when the conductivity within the material is equally distributed and the heat dissipation is only transferred by the surfaces. This implies that the maximum temperature is in the center of the material. Example: Cables.

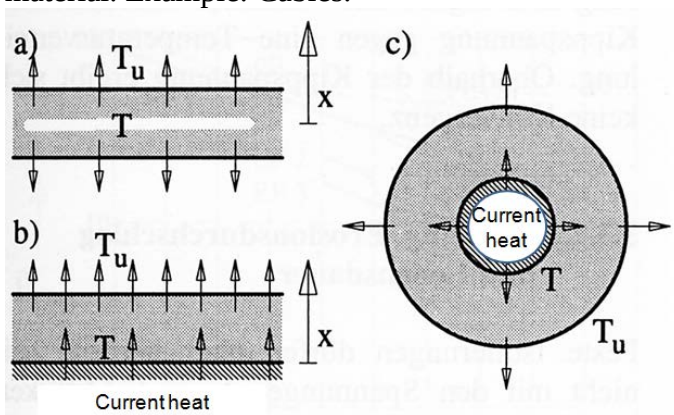


Fig. 4.22 Global Thermal Discharge

The “Forward Breakover Voltage” is not depending on the thickness:

$$U_K \neq f(d)$$

This means that an increase of the material thickness does not increase the voltage strength of a solid insulation system. It can be explained by the fact that an increase in thickness and in volume will lead to less efficient heat dissipation. The only measure to overcome this problem is to separate the solid insulation system in different layers and an increased cooling (i.e. decrease of the temperature of the environment)
For temperatures of the environment of 40°C and frequencies of f = 50 Hz, the “Forward Breakover Voltage” U_K results to:

Material	“Forward Breakover Voltage
Laminated Paper	80 kV
Porcelain	100 kV
Oil-Paper	250 kV
Epoxy Resin	300 kV

This breakdown mechanism is valid, when the heat transfer is mainly radial and not axial. This is given for **cable** and for **big drives**.

Local Thermal Discharge
In case of a local thermal discharge, the discharge is initiated by local areas of high conductivity. Consequently, the power losses are increased and lead to a locally generated breakdown.

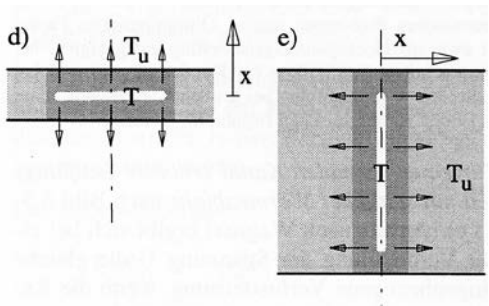


Fig. 4.23 Local Thermal Discharge

Such areas can be caused by irregularities within the structure of the solid insulating material or by pollution of the material.

The “Forward Breakover Voltage” U_K results to: $U_K = K \cdot \sqrt{d}$

In the case of a local thermal discharge, the “Forward Breakover Voltage” is

depending on the thickness, described by a root function, and a constant K.

The main difference between the global and the local thermal discharge is the additional axial heat dissipation.

Consequence: Insulating materials which provides K-factors of high magnitude or which generates significant heat losses (**ω*ε*tanδ** is high), cannot be improved by increasing the thickness > 5 - 10 cm!

For this reason, the dimensioning should be chosen such that the voltage is far away from the "Forward Breakover Voltage", since an aged solid insulating system would cause a breakdown at much lower voltages.

Discharge by Erosion (Partial Discharge):

Breakdowns caused by Partial Discharges are characterized by:

- Breakdown will occur **after some years**
- The Breakdown Voltage is far below the Breakdown Voltages of the Thermal Discharge

Discharges by Erosion are caused by external and internal Partial Discharges (PD). The external PD is affecting the insulating material at the surface and can be prevented by constructive countermeasures.
More critical are internal PD caused by gas voids or irregularities within the material. Due to field enhancement the dielectric strength is increased to a level, where the breakdown field strength is reached. Therefore locally limited discharges (PD) will occur.

Voids can occur during the casting process and are dangerous due to the following reasons:

- Gas voids reduce the breakdown strength drastically
- Gas voids provide a dielectric constant of $e_r = 1$, whereas solid materials covers a range of $e_r = 2$ up to 6. Since the displacement flux is constant, the highest field strength appears at the lowest e_r => at the gas void

The impact of damages of PD is depending on the voltage level. When PD appears far above the service voltage, then those PDs can only be generated by over voltages of short duration and are not critical. When PDs appear at or below the service voltage, then a continuous erosion of the solid insulation system happens, which lead to the following consequences:

- **Erosion** of the insulating material by collision of accelerated electrons

with solid material. Cause: mechanical removal of material

- **Chemical processes** => chemical "attack" on organic insulating material by O_3 (Ozone) generated by the PD
- **Injection of Charge Carriers** => the electron conductivity and the internal power losses are increased

PDs play an important role with respect to the life time of an insulating system. Therefore it is very important to measure the Partial Discharges regularly or continuously (Monitoring) at service voltages.
Another method of PD-detection is the tanδ-Measuring method. As soon as PDs are present, the tanδ will abruptly rise.

Estimation of lifetime of solid insulating materials:

Solid insulating materials withstand highest electrical field strength as compared to liquids or gases. For short term stresses, the limit of the electrical field strength is in the order of 50 kV/mm up to 1000 kV/mm. But we have to consider that solid insulating materials are aged by time and by the type of voltage stress.

According Fig. 4.19, the breakdown voltage will decrease over time. From this dependency, it can be derived:

- The breakdown voltage decreases with the duration of the voltage applied to the insulating system
- The insulating system should be stressed in such a way, that only the "Erosion, Humidity, Ageing"- Discharge can initiate the breakdown after very long time t > 30 years.

The dimensioning discharge process is given by the "Erosion, Humidity, Ageing"-Discharge. For new insulating materials, this process is not predictable. Therefore, it is mandatory to investigate the long term voltage strength of insulating materials experimentally.

Evaluation of the life time of a solid insulating material:

We consider insulating materials with negligible tanδ => the thermal discharge is obsolete. Then we prepare a reasonable amount of samples to start an investigation with respect to accelerated ageing.

- One measure is to investigate the samples at an increased temperature of i.e. 80° or 100° (depending on the application and the temperature limit of the material).

- The most critical decision is to choose the right initial voltage stress. The chosen voltage level is then applied continuously to a minimum amount of 5 to 10 samples, which are connected in parallel. Now the time until the samples will breakdown is fixed in a voltage/time-diagram of logarithmic order. The chosen voltage is perfect, when the breakdown times are in the order of minutes.
- Now the voltage should be reduced depending on the initial results. If the breakdown times of the initial samples are in the order of some minutes, then the voltage should be reduced by 20 %. If the breakdown times of the initial samples are in the order of hours, then the voltage should only be reduced by 10% (otherwise you will not get an breakdown in a reasonable time period). Perfect are breakdown times in the order of some 10 hours.
- Repetition of the step above. Perfect are breakdown times in the order of some months. Usual investigation time periods with regard to ageing is 6 to 7 months.
- All results should be taken into a diagram with both voltage axis and time axis of logarithmic order.
- To obtain a lift time curve representing all measuring points, a regression curve (based on an exponential function) should be calculated which provides a linearized match with the measuring points.
- Based on that regression curve, the life time can be estimated for time periods of 30 to 40 years.

Mathematically, the life time curve can be described by:

$$\frac{U_{d1}[kV]\,at\,breakdown\,time\,t_1}{U_{d2}[kV]\;\;at\,breakdown\,time\,t_2} = \left(\frac{t_1}{t_2}\right)^{-\frac{1}{k}} \quad with\ k = ageing\,coefficient$$

For epoxy resins, the ageing coefficient results to k = 8-12.

4.4 Partial Discharges (PD) in solid insulating material/ PD-measuring methods

Up to now, we considered only external Partial Discharges (i.e. caused by sharp edges or needle-plate configurations with field enhancement). Such external PDs can occur as well on surfaces of insulating materials. Then we

call them surface discharges or flashover discharge. Especially the transition from metal to insulators (interfaces) has to be carefully dimensioned.
The next figure 4.24 shows different principles of Partial Discharges. On the left side, homogeneous arrangements are demonstrated:

- When the breakdown field strength is reached, then the breakdown is initiated immediately, without showing any PD before.

In the center of Fig. 4.24, examples of **external** Partial Discharges are shown:

- Corona-/Glow-/Streamer-Discharges in case of strongly inhomogeneous arrangements (needle-plate set-up)
- Surface-/Flashover-Discharges in case of Interfaces (i.e. Cable End Termination)

The left side of Fig. 4.24 illustrates **internal** Partial Discharges, which are responsible for internal ageing and damaging of solid insulation systems:

- PDs are generated in voids (gas like air, ..)

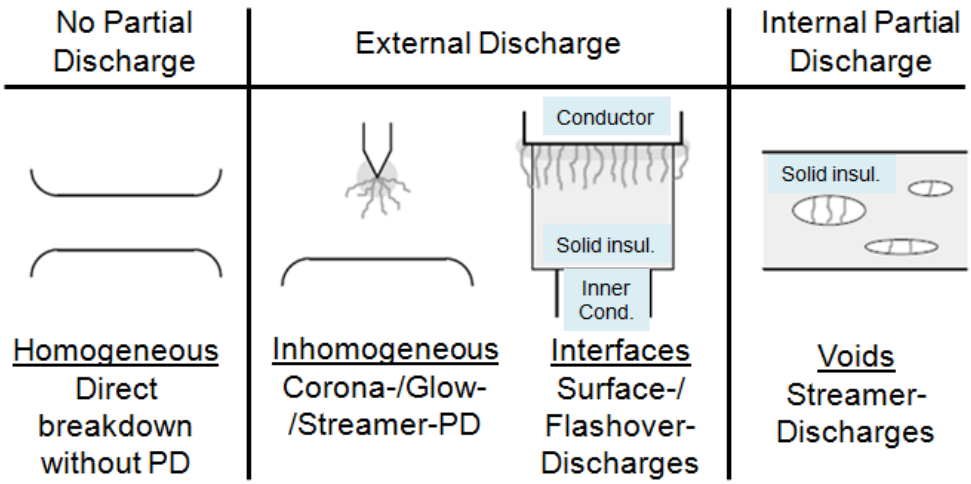


Fig. 4.24 Overview over different principles of Partial Discharges [Source: Küchler]

Assuming a complete homogeneous solid insulator without irregularities, no PDs would appear. Some critical voids are generated by the production process and are mainly filled with air or other production gases. Since the dielectric constant of gases is 1 and the dielectric constant of solids results to $\varepsilon_r = 2$ to 8, the electrical field in the void is enhanced by the factor ε_r.
Fig. 4.25 shows a scheme of a solid insulation system with an integrated void of air in the center of the insulation system. We apply a voltage and want to determine the electrical field strength in the different areas.

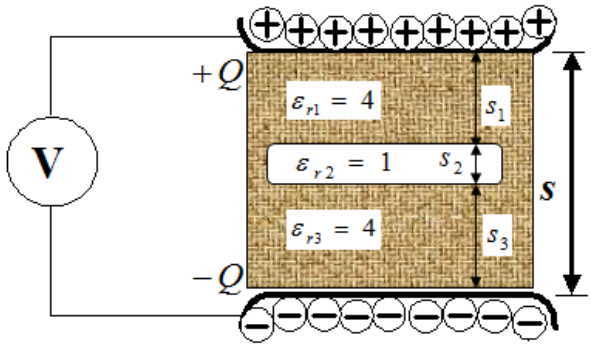


Fig. 4.25 Solid insulation system with void of air

We apply a voltage of 20 kV to the arrangement shown in Fig. 4.25. The related distances are given by: s_1 = 0.2 cm, s_2 = 0.1 cm, s_3 = 0.7 cm. Based on this geometry, we want to calculate the dielectric field strength of all three areas. The displacement flux is constant for this arrangement. It follows:

$$D = \varepsilon_1 \cdot E_1 = \varepsilon_2 \cdot E_2$$

Consequently, the electrical field strength inside the void is increased by the factor ε_r (solid insulation)/1. We start with the calculation of E_1:

$$E_1 = U \cdot \frac{\varepsilon_{r2} \cdot \varepsilon_{r3}}{N}$$

$$N = s_1 \cdot \varepsilon_{r2} \cdot \varepsilon_{r3} + s_2 \cdot \varepsilon_{r1} \cdot \varepsilon_{r3} + s_3 \cdot \varepsilon_{r2} \cdot \varepsilon_{r1} = 5.2cm$$

$$E_1 = U \cdot \frac{\varepsilon_{r2} \cdot \varepsilon_{r3}}{N} = 20kV \cdot \frac{1 \cdot 4}{5.2} = 15.4\frac{kV}{cm}$$

$$E_2 = U \cdot \frac{\varepsilon_{r1} \cdot \varepsilon_{r3}}{N} = 20kV \cdot \frac{4 \cdot 4}{5.2} = 61.5\frac{kV}{cm} \qquad E_3 = 14.4\frac{kV}{cm}$$

It is obvious, that the electrical field strength inside the void is four times higher than compared to the electrical field strength of the solid insulation.
Furthermore, E_2 exceeds the breakdown field strength of air by far (61.5 kV >> 30 kV/cm). Consequently, the voltage will break down within the void every time, when the alternating field strength reaches 30 kV/cm.
When this happens, the voltage breakdown will lead to short capacitive current, which charges the capacitive void again.
Such charges can be measured from the outside => the PDs can be measured and quantified.
The next scheme is illustrating the procedure to evaluate the PDs. For that reason, the equivalent circuit of such solid insulation systems is represented.

The model assumes that the process of breakdown, recovery, loading of charges to build up the new voltage is happening inside a channel (with the same basic area as the void and a length representing the complete height of the insulating system). This channel is responsible to recharge the capacitance of the void C_H through the resulting capacitance C_S of the insulating channel.

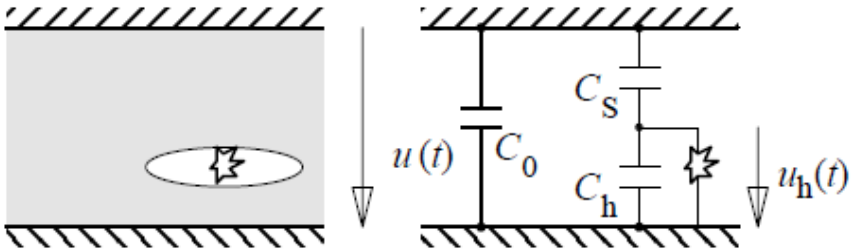


Fig. 4.26 **Model and equivalent circuit of an internal PD within a solid insulation system** (star represents the PD inside the void). C_S = resulting series capacitance of the channel, C_h = capacitance of the void. C0 = capacitance of the resuming area, Uh = sparking voltage at void

As soon as the breakdown voltage of the void is reached, a spark will occur and the voltage within the void is zero for short time. Afterwards the void will recover and will increase in voltage following the driving voltage u (t). This procedure will be repeated, when the voltage of the void reaches the breakdown voltage again.
The voltage u_h (t) is representing the un-influenced voltage, which would appear at the void without PD (depending on the ratio of the capacitive divider given by C_S and C_H.

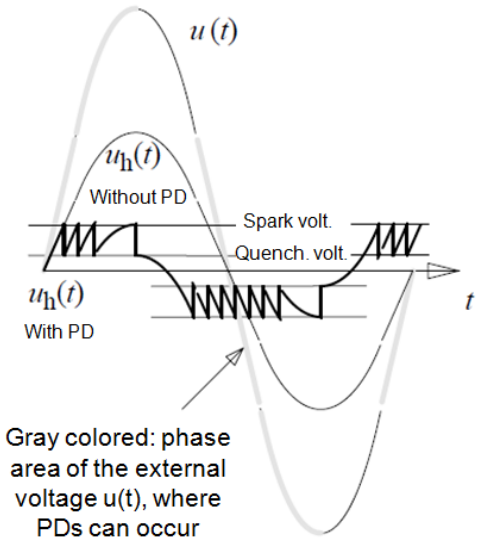


Fig. 4.27 **Illustration of PDs in a void [Source: Küchler]**

Every voltage breakdown and every recharging current can be measured from the outside. But the measurement only captures the virtual charge. Therefore, every test set-up has to be calibrated together with the test object.
This calibration procedure is done according IEC 60270. The measuring circuit is calibrated to 5 pC (pico Coulomb) according the following scheme:

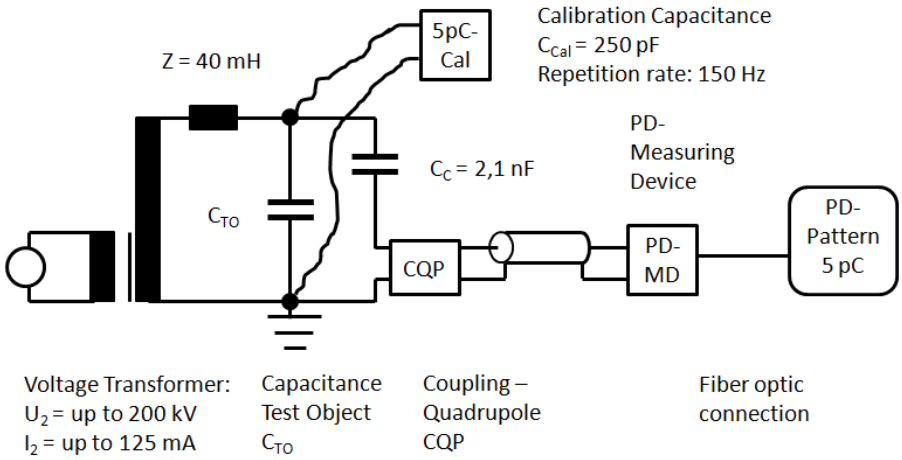


Fig. 4.28 Calibration of the PD-Measuring-Circuit acc. IEC 60270

The following figure shows an example of a test set-up (gas-insulated switchgear), which was tested in the HV-Laboratory in Darmstadt.

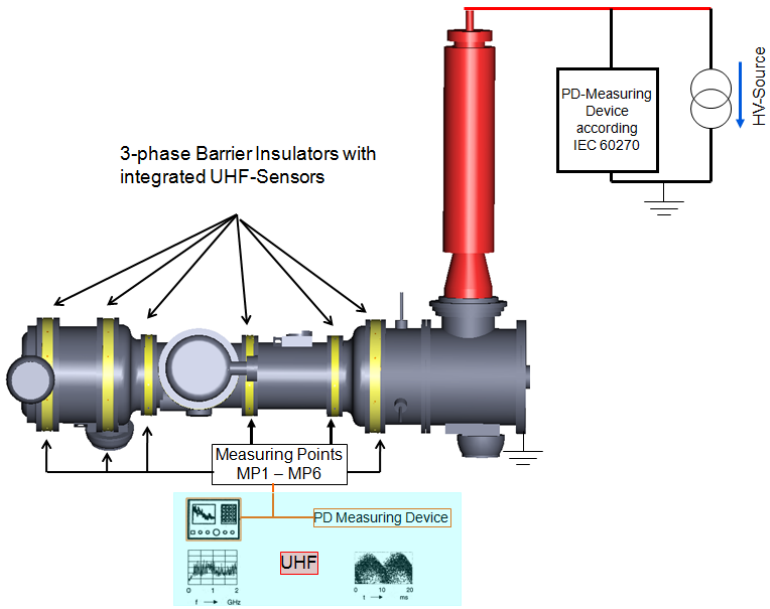


Fig. 4.29 Example of a test set-up (GIS) with related measuring points MP1/6

After calibration of the test circuit, PDs can be measured with high reliability and high accuracy. This method is proven since many years.
In the last years, a non-conventional method was introduced. This method is used for online-monitoring of PDs. The signals are detected by antennas, which are capacitive coupled to the inside of the GIS. The outcome is a frequency signal of the electromagnetic wave initiated by the partial discharge. The frequency response covers the range up to some GHz and therefore the method is called UHF-method (ultra high frequency). The procedure is explained within the next figure.

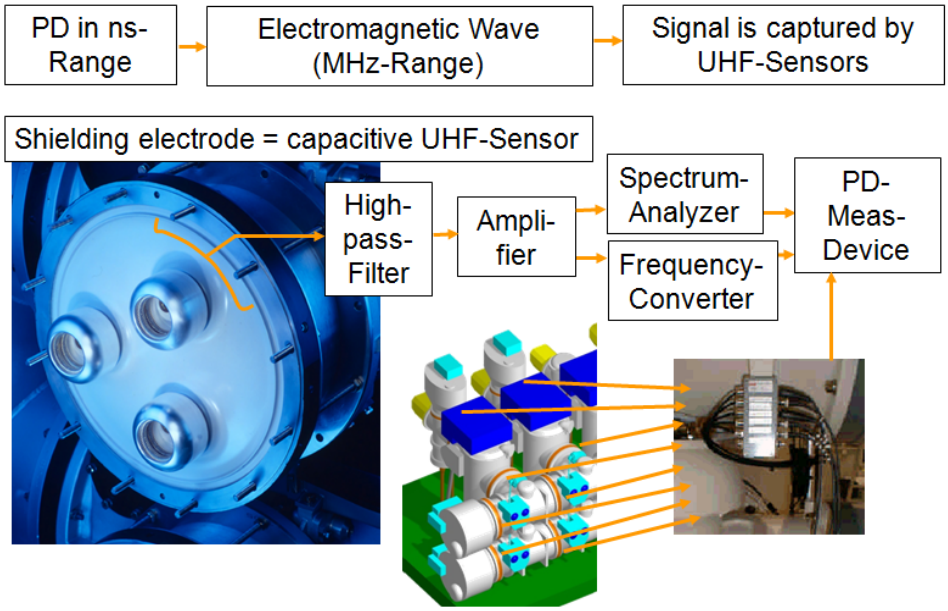


Fig 4.30 Principle of the UHF-Measuring Technique

It is possible to evaluate the frequency response (Spectrum Analyzer) or to visualize the time resolved partial discharge (in dB or V) in relation to the sinusoidal wave. The measuring chain is given by the next figure:

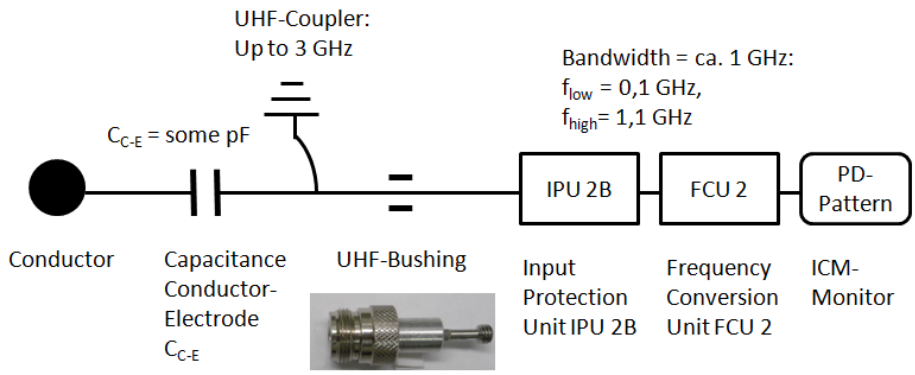


Fig. 4.31 UHF-Measuring chain according Fig. 4.30

Between the active part and the electrical shielding element, a capacitance in the order of some pF is present. Through this capacitive coupling the PD-signals are transferred to the outside. Afterwards the signal will be converted by a frequency converter unit (FCU) into a voltage signal (no reference to the related charge!).
The main difference with regard to the conventional method is that the interpretation of the resulting signals is much more complex and that the evaluation of the PD-pattern needs a lot of experience.
Furthermore, it has to be stated that the calibration has to be done in two steps: first of all a defined 5 pC-source should be applied both to a conventional system and a UHF-system. Afterwards, a reference voltage (Diriac Impulse) is applied to the sensor and the voltage has to be varied such that the same signal is indicated by the UHF-system as in case of the 5 pC-source. Now the UHF-system is calibrated indirectly.
After successful indirect calibration, the UHF-system delivers i.e. the following pattern:

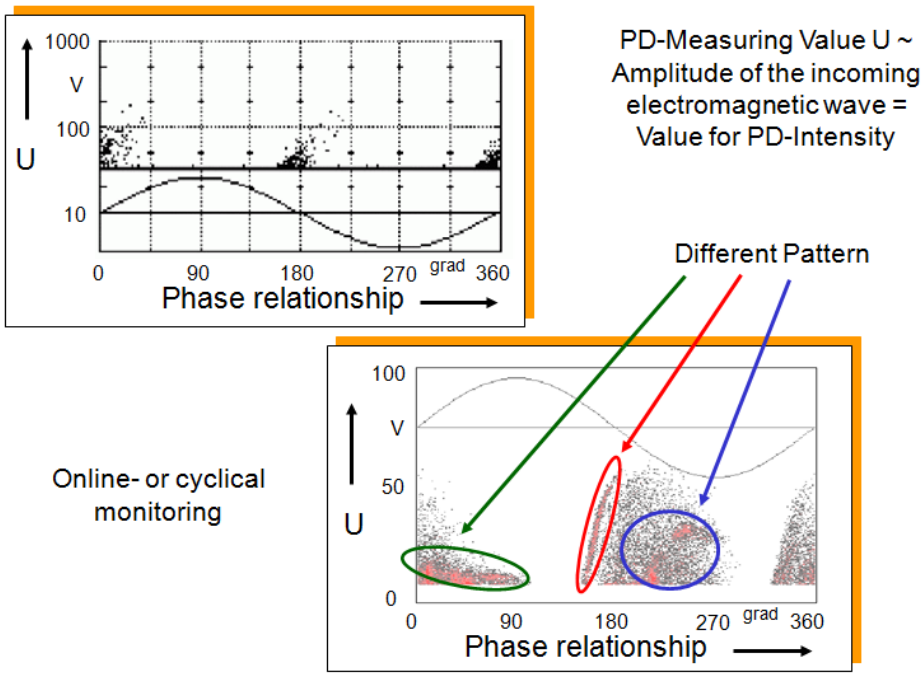
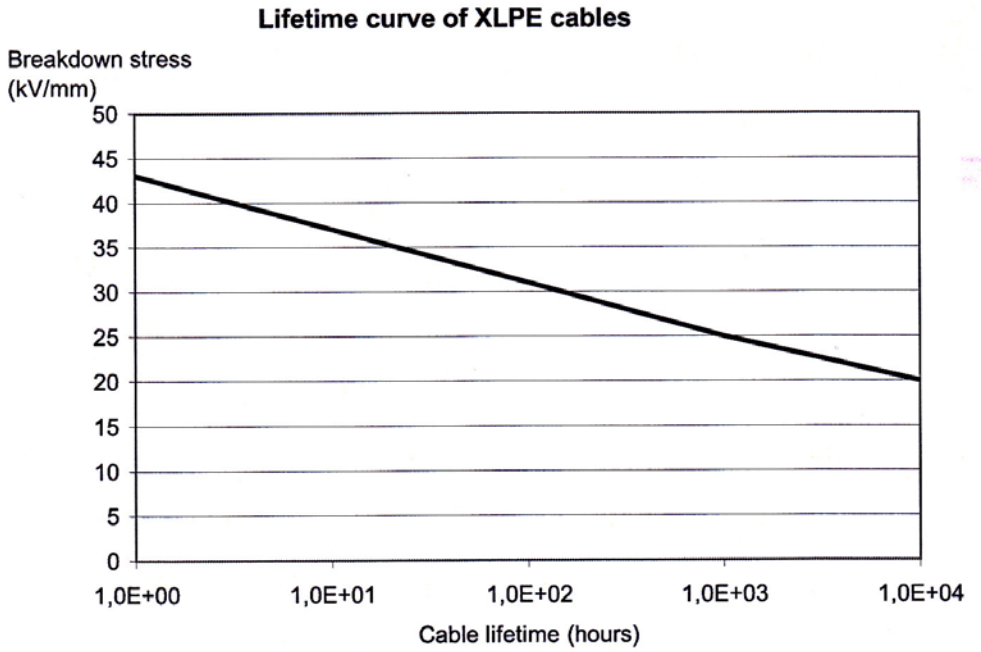


Fig. 4.32 UHF-Measuring System delivering voltage U and information about the phase relationship

In the figure above, a real measurement of an internal PD is illustrated. The main density is in the area around the voltage zero thus indicating an internal phenomenon related to a solid insulation.

Exercises

4.1 In chapter 3.4 a HV-Cable is introduced. For the given cable, the manufacturer [Brugg Cable] has provided the following ageing curve:



Please check –based on the given ageing curve- if the calculated maximum electrical field strength of 88 kV/cm of a 220 kV-Cable (see chapter 3.4) allows a life time of minimum 30 years. Please consider: you can extrapolate.

4.2 Which gases tend to absorb electrons and how we call such gases ?

4.3 Is it possible for ions to start an avalanche ? Which characteristic of ions could force an avalanche or prevent an avalanche, when we consider a lightning impulse voltage and a gap distance of 2 cm?

4.4 Which special effect of the streamer discharge allows very short breakdown times in the range of some nanoseconds ?

4.5 Please list up the influencing factors, which leads to the reduction in the breakdown strength in case of solids.

4.6 Please indicate, for which arrangements the polarity will appear: homogeneous arrangements, measuring sphere gaps, strongly inhomogeneous arrangements.

4.7 You apply an AC-voltage to an inhomogeneous arrangement. Locally the breakdown field strength is passed. What will happen in the first step? In a second step you increase the voltage further and further. In which half wave of the sinusoidal voltage the breakdown will appear?

4.8 We look at the example of Fig. 4.25 and we assume now that the length $\mathbf{s_2}$ of the given insulating system will tend to **zero**. Please calculate the electrical field strength E_1, E_2, E_3 and the related voltages U_1, U_2 and U_3 for that case.

4.9 Which measuring unit will be given in the display in case of a UHF-PD-measurement system?

5 Dimensioning of HV-Equipment and HV-Systems

For High Voltage Equipment, many dimensioning rules have to be considered. Beneath the dielectric criteria covered deeply in chapter 4, the mechanical requirements, the thermal criteria and -of course- the arc quenching rules are relevant for the development of HV-Components and HV-Systems.
Within this chapter, we will learn about the interaction between the different stresses.

5.1 Dimensioning rules based on Chapter 4

In the voltage range from medium voltages up to 170 kV/245 kV (Transition from distribution level to transmission level), the dimensioning dielectric stress is the lightning impulse voltage. For that reason, all HV-Parts have to provide field strength below the experimental limits of the electrical field strength. This can be ensured by 2D or 3D-field calculations or by experimental tests. It is wise to start with theoretical simulations before you start expensive practical tests.
The limits of the electrical field strength are depending on the insulating material, the environmental conditions and the construction. In case of gases, the condition of the active parts is of high importance. Especially the surface condition like roughness or irregularities due to protrusions, can initiate a discharge. Painting of the active parts will increase the dielectric performance by 5 to 10 %.
In case of solid insulating materials, the homogeneity of the material and the long term stability is decisive. Irregularities like voids, pollutions or other material irregularities have to be prevented during the production process.
Let us look to the dimensioning example of the HV-Cable from chapter 4: the life time curve shows that the maximum tolerable electrical field strength is depending on the required life time. For example, a 30 year life time limits the maximum tolerable electrical field strength to ca. 15 kV/mm.
In practice, the manufacturer can make his dimensioning choice: in the homogeneous part of the extruded cable, an electrical field strength of 7 to 14 kV/mm applied as service stress should be not a problem. At the cable end termination the tolerances will deviate much more and the assembly process will require more safety margin: therefore it is wise to reduce the applied electrical field strength to a range of 5 to 10 kV/mm. Even when protrusions lead to a local field enhancement, the limit of 15 kV/mm should be not

reached. It has to be mentioned, that the life time strongly depends on the temperature (the higher the temperature, the life time will be shortened) and of course on the voltage level.
As a **thermal** dimensioning basis for solid insulation systems: when the temperature is continuously 8 to 10 % above the rated temperature (given by the maximum allowed current), then the life time will be reduced by factor 2 !

If we consider gaseous insulating systems, the limits of the electrical field strength are lower and much easier to determine. An ageing effect of gases does not take place. Although physical effects like humidity or contamination are influence the dielectric strength. When we consider an encapsulated system like a gas-insulated switchgear (GIS), the influence of the environment and the influence of humidity are prevented. In these cases, it is possible to reduce the safety margin with respect to breakdown limit. In case of GIS, it is very important to prevent micro-protrusions on the active parts (field enhancement). One countermeasure to avoid micro-protrusions is to apply a vibratory grinding of i.e. casted parts. Additionally, a painting of casted parts increases the electron release energy and therefore improves the dielectric performance by 10 %.
SF_6 as insulating medium is able to withstand 70 kV/cm (7 kV/mm) in practice with high reliability (theoretical limit is 89 kV/cm). By increasing the gas pressure, the limit can be increased nearly linearly.

5.2 Introduction into the development process of HV-Components

Nowadays, a development process should be initiated by the product management. Based on a detailed market study, weak points of the portfolio of a company can be detected and these open points can be filled by a new product. Even a modification could be successful to step in a new market segment. The market study should provide a clear technical requirement! Since the development time can vary between 1 year and 3 years, it is very important to choose the right technical ratings (!) to guarantee that the product is competitive in the market after such a long period.
But in practice, the ideas for new products have different origin: creative ideas from development engineers or customer requests or –not seldom- the pressure from the competitor, which introduces a new product in the market.
Independent from the origin of the idea, the development process has to be managed in a defined way to guarantee a high quality product. This process

usually is defined in milestones, which can be tracked during the process:

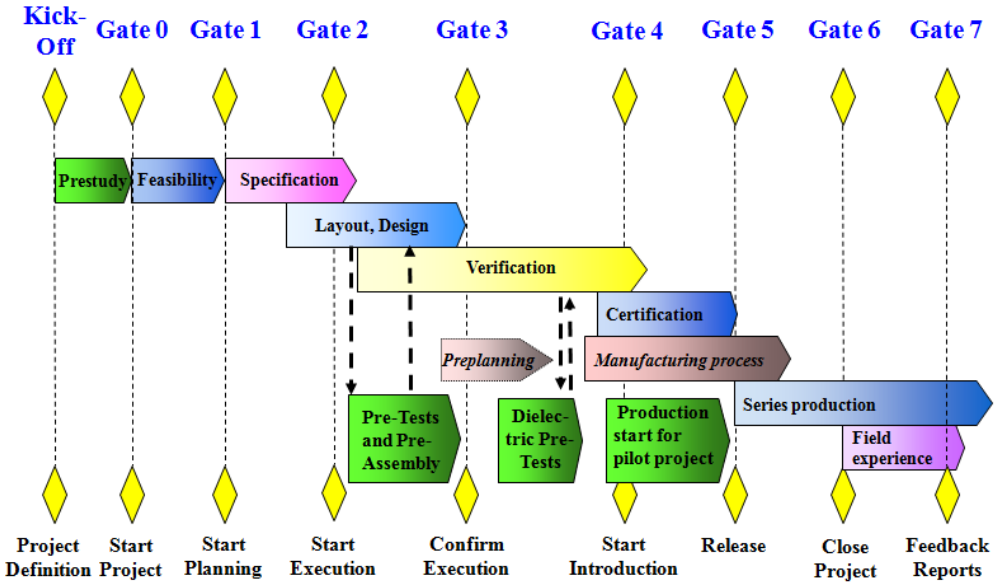


Fig. 5.1 Development Process according „Gate-Model“

The Kick-Off-Meeting is the official start of the project. Within this meeting, the project idea and the first steps are introduced usually by the pre-named project leader, which will propose a project team. When the management decides to start the project, then the project team will prepare a pre-study.

The next milestone is **Gate 0.** Until Gate 0 the following tasks have to be fulfilled by the project team and/or the product manager:

- Evaluation of a Pre-Study comprising schemes, calculations, theoretical analysis, 3D-vision of the idea
- Project Proposal will be extended by a concrete time planning up to Gate 2 and a first estimation of the development costs (Product manager and Project Team)

If the decision is positive at Gate 0, then the project team (which is possibly enlarged) has to prepare a detailed **Feasibility Study** up to **Gate 1** which contains some specific aspects:

- Simulations of any necessary kind, detailed 3D-constructions, proof how to fulfill the technical requirements, first product cost estimation
- Draft of a Technical Specification (close with product management)
- Cost estimation for development cost and the main parts of the product
- It is very important, that at this early stage a **Patent Research** is introduced to monitor the existing patented ideas in detail. If the idea already exists, then it has to be clarified immediately, if the new idea

can be patented in a modified way. If the proposed idea will lead to a patent infringement, this will lead to a project stop. If there are no patent obstacles, it is recommended to discuss the first schemes of the idea with the patent attorney to ensure the best protection of the idea.

The results are presented and discussed at **Gate 1.** In case of a positive decision, the team will get the release to detail the product ideas. Based on this detailed construction, it has to be checked, if the requirement specification can be fulfilled. The following additional tasks have to be done until **Gate 2**:

- Check and confirmation of the Requirement Specification
- Project Plan comprising a resource planning and a time schedule
- Possibly first pre-tests
- Draft for payback calculation (ROI: return of investment) based on the development and product cost estimation

These steps have to be done in close cooperation with the product management, who usually supervises the progress of the project.

Gate 2 is the most important Gate which decides about the release of the project or the end of the project. In case of a project end, the damage with regard to cost, manpower and management effort is comparable low for the company. On the other hand: in case of a release decision, the project leader has a strong position to push the project based on a company decision.
For this reason, the preparation work for Gate 2 should be very detailed and reliable to ensure the success of the project. At this moment, usually the operating unit takes over the responsibility for the next steps. This has two advantages: the operating unit wants to gain money from the new product as soon as possible and is interested in a short development time. Secondly, the development team is relying on the unit for first assembly tests and other support.

Starting with Gate 2, the project leader will be equipped with a defined target, a clear budget, a defined team and usually with management power. The project leader is from now the key person and is responsible to solve the tasks in a reasonable time. The required tasks for **Gate 3** are:

- Planning of all development tests, risk analysis
- Comparison and evaluation of different design variants, selection of the most promising variant, cost evaluation of this design variant
- Realization of the most promising variant from constructive point of view. Creation of part list, Review Process of the design. Finally the design will be inserted into a data management system like i.e. SAP
- Based on the production drawings, the product cost are checked by the supply chain management (SCM) and possibly corrected

- Selection of the possible suppliers inclusive contracting (done by SCM)
- The production process will be developed together with the operating unit. The assembly times and the assembly procedure will be created, discussed and confirmed.

With positive decision at **Gate 3** the layout/design is released and fixed. The project leader is now encouraged to build up the prototypes and to start with the development tests. At **Gate 4** the following tasks have to be completed:

- All drawings, part lists, procedures have to be available in SAP
- All development tests have to be successfully completed
- No technical risks should remain
- A zero series of the new product is successfully assembled by the operating unit, the process is checked
- All tools for series production are described and construction drawings are ready, production plans are available in electronic format
- Schedule and planning for type tests (usually the laboratories are highly loaded => the pre-planning about the availability of the test laboratories is very important
- Market introduction plan is done by product management
- Overview about chosen suppliers for series production is available
- Product cost estimation is checked, possibly corrected and confirmed
- A pilot project is defined, first market brochures are finalized
- Sales department has created a tool for product offers

Up to now, mainly resource cost and material cost for the prototypes have been spent. With a positive decision at **Gate 4** the budget for very expensive type tests (usually in external laboratories) will be released. To obtain a final product release at **Gate 5** the following tasks have to be successfully finalized:

- Final product cost calculation
- All type tests are completed successfully. For better understanding, here an example for the main tests for a 145 kV-design:
    - Dielectric tests according IEC 60694:
        - AC-Test-Voltage: 275 kV
        - Lightning Impulse Test Voltage: 650 kV
        - Partial Discharge Intensity Test: < 5 pC (acc. IEC), usually < 2 pC or PD-free
    - Mechanical Tests: 2000 On-Off-operations (or 10.000)
    - Short time current: i. e. 50 kA for 3 seconds
        - Temperature Rise Test: i.e. 3150 A. Temperature is measured under steady state conditions
    - Power Tests (extract from IEC 62271-100)
        - Disconnector/Isolator

        - Busbar commutation current test
        - Small inductive/capacitive currents
    - Circuit Breaker
        - 100%/60%/30%/10%-Short Circuit Current
        - Capacitive/Inductive Switching
        - Short Line Fault
        - Phase opposition
        - Switching of capacitor banks

- All engineering tools are available
- Contracts with suppliers are signed
- Production plan is confirmed, all assembly tools are available and proven
- Documentation of all production processes are completed (Manual, instructions, service concept)
- Education of all departments is done
- Environmental Product Declaration is written

If the above mentioned requirements are fulfilled, the **general product release** will be given at **Gate 5**.

At a final Gate 6, the experience of the pilot projects and the first series installations will be evaluated and assessed. For future projects these learnings can further improve the quality of the products and the development processes.

This overview about a development process is covering the practical methods for High Voltage Equipment development and especially at High Voltage Substations in a high grade. On the other hand, Fig. 5.3 shows parallel paths and deviations, which are the responsibility of the project leader. Such additional or parallel paths can improve the development time or can be a measure to adapt to special requirements during the project duration. Sometimes it makes sense to execute critical tests in an early stage or one Gate earlier to minimize the technical risk.
Even if some early tests are not successful, it gives a lot of early information. This gains time to improve the idea and to keep the targeted project end.

6 Origin of lightning and lightning protection measures

Atmospheric lightning discharges can cause significant **damage**. It can result to failures and damages in electro technical systems caused by external over voltages.
When we consider important systems like 3-phase power networks or communication and data transfer-systems, such systems have to be protected efficiently.
For security reasons high voltage equipments are usually tested with standardized lightning impulse voltages to prove the dielectric withstand ability. The probability that a lightning discharge will occur in Germany is given in the range of 2 lightning's every km^2 and every year. Considering the large extend of the German power grid, lightnings are occurring regularly. Due to the climate change effects, this probability will rise in the near future.

6.1 Origin of Lightning

The **appearance of a thundercloud** is initiated by strong up winds and humid air.

Two main types of thunderstorms are observed:

1. In case of thunderstorms created by heat effects, the earthy air will be heated up and this lead more and more to a non-stable lamination of **heated air at the bottom and cold air at the top**. Local irregularities like earth unevenness will lead to rise of the humid and heated air in a short time and with high velocity in the sky. The more the heated air is growing in height, the more it will be cooled. This leads to the effect that the humidity is condensing thus creating big swelling clouds reaching up to 10 km into the troposphere. Thunderstorms correlated with heat are usually appearing in the summer months with earthy temperatures above 30° C. They are present mainly in the afternoon and mainly at the edges of the mountains.

2. In case of thunderstorms created by air frontiers, a **cold front** is moving below a layer of humid and heated air. Consequently, an upward directed air flow is created comparable to the effect 1. Frontier thunderstorms tend to move with the cold front in eastern direction towards the low pressure area (northern

hemisphere: zone of western winds). Such thunderstorms are often present in the unstable times of the year => during change of the seasons.
The chimney-like up winds reaches velocities of 5 to 30 m/s. Due to this relative movement combined with physical effects, such up winds will separate positive and negative charges.
On the way upside, the humid air will condense and will result to ice crystals at temperatures below 0 degree (at a height of ca. 5 km, the temperature of the thundercloud will go below 0° C) thus representing a positive charge. The generated water droplets are heavy and move downside thus representing a negative charge.
The charge separation can be caused by different processes: sputtering of the droplets, separation of the ice crystals, the dipole effect of droplets, which are separated in positive and negative droplets or simply the relative movement of chargeable air.

A typical charge distribution within a thundercloud is represented by a zone of positively charged ice crystals (+20 As) which is located at the top of the thundercloud in a height of 7 to 10 km.
The center of the negative charge (-24 As) can be found in a height of ca. 5km.
In many cases a further limited zone (+ 4 As) of positive charge will occur, which is created by “graupel”- pellets or snow-pellets on their way down to earth. It is called “positive rain”.

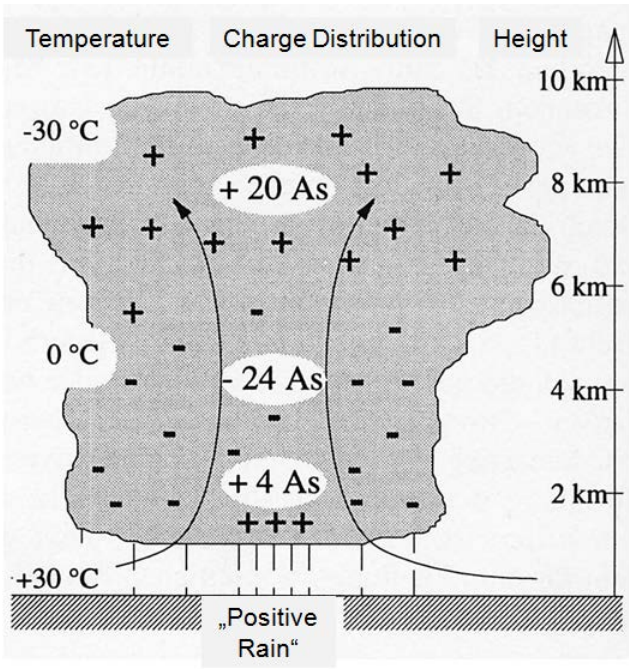


Fig. 6.1 Typical charge distribution within a thundercloud [Source: Küchler]

The development of a thundercloud is completed usually within 30 to 45 minutes. After that period, the up winds will decrease and cold fall winds are created, which are acting like thunder squalls at the ground. After ca. 30 minutes the rain will be initiated.
The overall process of thunderclouds can be repeated by generation of new thundercloud cells thus keeping a long duration of the lightning.

Which types of lightnings are known in practise?

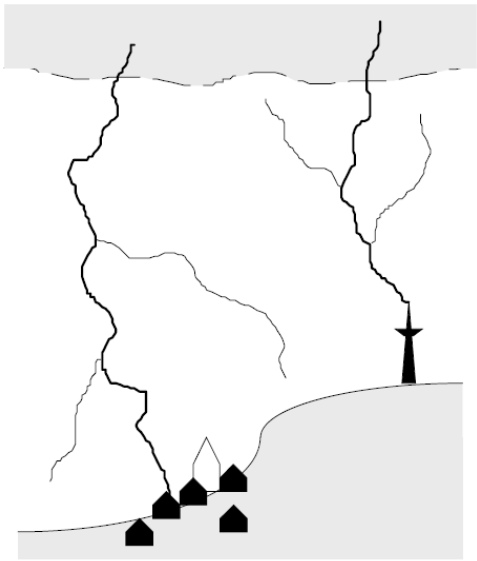


Fig. 6.2 Upward and downward directed lightning‘s [Source: Küchler]

We can divide into downward directed lightning of a high probability (left side of Fig. 6.2) and -less likely- upward directed lightning as illustrated in the right side of Fig. 6.2. The appearance of upward directed lightning requires a field enhancement as given by geometries of high magnitude. This field enhancement can initiate a starting point located at the edge of such a geometry leading to an upward lightning.

Please note: most of the lightning’s (not illustrated in Fig. 6.2) appear inside a cloud itself or between two thunderclouds (Cloud-cloud lightning), but these lightning’s does not influence equipment or human beings and therefore are not considered here.

The polarity and magnitude of the related charge of a lightning is depending on the point of origin. The most probable lightning is negatively charged and driven by the zone of -24 As according Fig. 6.2. The reason behind this is that

-24 As is covering a big area near to the earth.
How does the lightning find his way down to earth? This question will be answered by the following scheme of a step-by-step-model. Figure 6.3 shows the example of a negative thundercloud-earth lightning which is directed downwards:

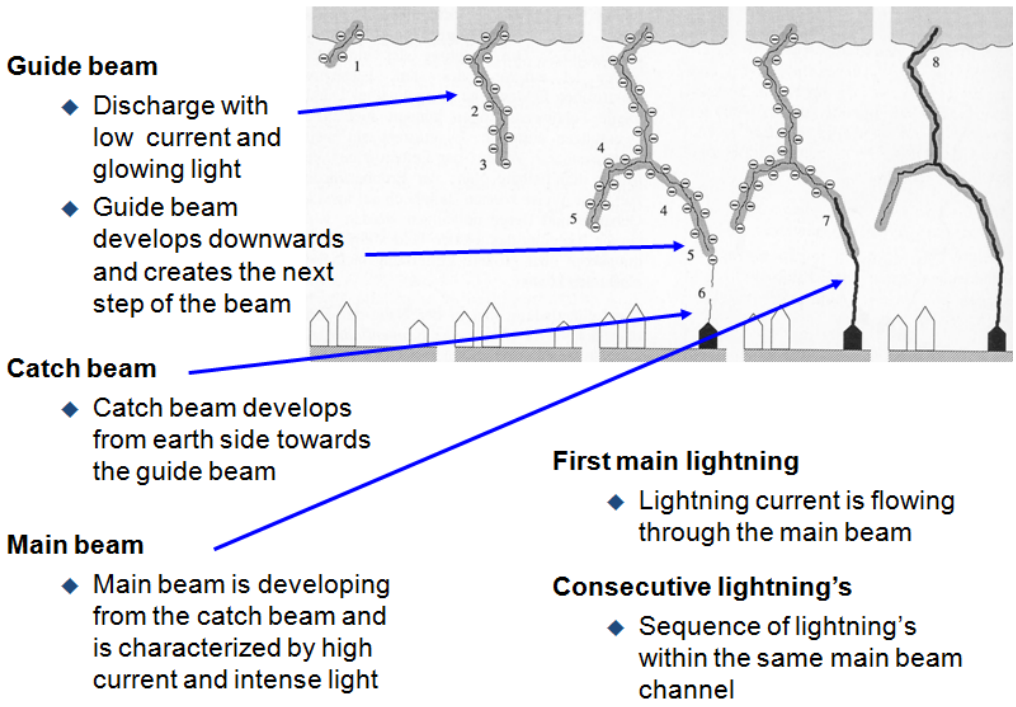


Fig. 6.3 Scheme of a step-by-step-model and definition of terms

According Figure 6.3 the development of the negatively charged lightning can be divided into a **guide beam** starting at the thundercloud with a duration between 300 and 1000 µs, the **catch beam** starting from earth and developing towards the main beam, the **main lightning** (main beam) with a duration of ca. 10 to 100 µs and the **consecutive lightning's** in the range of 10 ms to some 100 ms.
The guide beam is initiated when the breakdown electrical field strength is crossed. Then the beam starts as streamer discharge which builds up a long first channel comparable to a leader. The length of the first channel is limited by the amount of available charge. After ca. 15 to 100 µs, the needed amount of charge is reloaded into the existing channel. When the electrical field strength at the head of the first channel is high enough, then the second channel is created in one step. Therefore the lightning is progressing step-by-step.
The guide beam is progressing in ca. 50 m long steps (step-by-step lightning).

The direction of the single beam steps is not regular due to the strongly distorted electrical field. Every kind of junctions (y, T etc.) can occur due to local field enhancement thus creating a non predictable distribution of the guide beam and side beams. The distribution occurs during the way towards the earth. The influence of protrusions at the earth (tower, mountains, skyscraper) on the guide beam is not given when the distance to earth is high, because the guide beam is driven by the local field enhancement at the head of the channel.

If the guide beam is in the near of the earth, a ca. 10 m long catch beam is initiated, which tend to move towards the guide beam. As soon as the catch beam meets the guide beam, the lightning is started.
The limited radius of the catch beam is the reason that lightning's can occur directly beneath towers or mountains. Furthermore, the limited radius is limiting the protection area of protection measures.

The discharge is growing in the up to now semi conductive channel towards the thundercloud. By reaching the cloud, the channel is now heated and therefore becomes conductive and carries the charge of the cloud segment to earth in an intense short time current (impulse current).
This main lightning is the phenomenon that we can recognize as thunder and lightning. The current curve reaches his peak value of some 100 kA after some microseconds and will decline in a range of some 100 microseconds.

The first lightning leads to a high ionization of the channel, which can initiate a sequence of consecutive lightning's, which are usually lower with respect to the peak value.

Please note: in reality the current curves deviates strongly. For test purposes, the standards define a lightning impulse test voltage with a rise time of 1,2 µs and a tail time of 50 µs to test electro technical equipment. Every new product has to demonstrate the lightning impulse withstand ability by applying test voltages depending on the voltage level.

To judge possible damages of lightning, we distinguish between direct and indirect impact on electrical/electronic systems, human beings, animals, buildings, trees etc.

Direct influences are given i.e. by electromagnetic waves, over voltages or the impact of electromagnetic forces on overhead lines of power grids.
Additionally, thermal effects can damage active parts when affected by a

lightning impulse current.

Indirect influences are created by voltage drops appearing at ohmic and inductive impedances. Such voltages across a resistance or an inductance lead to potential jumps in earthed networks and possibly can cause “backward flashover voltages” of earthed conductors into active conductors of electronic systems.
When electrical loops are present, the induction principle of the magnetic field of a lightning impulse current can cause high induced voltages, which can damage electronic systems. Furthermore it is possible, that such induced voltages lead to flashover effects between conductors thus generating a short circuit current.

The Lightning is characterized by four parameters with regard to the lightning impulse current, which allows the estimation about the possible damage:

1. The **peak value** of the current determines the evaluation of the maximum voltage appearing at ohmic earthing resistances and secondly the calculation of over voltages due to traveling waves.
   The peak value of the lightning current is varying between 5 and 100 kA, in seldom cases even some 100 kA can occur.

2. The **rate of rise** of the current di/dt is responsible for voltage drop at inductive parts of a lightning arrester and is responsible for induced voltages in neighbored loops.
   For this reason, the rate of rise of the current is the most important parameter to evaluate the coupling effect into electronic systems. Usually the values for the rate of rise of the current di/dt are ranging between l and 100 kA/ µs.

3. The **charge** of lightning currents $\int i \cdot dt$

   The charge is an indication for the energy converted at the arcing foot point, when we assume a constant arcing voltage. This energy is responsible for the melting of metallic conductors at the foot point. The bandwidth of the charges is varying between 0.5 und some 100 As.

4. The **integral of the current square** $\int i^2 \cdot dt$

   is responsible for the heat energy applied to the conductors $R \cdot \int i^2 \cdot dt$

   $\int F \cdot dt$

On a conductor a electromagnetic force is active

Typical values are 10^3 up to 10^7 A^2s.

The development of a lightning discharge over time will be explained for a downward directed lightning.

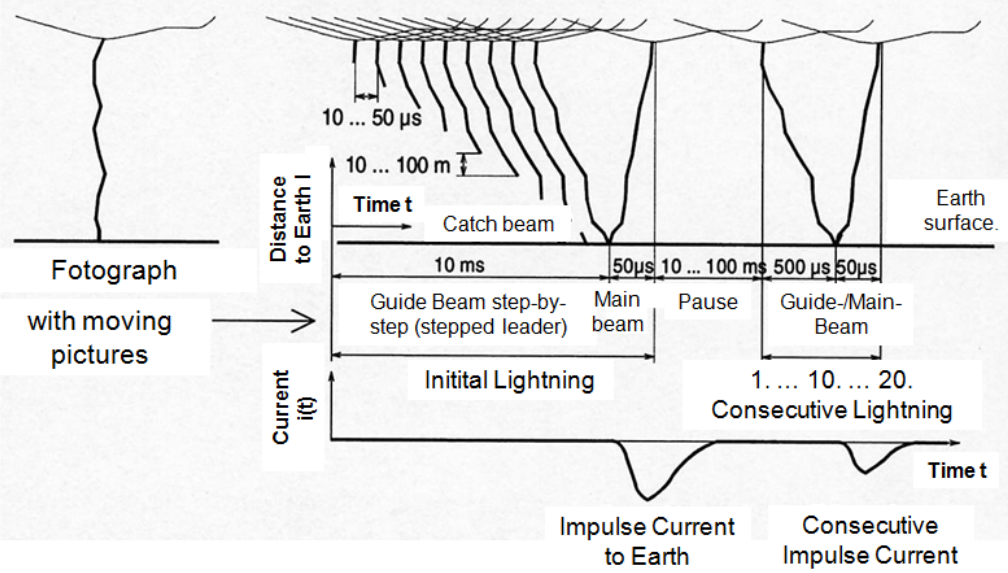


Fig. 6.4 Development of a negative lightning over time directed from thundercloud to earth [Source: Hasse/Wiesinger]

The human eye recognizes the lightning as one event showing a bright lightning channel, which usually is characterized by many junctions. The real character of a lightning can be demonstrated by a camera with moving pictures. Starting at the thundercloud, leader discharges are generated in time steps of 10...50µs using the same channel and growing ca. 10 to 100m every step forward, thus representing **stepped leader**.

At the last step of the leader discharge, a catch beam is growing from earth towards the guide beam. After ca. 10ms the **guide beam** is connecting with the **catch beam** thus initiating the **main beam**. This main beam is carrying some Ampere seconds (As) over a time of ca. 50 µs. The resulting lightning currents are characterized by a steep rise of current, so called impulse currents to earth. These currents can reach amplitudes usually of some 10kA, which heat up the lightning channel up to some 10.000K. The sudden thermal expansion of the air (before and after the lightning) creates the sound of the thunder.
The initial lightning usually does not carry the complete charge of the

thundercloud. During a pause of 10 to 100ms the local lack of charge is compensated within the cloud and some consecutive lightning's follows using the same high conductive channel. The main difference between the initial lightning and the consecutive lightning's is that the last ones do not need stepped discharges, but forming the main beam in one step. The amplitudes of the consecutive lightning are smaller than compared to the initial one.

The typical sequence of a consecutive lightning is usually 10, but can reach even 20 to 40.

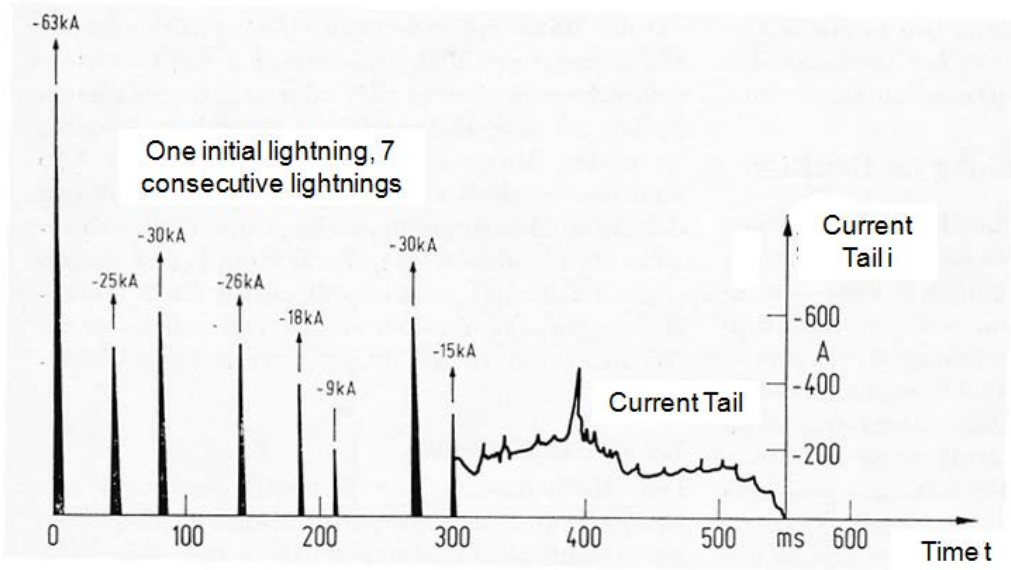


Fig. 6.5 Main Beam and Consecutive Beams in case of negative lightning [Hasse, Wiesinger]

Fig. 6.5 shows one initial lightning and 7 consecutive beams. The remaining charge of the thundercloud is carried by a current tail of some hundred A and a duration of some 100ms.

Positive Lightning's usually are not characterized by consecutive beams, but provide extremely high amplitudes up to 200kA. The related rate of rise of the current is high. Comparable to the negative lightning, the remaining charge of the thundercloud will be carried by a current tail.

6.2 Lightning Protection

The main task of the lightning protection is to prevent buildings and systems

from the impact of a lightning. The protection can be divided into an external protection, which leads the lightning current across a defined way, and the internal protection against the impact of currents and over voltages.

The external lightning protection to protect an object consists of catching elements called Lightning arresters with conducting connections to an earth system.

The catching element is an exposed conductive element, which creates a local field enhancement thus initiating a catching beam directed towards the guide beam.
During this process, radial distances of r = 20 to 45m will be crossed by the final stepped leader. Using the geometrical model according Fig. 6.6, the guide beam is approaching the earth **or** the catching element (here catching rod) exactly up to the radial distance of r.

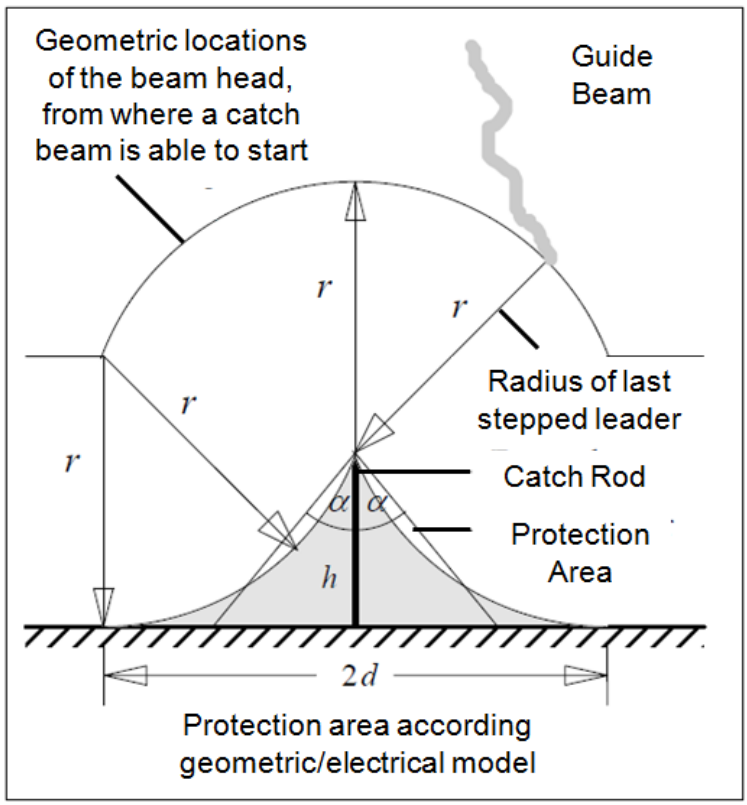


Fig. 6.6 Protection area of a catching rod (geometrical model) [Source: Küchler]

The physical basis of that geometric/electrical model is the step-by-step leader development, which finally initiates the main beam thus representing the

lightning.
The step length is depending on the amplitude of the lightning current. For protection reasons, only the last step of the lightning development is important. When assuming the most critical case, a radius of 20 m has to be chosen. Around the edge of the protection catching rod, a circle with radius 20 m is applied. By superposing this circle with a 20m-Parallel to earth, the resulting contour represents the decision line of the last stepped leader! Directly at the crossing point, the beam can cross directly to earth or directly into the protection rod.
The protected area is marked in gray in Fig. 6.6. This zone cannot be reached by the beam! A simplified construction of the protection area is to apply a 45° protection angle from the edge of the catching rod down to earth (more risk).

If complex constructions or large-scaled areas have to be protected, the position of the catching rod has to be evaluated in 3D. This can be easily done by scrolling a “lightning sphere” with the radius r through the complete layout. The protection volume is characterized by the lowest 3D-curve of the sphere. Any structure touched by the sphere is not protected!
In case of medium requirements, a sphere with a radius of 45 m is used, in case of high grade of protection, a radius of 20m should be chosen.

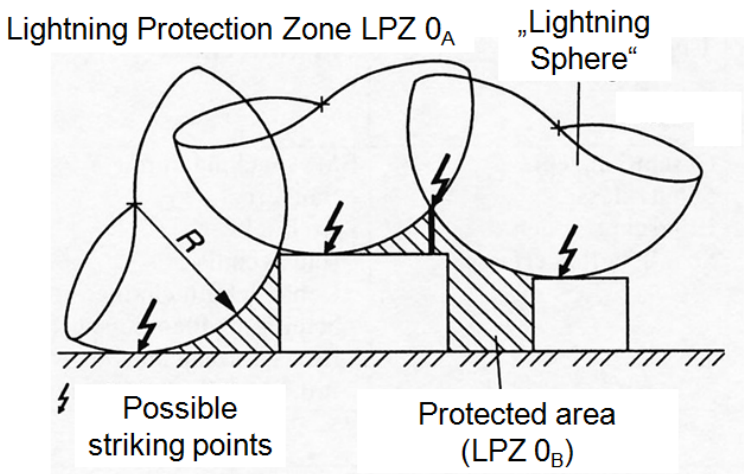


Fig. 6.7 Sphere model “Scrolling of a Lightning Sphere“ [Source: Küchler]

An application example for this sphere-model is a large solar plant. Such plants are preferably protected with catching rods. To design the catching rods in optimal way, the complete facility is built up in 3D and scaled i.e. by the factor 1:100. Then the catching rods have to be defined in the height and their locations such that a sphere can be scrolled above without touching any

protecting device.
Please note: In case of skyscrapers a protection device at the top of the building cannot prevent a lightning to the side of the building. Therefore i.e. additional catching rods have to be installed along the side of the building.

Lightning Protection for Buildings:

Roof ridges are protected by catching conductors applied in parallel to the roof. Inclined roofs have to be covered by conductive knitting. To prove the protection of the knitting, a sphere scrolling should not create a contact to the roof.
Practically, a knitting of 5m x 10m is seen as sufficient protection.
It is of high importance to connect all metallic parts (roof gutter, roof windows) to the catching conductors. Chimneys have to be protected by catching rods.

The current path to earth has to be designed low ohmic and low inductive. The maximum length of the current path is considered to be below 20m.
The practical experience shows that the current path should be conducted across building edges and roof edges to ensure a mashed cage.

Please note: Optimal mashing of the conductors can be reached in buildings built up with armored concrete. But in this case the armoring of the single elements has to be connected before casting. Additionally, the armoring conductors have to be integrated into the lightning protection system.

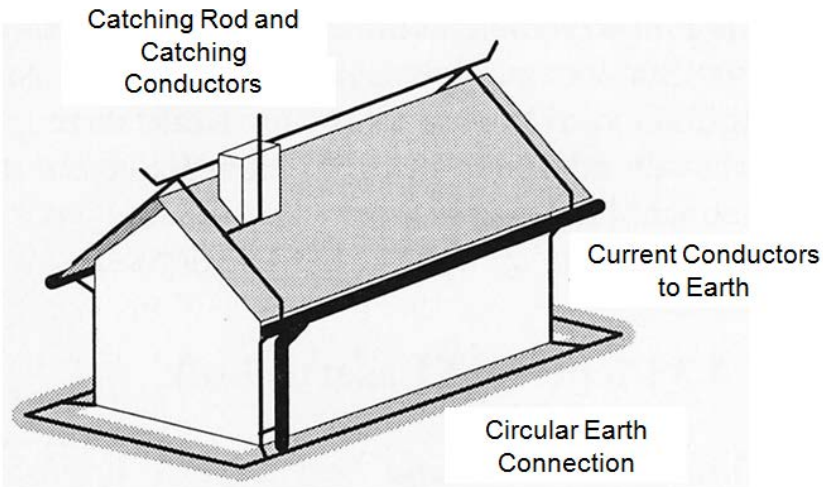


Fig. 6.8 Connecting Scheme of Lightning Protection Elements

Please note: Overhead Lines are protected by earth conductors at the top of the pylon. The pylon itself acts as current carrying path. Therefore it is of high importance to have a good connection to earth (low earth resistance). The earth system can be realized by a large-scale fundament earth.
As alternative, a closed loop of conductors around the building can be the earth system in combination with a single earth element of high deepness at each edge.
If only single earth elements of high deepness are used, the danger of an unsymmetrical potential distribution is given.
It is recommended to ensure a symmetrical current conducting system in order to reduce the magnetic field component (parallel current path at each edge of the building) and the amplitude of the induced over voltage.

Please note: the earth system has to be checked periodically by measuring the earth resistance. Therefore the connection of the current conducting system has to ensure that it can be separated for short measuring time. During this check, special security measures have to be done: i.e. protection sphere gaps should be connected in between the separated systems to avoid over voltages! Such over voltages could appear when a short circuit current is present in one of the two systems.

When different metals are connected to each other, which are located in the ground, electro chemic corrosion will take place. A countermeasure could be i.e. silver plating of the contacts. In extreme case, the earth systems can be separated and connected via a disconnecting sphere gap.

The internal lightning protection requires a potential equalization usually in the basement of a building. With aid of this equalization, dangerous potential differences between lightning protection system and earth system of the building are prevented.
The potential equalization comprises all galvanic components like heating installation, gas, water installation and lightning protection elements.

Please note: Lightning current arrester are activated by over voltages and should be able to carry highest currents. Sparking gaps are used as rude protection, electronic devices act as fine protection.

Flashover voltages between the lightning protection system and building installations have to be avoided either by a reliable distance or by galvanic connection. The necessary distances are increased with increasing loop area between the current path and the second earth system. In case of large

buildings, the potential equalization should be done on several levels.
Sensitive devices (computing center) have to be protected by several measures. These measures are can be described by **protecting zones**, comprising each zone with different security standards:

- The external lightning protection system prevents from direct lightning and this area is classified as zone 0.
- Under this external protection system, electrical shielding is provided (i.e. shielding of the building, shielding of the room or shielding of the device). All conductors and cables are shielded and this shielding will be connected inside this zone 1 with the potential equalization. Additionally, current and voltage arresters are applied thus resulting in an effective protection against magnetic and electric fields and against over voltages coming from the cable.

The different "Lightning Protection Zones" LPZ are explained in Figure 6.9.

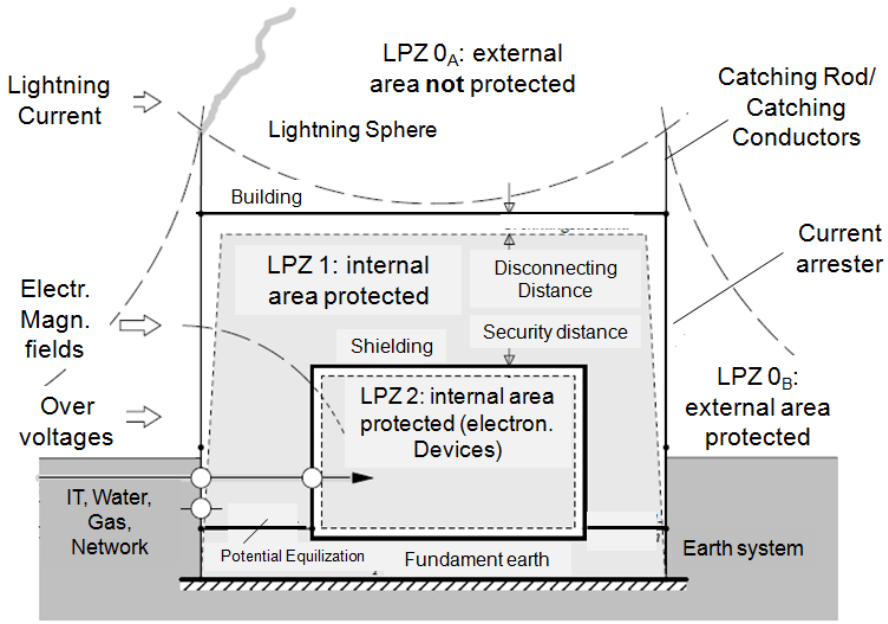


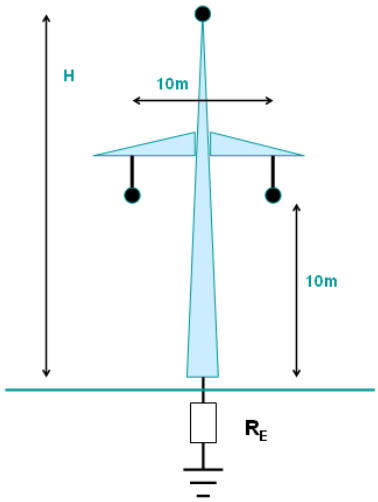
Fig. 6.9 Internal and external Lightning Protection and related Potential Equalization [Source: Küchler]

Zone LPZ 0 is representing the area which is not protected. Zone LPZ 0_B is only protected by direct lightning, but not protected against the electromagnetic influence.
Zone LPZ1 is the protected area. Induced voltages and potential drops are avoided by disconnecting distances between installations and the shielding of the building. Sensitive electronic devices are protected within zone 2, providing a closed shielding cage. Such shielding cages have to be connected

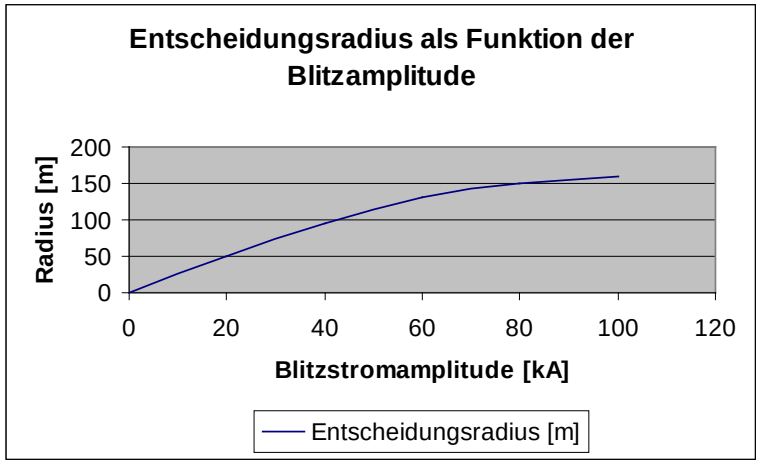
by low inductive conductors to the potential equalization.

Exercises

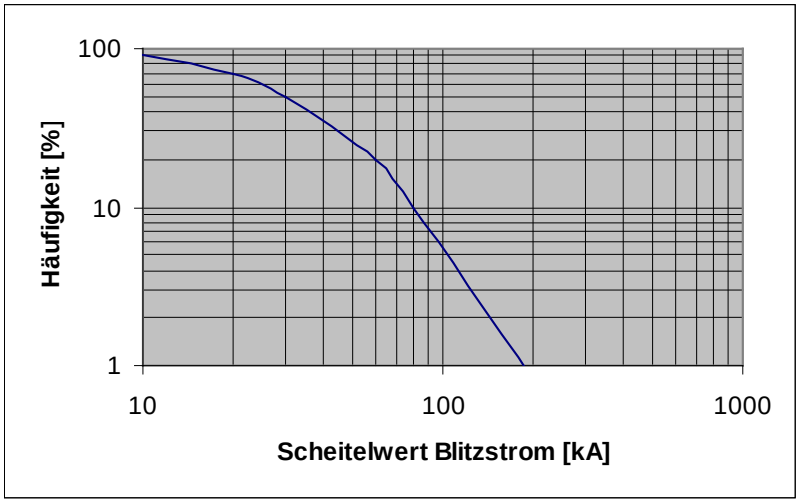
6.1 Example of a lightning protection of an overhead line to avoid a direct lightning into the phase line:
Assuming an alpine environment, an AC-123 kV-Railway-Overhead Line shall be installed over a length of 100 km. The design of the pylons is shown in the following scheme. The distance between the two phase lines and the distance to earth are given. The impulse voltage strength of the line insulators is 600 kV. It is only one earth line foreseen at the top of the pylon with a unknown height H.



1. Please evaluate the necessary height of the earth line, when a direct lightning into the phase line should be prevented for lightning currents of $\hat{i} > 20$ kA? Help: please use the following diagram.
2. Please check, if the protection is given for lightning currents of $\hat{i} > 15$ kA and $\hat{i} > 40$ kA.



3. Please calculate the maximum value of the earth resistance R_E, when only every 10. Lightning into the earth line will lead to a backward flashover into the phase line.



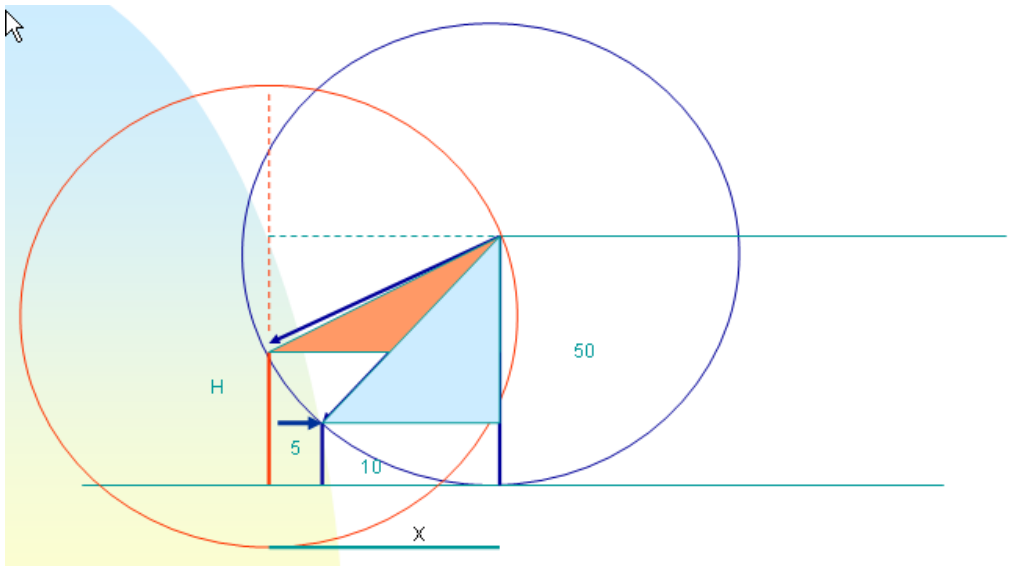
Details to evaluate the solution:

The geometric solution can be simplified by considering the fact that the arrangement is symmetrical to the pylon center. For this reason it is allowed to focus on the right side of the pylon.
Procedure: first a parallel to the earth should be applied in a distance equivalent to the decision radius „r“. Now we create a circle with radius “r” around the phase line which has to be protected. At the crossing point, an additional circle with radius “r” has to be made. Below this important circle, the area is protected against a direct lightning.
If you extend now the circle until it meets the axis of the pylon, the crossing point is representing the minimum height of the earth line.
Please answer: when the guide beam meets the parallel line to earth, the lightning will hit into _____. When the guide beam is meeting the circle above the earth line, the lightning will hit into _____.

Concrete solution:

1. For 20 kA a decision radius of 50m results. This means that the protection area of the earth line must cover the line of the phase voltage.

Geometric Solution:



Solution for the first triangle:

$$40^2 + (x-5)^2 = 50^2 \qquad \Rightarrow \qquad (x-5)^2 = 900 \Rightarrow x = 35$$

Solution for the second triangle:

$x^2 + (50-H)^2 = 50^2$ [redacted] $\Rightarrow H =$ [redacted]

3. The earth resistance R_E can be calculated by:

$$U_{Isolator} = R_E \cdot \hat{I} + L \cdot \frac{di}{dt} + \hat{U}$$

with $U_{Isolator} = 600\,kV$

From 10%-probability follows:

$\hat{I} =$ [redacted]

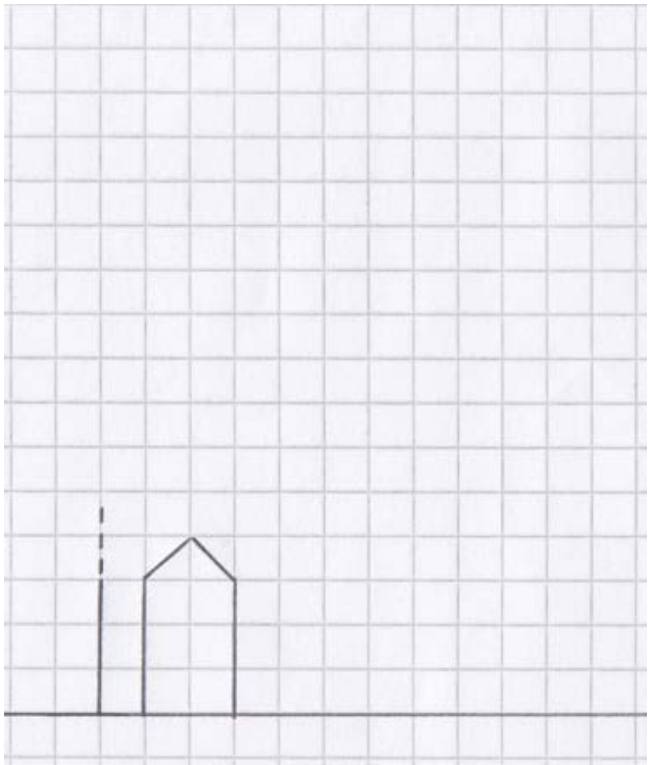
Thus leading to the earth resistance: $R_E =$ [redacted]

Question: Which simplification is made and why is it tolerable?

6.2 A building shall be protected against lightning. The external contour of the building is given by the following scheme. The used catching rods are located 5m left of the building.

Please evaluate the minimum height of the catching rod, when the building shall be protected against lightning currents $\hat{i} > 30$ kA.

Detail: one casket is representing 5m



List of References

Special Literature „High Voltage Engineering“

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Andreas Küchler	**High Voltage Engineering,** 1. Edition, Springer publishing company, Berlin Heidelberg, 2011, ISBN: 978-3-642-11992-7
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Hints for protection measures, when human beings are outside in nature

Short excursus: how can human beings can be protected against lightning's?

The most critical danger is present, when human beings are in nature. Two types of danger can be divided:

- Direct lightning protection outdoor:

When the human is small compared to the environment, the grade of danger is low. Therefore the following measures lower the probability of a direct lightning:

  - **Crouch down**
  - Be far away (> 3m) from trees or similar objects of increased height

- Danger caused by voltage drop in the ground:

When a lightning hits the ground, a potential cone is resulting. This voltage cone is decreasing with distance from the lightning location. The potential difference is the critical danger. Therefore you should minimize the length which is covered by your body to minimize the potential difference.

  - **Close your legs**
  - Hands close to the corpus, not touching the ground!
  - Crouch down instead of lying down!

The best protection is a Faraday cage, i.e. within a car (car body should be of metal, carbon fiber has to be checked).